Running least common multiples: distinct values and multiplicities

Erdős Problem #269

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July 2026; revised 5 October 2026

Abstract

We study the running least common multiples of integers whose prime factors belong to a fixed finite set. For $\{2,3,5\}$, a five-map argument proves irrationality of the sum over distinct running LCMs. Two-prime sums reduce to a transcendental Hecke–Mahler value. For the repeated three-prime sum we give an equivalent residue criterion, whose arithmetic condition remains open. Complete ordinary arguments for general sums over distinct running LCMs and prime-power subseries are retained as remarks awaiting formalisation.

1 Two sums from running least common multiples

Let P be a finite set of at least two primes, and let $\mathcal{S}_P$ consist of 1 and all positive integers whose prime factors belong to P. For $x \geq 1$, put

$$\mathrm{L}(x) = \operatorname{lcm}\{u \in \mathcal{S}_P : u \leq x\}, \qquad \mathrm{H}_P(x) = \prod_{p \in P} p^{\lfloor \log_p x \rfloor}.$$

Every integer in this finite set divides $\mathrm{H}_P(x)$, and the set contains the largest power of each prime in P below or equal to x. Their product therefore equals the least common multiple: $\mathrm{L}(x) = \mathrm{H}_P(x)$. The value changes only at a positive prime power. Summing its distinct reciprocals gives

$$\mathcal{D}_P = 1 + \sum_{\substack{t = p^n \\ p \in P,\ n \geq 1}} \frac{1}{\mathrm{H}_P(t)}.$$

We begin with an irrationality result for three primes.

Theorem 1.1 (the sum over distinct running LCMs at $\{2,3,5\}$)**.** *The sum $\mathcal{D}_{\{2,3,5\}}$ is irrational.*

Authorship and AI use. Will Cook built and directed the research infrastructure and reviewed the claims when he could. AI agents did most of the research and drafting. Cook did not independently verify every claim.

The proof in Section 2 gives the primes three different roles. Powers of 2 delimit the blocks; powers of 5 occur often enough to separate the five affine tail images; powers of 3 have an irrational block frequency. Separation makes each tail determine the next block and tail. Under rationality there are only finitely many tails, so repetition would force an impossible periodic block sequence.

Erdős asserted the general conclusion for every finite P with $|P| \geq 2$ in his letter of 1 January 1973, without printing a proof [3, p. 335]. The theorem above is a special case of that assertion. The general argument in Section 5.1 compares nearby tails and finds an isolated interchange of jumps on a torus. We retain it as an ordinary argument awaiting formalisation, with its intermediate claims stated as remarks. The specialised five-map proof is independent of that argument.

The question in Erdős Problem #269 retains multiplicities. If $a_1 < a_2 < \cdots$ enumerates $\mathcal{S}_P$, it asks about

$$\mathcal{R}_P = \sum_{n\geq 1} \frac{1}{[a_1, \ldots, a_n]} = \sum_{u\in\mathcal{S}_P} \frac{1}{\mathrm{L}(u)}$$

[1, p. 65][2, p. 106]; the numbering is Bloom's [4]. For $P = \{2, 3, 5\}$, the integers 5 and 6 both have running LCM 60. They contribute $2/60$ to $\mathcal{R}_P$ and $1/60$ to $\mathcal{D}_P$. Thus Theorem 1.1 concerns the sum over distinct values, whereas the problem concerns the sum with multiplicity. The arguments in this paper leave $\mathcal{R}_P$ unresolved when P is finite and $|P| \geq 3$.

For two primes the conclusion is stronger.

Theorem 1.2 (two-prime transcendence)**.** *Let p and q be distinct primes. Then $\mathcal{R}_{\{p,q\}}$ and $\mathcal{D}_{\{p,q\}}$ are transcendental.*

The Lean proof assumes the transcendence theorem of Bugeaud and Laurent.

Fan's post of 26 June 2026 gave the finite-prime running-LCM identity, the factorisation of $\mathcal{R}_{\{p,q\}}$, and the Hecke–Mahler reduction with its transcendence conclusion [17]. We reproduce that calculation in Section 4. Both values are nonconstant polynomials over $\mathbb{Q}$ in one boundary sum, whose transcendence follows from Loxton and van der Poorten [16, Theorem 8, p. 40], in the formulation of Bugeaud and Laurent [15, Theorem 1.1]. No priority is claimed for this deduction.

Retaining multiplicities preserves the bases but changes the numerators. The repeated tails satisfy $X_{a+1} = b_a X_a - m_a$, with $b_a \in \{2, 6, 10, 30\}$, while both m_a and X_a grow quadratically. The preceding finite-tail repetition argument no longer applies. Rationality would still make every sufficiently late BX_a integral, where B is the denominator part coprime to 30. A finite segment excludes such an integer tail when its least positive residue exceeds the quadratic bound. Irrationality is equivalent to finding these segments for every B beyond every prescribed onset. The finite computations do not establish either unbounded quantifier.

The five-map proof comes first. Sections 3 and 4 then supply the running LCM geometry and two-prime formulae. For general distinct sums, Section 5.1 uses comparison of two tails rather than recovery from one; the finite-state, subseries and linear-flow extensions follow as ordinary arguments pending formalisation. Section 7 separates exact kernel rank from approximation. For the repeated target, Sections 8–10 derive the

coefficients, bounds and residue criterion; the remaining sections distinguish finite certificates from missing arithmetic inputs. Section 12.5 records formal coverage. The short paper concentrates on the five-map theorem and the repeated-tail criterion.

We use $\mathbb{N} = \{0, 1, 2, \ldots\}$ for exponents and braces for fractional parts. In the finite-geometry section and in the kernel calculation, $P = \{p, q, r\}$ consists of three pairwise distinct primes, $\mathrm{H} = \mathrm{H}_P$, and

$$\mathrm{K}(i, j, k) = \mathrm{H}(p^i q^j r^k)^{-1}.$$

The other sections specify their prime sets separately. For $\{2, 3, 5\}$, $\mathcal{S}_P$ consists of the usual 5-smooth integers. This fixed support differs from a varying smoothness bound in $\Psi(x, y)$ [11]. Tijdeman and Meijer [28, Sections 1–4] and the two-prime study of Languasco, Luca, Moree and Togbé [29] give the relevant lattice-counting context; the estimates below use no smooth-number density asymptotic or gap theorem. Kovač and Tao [13] study other unit-fraction series by elementary methods, without providing an input to these proofs.

Keywords. irrationality; transcendence; least common multiple; smooth numbers; Cantor series; separated rank. **MSC 2020.** 11J72 (primary); 11A05, 11N25, 68V20 (secondary).

2 Recovering the prime-power jumps

As in long division, periodicity requires both finitely many remainders and a deterministic rule for the next step. The varying bases make the second requirement nontrivial. We obtain it by separating the five possible block images, using a bound on the actual arithmetic tails.

Proof of Theorem 1.1. Finitely many tails under rationality. Write $\mathrm{H} = \mathrm{H}_{\{2,3,5\}}$ in this proof and put $P_a = \mathrm{H}(2^a)$. Every running LCM at or before 2^a divides P_a. We therefore normalise the tail after 2^a by

$$Y_a = P_a \sum_{\substack{t > 2^a \\ t = 2^n, 3^n \text{ or } 5^n,\ n \geq 1}} \frac{1}{\mathrm{H}(t)}. \tag{1}$$

Each subsequent jump multiplies the running LCM by at least 2, and a jump by 3 eventually occurs. Comparison with $\sum_{j \geq 1} 2^{-j}$ gives $0 < Y_a < 1$.

Suppose that $\mathcal{D}_{\{2,3,5\}} = N/K$, where N is an integer and $K \geq 1$ is an integer. Clearing the finite prefix gives

$$KY_a = P_a N - K \left(P_a + \sum_{\substack{t \leq 2^a \\ t = 2^n, 3^n \text{ or } 5^n,\ n \geq 1}} \frac{P_a}{\mathrm{H}(t)} \right) \in \mathbb{Z}.$$

Thus the Y_a lie in the finite set $(K^{-1}\mathbb{Z}) \cap (0, 1)$. It remains to show why equality of two tails propagates.

Recovering a block from its tail. Consider the jumps between 16 and 32, including the last endpoint. They occur at $25, 27, 32$, multiplying the running LCM successively by $5, 3, 2$. After normalisation at 16, their contribution is $1/5 + 1/15 + 1/30$. The remaining tail is normalised at a running LCM 30 times as large, so

$$Y_4 = \frac{1}{5} + \frac{1}{15} + \frac{1}{30} + \frac{Y_5}{30} = \frac{9 + Y_5}{30}.$$

In the block $(64, 128]$, the interior jumps occur in the opposite order, at $81, 125$. Its constant term is $1/3 + 1/15 + 1/30 = 13/30$. The multiplier 30 records the primes, but not their order; the numerators 9 and 13 retain the order. The remaining question is whether the unknown following tail can conceal this difference.

In general, $(2^a, 2^{a+1}]$ contains its terminal power of 2, at most one power of 3 and at most one power of 5. Let τ_a denote the sequence of interior prime labels, omitting the terminal 2. Splitting off the block gives

$$Y_a = G_{\tau_a}(Y_{a+1}), \qquad G_\tau(y) = \frac{\mu_\tau + y}{b_\tau}, \qquad \begin{array}{c|ccccc} \tau & \emptyset & 3 & 5 & 35 & 53 \\ \hline b_\tau & 2 & 6 & 10 & 30 & 30 \\ \mu_\tau & 1 & 3 & 3 & 13 & 9 \end{array} \tag{2}$$

Here b_τ is the factor by which the running LCM increases across the block, and μ_τ / b_τ is the block's normalised contribution.

The five formulae alone do not recover τ_a from Y_a. The coarse images $G_{53}([0,1]) = [3/10, 1/3]$ and $G_5([0,1]) = [3/10, 2/5]$ overlap. We need a smaller interval known to contain the next *arithmetic* tail. The table gives the lower bound $Y_a \geq 3/10$. With that lower endpoint, separating these two images requires an upper endpoint U satisfying

$$G_{53}(U) < G_5(3/10), \qquad \frac{9 + U}{30} < \frac{33}{100}, \qquad U < \frac{9}{10}.$$

This specifies the improvement needed from the arithmetic sequence.

A power 5^f belongs to block $\lfloor f \log_2 5 \rfloor$. These indices start at 2 and their successive differences are 2 or 3. Every three consecutive blocks therefore contain a power of 5. Given a, let $j \in \{a, a+1, a+2\}$ be the first such block. The maps for blocks containing 5 give $Y_j \leq 7/15$, since $Y_{j+1} < 1$ and $G_{35}(1) = 7/15$ bounds $G_5(1)$ and $G_{53}(1)$ as well. We now work *backwards* from j to a. Each intervening block has map $G_\emptyset$ or G_3, and $G_3(y) \leq G_\emptyset(y) = (1 + y)/2$ for $y \geq 0$. At most two backward steps give the successive bounds $7/15, 11/15, 13/15$. Hence

$$Y_a \in I = \left[\frac{3}{10}, \frac{13}{15}\right], \qquad G_\emptyset(G_\emptyset(7/15)) = \frac{13}{15} < \frac{9}{10}. \tag{3}$$

The bound uses the occurrence of a power of 5 within three blocks. It does not assert invariance under arbitrary words in the five maps: indeed, $G_\emptyset(13/15) = 14/15$ lies outside I.

The following table gives the exact images of I. They are pairwise disjoint, with successive gaps $1/900, 17/300, 79/900, 1/180$. Writing $L = 3/10$ and $U = 13/15$, the two narrow gaps are

$$G_5(L) - G_{53}(U) = \frac{3L - U}{30} = \frac{1}{900}, \qquad G_\emptyset(L) - G_3(U) = \frac{3L - U}{6} = \frac{1}{180}.$$

Both are positive because $U < 3L = 9/10$.

τ	$G_\tau(I)$
53	$[31/100, 74/225]$
5	$[33/100, 29/75]$
35	$[133/300, 104/225]$
3	$[11/20, 29/45]$
$\emptyset$	$[13/20, 14/15]$

Since $Y_{a+1} \in I$, the actual tail Y_a belongs to exactly one of these five image intervals, identifying τ_a. For instance, $Y_a \in [31/100, 74/225]$ forces $\tau_a = 53$ and gives $Y_{a+1} = 30Y_a - 9$. In general we calculate $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$, identify its image interval, and continue. Each recovered value is the next arithmetic tail, so the same bound I remains available at every step.

Repetition would give a rational frequency. The finitely many tail values obtained above include a repeated value $Y_u = Y_v$ with $u < v$. Since the image intervals are disjoint, this forces $\tau_u = \tau_v$. Applying the corresponding inverse map gives $Y_{u+1} = Y_{v+1}$. Induction yields $\tau_{u+j} = \tau_{v+j}$ for every $j \geq 0$. Thus the entire later block word has period $v - u$.

To exclude this possibility, let $\alpha = \log 2 / \log 3$. The number of powers of 3 in block a is $\lfloor (a+1)\alpha \rfloor - \lfloor a\alpha \rfloor$, and is determined by τ_a. If the block word has eventual period $\ell \geq 1$, let c be the number of 3-powers in one period. For every sufficiently large fixed a and every $n \geq 1$ we then have

$$\lfloor (a + n\ell)\alpha \rfloor = \lfloor a\alpha \rfloor + nc.$$

Dividing by n and taking the limit gives $\ell\alpha = c$, contrary to unique factorisation, since it would imply $2^\ell = 3^c$. □

The Lean proof follows this route. It identifies $\mathrm{H}(t)$ with the running least common multiple, shows that the sum counts each distinct value once, identifies the contribution of each dyadic block with one of the five pairs (b_τ, μ_τ), and proves that every three consecutive blocks contain a power of 5 and that the blocks containing a power of 3 are not eventually periodic. It then applies an irrationality criterion for block series with these five pairs, whose proof contains the bounds, the separation and the return above.

The first twelve block types are $\emptyset, 3, 5, 3, 53, \emptyset, 35, 3, \emptyset, 53, \emptyset, 35$. The saved numerical calculation over the first 3000 blocks reports tails in $[0.3186, 0.7799]$. These approximations are separate from the rational bounds used in the proof.

Grouping the jumps into dyadic blocks matters. One jump at a time, the tails $x_k = (q_1 \cdots q_k) \sum_{j>k} (q_1 \cdots q_j)^{-1}$ satisfy $x_k = (1 + x_{k+1})/q_{k+1}$ and $1/4 \leq x_k \leq 1$, and the images of $[1/4, 1]$ under $x \mapsto (1+x)/2$ and $x \mapsto (1+x)/3$ overlap on $[5/8, 2/3]$. A tail before a jump by 2 cannot be told from a tail before a jump by 3 from these bounds alone.

Listing the jumps one at a time writes $\mathcal{D}_{\{2,3,5\}}$ as a Cantor series with every digit equal to 1 and every base in $\{2, 3, 5\}$. The criteria of Erdős–Straus [18, Theorem 2.1] and Hančl–Tijdeman [19, Theorem 3.1], recalled in Section 9, assume that each digit divided by the product of the last two bases in its denominator tends to 0. Here that quotient is at least $1/25$, so neither criterion applies.

3 Intervals of constancy of the running LCM

We first describe which smooth integers contribute the same running value. The LCM is determined by the largest exponent of each prime in the prefix. For the unrestricted prefix this gives $\operatorname{lcm}(1,\dots,N) = \prod_{t\le N} t^{\lfloor \log_t N\rfloor}$, with t running over primes, and hence $\log \operatorname{lcm}(1,\dots,N) = \psi(N)$; see Apostol [10] and Montgomery and Vaughan [12]. We apply the same exponent rule to a fixed prime support, keeping track of the intervals on which the running LCM is constant.

The smooth numbers up to x are indexed by the exponent triples (i,j,k) with $i \le \lfloor \log_p x\rfloor$, $j \le \lfloor \log_q x\rfloor$, $k \le \lfloor \log_r x\rfloor$ and $p^i q^j r^k \le x$. The coordinate bounds alone need not describe the prefix: each prime-power factor may be at most x while their product exceeds x. For instance, at $x = 6$ the factors 4, 3 and 5 satisfy their coordinate bounds, but the corresponding smooth number is 60.

Theorem 3.1 (the running least common multiple)**.** *Let p,q,r be pairwise distinct primes and $x \ge 1$. Then $\mathrm{L}(x) = \mathrm{H}(x)$.*

Proof. Every smooth $n \le x$ has exponents bounded by the corresponding integer logarithms, so $n \mid \mathrm{H}(x)$ and hence $\mathrm{L}(x) \mid \mathrm{H}(x)$. Conversely the three pure powers $p^{\lfloor \log_p x\rfloor}$, $q^{\lfloor \log_q x\rfloor}$ and $r^{\lfloor \log_r x\rfloor}$ are themselves smooth numbers not exceeding x, so each divides $\mathrm{L}(x)$, and distinct primes have coprime powers, so their product divides $\mathrm{L}(x)$ as well. □

At $(p,q,r) = (2,3,5)$ the first ten values are

x	1	2	3	4	5	6	7	8	9	10
$\mathrm{L}(x)$	1	2	6	12	60	60	60	120	360	360

So $\mathrm{L}(6) = 4 \cdot 3 \cdot 5 = 60$, which exceeds 6: the running value at a smooth cutoff already contains powers of the other two primes that the cutoff itself does not. The kernel must therefore account for all three maximal powers, not just the factorisation of the cutoff. Also $\mathrm{H}(x) \le x^3$, since each factor is at most x. The exponent is the number of generating primes.

Say that x and y lie in the same *logarithmic cell* when $\lfloor \log_b x\rfloor = \lfloor \log_b y\rfloor$ for each of $b = p,q,r$. By Theorem 3.1 the running value depends on x only through the three integer logarithms, so it is constant on cells and moves only where one logarithm moves.

Proposition 3.2 (constancy and jump ratios)**.** *If $x,y \ge 1$ lie in the same logarithmic cell then $\mathrm{L}(x) = \mathrm{L}(y)$, and the same holds for the kernel at two smooth points of one cell. If $\lfloor \log_p y\rfloor = \lfloor \log_p x\rfloor + 1$ while the other two logarithms agree, then $\mathrm{L}(y) = p\,\mathrm{L}(x)$, and similarly with q or r in place of p.*

Proof. Both parts are immediate from Theorem 3.1: the running LCM depends on x only through the three integer logarithms, and advancing one of them multiplies exactly one factor by its base. □

Proposition 3.3 (jump count)**.** *Let $n \geq 0$. The set of the first n positive powers of p, of q and of r has exactly $3n$ elements, and adjoining the common origin 1 gives exactly $3n + 1$.*

Proof. For a fixed prime b, the powers $b, b^2, \dots, b^n$ are distinct. A common value for two different primes would contradict unique factorisation. Finally 1 is not a positive power of any prime. □

The later tail estimates use two elementary counting facts. Write $\mathcal{B}(h_p, h_q, h_r)$ for the exponent triples with $i \leq h_p$, $j \leq h_q$ and $k \leq h_r$, and $F(H)$ for the points of this box whose running LCM equals H.

Proposition 3.4 (grouping equal running LCMs)**.** *For every box $\mathcal{B}$,*

$$\sum_{(i,j,k)\in\mathcal{B}} \mathrm{K}(i,j,k) = \sum_{H} \#F(H)/H,$$

the outer sum ranging over the running LCMs attained on $\mathcal{B}$.

Proof. Partition $\mathcal{B}$ into the fibres of the running LCM map. On $F(H)$ every summand is $1/H$, so the fibre contributes $\#F(H)/H$. □

Now fix an interval $[\lambda, \eta)$ with $0 \leq \lambda < \eta$ and write $\mathcal{S}$ for the exponent triples of $\mathcal{B}(h_p, h_q, h_r)$ whose value $p^i q^j r^k$ lies in it. In the next lemma, the exponents a, a' are nonnegative integers, and $w \geq 0$ is the product of the fixed factors.

Lemma 3.5 (uniqueness in a short interval)**.** *Let $b \geq 1$ and $\eta \leq b\lambda$. If $b^a w$ and $b^{a'} w$ both lie in $[\lambda, \eta)$ then $a = a'$.*

Proof. If $a < a'$ then $\eta \leq b\lambda \leq b^{a+1} w \leq b^{a'} w < \eta$, which is impossible; the case $a > a'$ is symmetric. □

The short-interval condition says that multiplying by the omitted base moves a point beyond the interval. It holds for $[L, 2L)$ when that base is at least two. A wider interval need not have this property: 1 and 2 both lie in $[1, 3)$ and have the same odd part. Thus the ratio bound, not just finiteness of the interval, permits the injective projection.

Proposition 3.6 (counting a block by two coordinates)**.** *If $\eta \leq r\lambda$ then $\#\mathcal{S} \leq (h_p + 1)(h_q + 1)$, and if $\eta \leq p\lambda$ then $\#\mathcal{S} \leq (h_q + 1)(h_r + 1)$. If moreover $\eta \leq r\lambda$ and $h_p \leq h_q \leq h_r$ with $h_p + h_q + h_r = j$, then $9\#\mathcal{S} \leq (j + 3)^2$.*

Proof. Suppose $\eta \leq r\lambda$. If two triples of $\mathcal{S}$ agree in their first two coordinates, Lemma 3.5 with $b = r$ and $w = p^i q^j$ forces their third coordinates to agree, so the projection forgetting the third coordinate is injective on $\mathcal{S}$ and its image lies in a rectangle with $(h_p + 1)(h_q + 1)$ points. The other case is the same with the first coordinate projected away. Under the sorting hypothesis the two surviving coordinates are the two smallest, so it suffices that $a \leq b \leq c$ with $a + b + c = j$ gives $9(a + 1)(b + 1) \leq (j + 3)^2$. From $a \leq b \leq c$ we get $a + 2b \leq j$, so it is enough that $9(a + 1)(b + 1) \leq (a + 2b + 3)^2$; writing $b = a + d$ with $d \geq 0$, the difference of the two sides is $d(3a + 4d + 3) \geq 0$. □

4 The two-prime Hecke–Mahler reduction

Let $P = \{p, q\}$ with $p < q$, and put $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$. The proof of Theorem 3.1, with one coordinate omitted, identifies this with the running LCM. The two sums are therefore

$$\mathcal{D}_{\{p,q\}} = 1 + \sum_{t \in \{p,p^2,\dots\} \cup \{q,q^2,\dots\}} \frac{1}{L_{p,q}(t)}, \qquad \mathcal{R}_{\{p,q\}} = \sum_{i,j \geq 0} \frac{1}{L_{p,q}(p^i q^j)}.$$

We first express the contribution from powers of q in terms of that from powers of p. The repeated kernel then factors, and this relation puts both factors in a single transcendental parameter. This is the factorisation and Hecke–Mahler reduction credited to Fan in the introduction.

Proof of Theorem 1.2. Set

$$\theta = \frac{\log p}{\log q}, \qquad x = \frac{1}{p}, \qquad y = \frac{1}{q}, \qquad m_n = \lfloor n\theta \rfloor, \qquad \delta_n = m_{n+1} - m_n.$$

Here $0 < \theta < 1$, and θ is irrational, since a rational value would give $p^b = q^a$ for positive integers a, b. Consequently $\delta_n \in \{0, 1\}$. Put

$$A = \sum_{n \geq 0} x^n y^{m_n}, \qquad B_* = \sum_{n \geq 0} \delta_n x^n y^{m_n + 1}.$$

The initial value and the positive powers of p contribute A, since $L_{p,q}(p^n) = p^n q^{m_n}$. A power of q lies strictly between p^n and p^{n+1} exactly when $\delta_n = 1$, it is then q^{m_n+1}, and its post-jump reciprocal is $x^n y^{m_n+1}$; so the positive powers of q contribute B_* and $\mathcal{D}_{\{p,q\}} = A + B_*$. All these series converge absolutely.

Since $y^{m_{n+1}} - y^{m_n} = \delta_n y^{m_n}(y - 1)$, an index shift gives $A - 1 - xA = x(y-1)B_*/y$, so $B_* = ((p-1)A - p)/(1-q)$ and

$$\mathcal{D}_{\{p,q\}} = \frac{(q-p)A + p}{q-1}. \tag{4}$$

At a smooth point, $\log_p(p^i q^j) = i + j/\theta$ and $\log_q(p^i q^j) = j + i\theta$, so $L_{p,q}(p^i q^j) = p^{\,i + \lfloor j/\theta \rfloor} q^{j + m_i}$ and absolute convergence permits the factorisation

$$\mathcal{R}_{\{p,q\}} = A \sum_{j \geq 0} y^j x^{\lfloor j/\theta \rfloor}.$$

For $j \geq 1$, the index $n = \lfloor j/\theta \rfloor$ is precisely the one for which $p^n < q^j < p^{n+1}$; the inequalities are strict because distinct primes have no common positive power. This interval contains at most one power of q, since $p < q$. Thus $\delta_n = 1$ and $j = m_n + 1$. Conversely, every n with $\delta_n = 1$ contains that unique power of q. The second factor is therefore $1 + B_*$, giving

$$\mathcal{R}_{\{p,q\}} = \frac{(p+q-1)A - (p-1)A^2}{q-1}. \tag{5}$$

It remains to prove that A is transcendental. For the Hecke–Mahler series

$$F_\theta(x,y) = \sum_{n\ge 1} \sum_{k=1}^{\lfloor n\theta \rfloor} x^n y^k$$

a finite geometric sum gives $((1-y)/y)F_\theta(x,y) = x/(1-x) - (A-1)$, that is

$$A = \frac{1}{1-x} - \frac{1-y}{y} F_\theta(x,y). \tag{6}$$

Bugeaud and Laurent's Theorem 1.1 states, in particular, that $F_\theta(\beta,\alpha)$ is transcendental when $\theta \in (0,1)$ is irrational, α and β are nonzero algebraic numbers, $|\beta| < 1$ and $|\beta\alpha^\theta| < 1$ [15, Theorem 1.1]; the $\rho = 0$ case used here goes back to Loxton and van der Poorten [16, Theorem 8, p. 40]. Take $(\beta,\alpha) = (x,y)$: then $|\beta| = 1/p < 1$ and

$$|xy^\theta| = \frac{1}{p}\left(\frac{1}{q}\right)^{\log p/\log q} = \frac{1}{p^2} < 1.$$

So $F_\theta(x,y)$ is transcendental, and (6) makes A transcendental. The coefficient of A in (4) is $(q-p)/(q-1) \neq 0$ and the coefficient of A^2 in (5) is $-(p-1)/(q-1) \neq 0$, both rational. If either value were algebraic, its identity would exhibit A as a root of a nonzero polynomial over the algebraic numbers. □

A product of two transcendental numbers need not be transcendental. What proves the repeated sum transcendental is its nonconstant quadratic expression in the single value A, not the factorisation by itself. The value theorem adds no unverified hypothesis in this two-prime case: distinct primes give an irrational slope and the displayed reciprocal arguments satisfy its size conditions. Primality is stronger than the calculation needs. For coprime integers $1 < p < q$, enumerate the monoid $\{p^i q^j : i,j \ge 0\}$, not all integers supported on the prime factors of pq. Unique exponent pairs, the running-LCM product and the irrationality of $\log p/\log q$ still hold, so both identities and transcendence conclusions remain valid. For example, this applies to generators $4, 9$. By contrast, the monoid generated by $4, 8$ has running LCM 8 at the cutoff 8, not $4^{\lfloor \log_4 8 \rfloor} 8^{\lfloor \log_8 8 \rfloor} = 32$; its slope is rational as well. The coprime extension was already noted in the forum discussion (Section 13.1).

A third prime introduces an additional floor term that cannot be separated in this way; Section 7 makes that obstruction precise. For the sums over distinct running LCMs, the next section proves irrationality for every finite set of primes by an elementary argument that uses no value theorem.

5 The sums over distinct running LCMs

We record the ordinary arguments for general finite prime supports. Their assertions, including the block and finite-state criteria below, are presented as remarks because this part has not been formalised. The complete proofs are retained. The five-map theorem in Section 2 has a separate formal proof. No argument here gives transcendence for more than two primes.

Let P be a finite set of at least two primes, with least element p, and put $\mathrm{H}_P(t) = \prod_{r\in P} r^{\lfloor \log_r t\rfloor}$. The proofs of Theorem 3.1 and Propositions 3.2 and 3.3 apply unchanged to any finite set of primes: the running least common multiple of the P-smooth integers up to t is $\mathrm{H}_P(t)$, it changes only at the prime powers r^n ($r \in P$, $n \geq 1$), where it is multiplied by r, and distinct prime powers are distinct numbers. List the prime powers as $t_1 < t_2 < \cdots$ and let q_k be the prime of t_k. The distinct running values are then 1 and $\mathrm{H}_P(t_k) = q_1 \cdots q_k$, so

$$\mathcal{D}_P = 1 + \sum_{k\geq 1} \frac{1}{\mathrm{H}_P(t_k)} = \sum_{k\geq 0} \frac{1}{q_1 \cdots q_k}. \tag{7}$$

This is the sum Σ' of Erdős's letter [3], which counts each distinct value of $[a_1, \ldots, a_n]$ once. For $P = \{2,3,5\}$ the prime powers $2,3,4,5,8,9,16,25,27,32$ give the following terms, including the initial 1

$$1, \frac{1}{2}, \frac{1}{6}, \frac{1}{12}, \frac{1}{60}, \frac{1}{120}, \frac{1}{360}, \frac{1}{720}, \frac{1}{3600}, \frac{1}{10800}, \frac{1}{21600}, \cdots.$$

Numerically, $\mathcal{D}_{\{2,3,5\}} = 1.779616983294749050459190649316983 27\ldots$. Put $Q_k = q_1 \cdots q_k = \mathrm{H}_P(t_k)$.

5.1 Proof for an arbitrary finite prime set

Remark 5.1 (the sums over distinct running LCMs). For every finite set P of primes with $|P| \geq 2$, the sum $\mathcal{D}_P$ is irrational.

Here we compare two tails rather than recover a word from one tail. Discreteness is used twice. Under rationality with denominator K, the tails lie in $K^{-1}\mathbb{Z}$; a long common prefix makes two of them closer than $1/K$ and hence equal. Next, corresponding blocks contain the same prime labels in possibly different orders. Their denominator products agree. Clearing that product leaves an integer difference of block numerators equal to a difference of tails in $(-1, 1)$. Both must vanish, and this conclusion propagates through the blocks.

We therefore need one rearranged block with a changed numerator. For two or three distinct primes, every change of order does this; for four or more, some do not. The torus construction isolates a pair of neighbouring jumps and reverses it. Other blocks may change too, but equality constrains every block separately, so one changed two-letter numerator suffices. We first establish the word identity, then construct the two arithmetic tails.

Words and tails. For an infinite word $u = u_1 u_2 \cdots$ over P, define $V(u) = \sum_{j\geq 1} (u_1 \cdots u_j)^{-1}$. For a nonempty finite word $\sigma = \sigma_1 \cdots \sigma_n$, put

$$\Pi(\sigma) = \sigma_1 \cdots \sigma_n, \qquad f(\sigma) = \sum_{j=1}^{n} \sigma_{j+1} \cdots \sigma_n.$$

The final term in f is 1. Thus f is the numerator of the block contribution when its denominator is the product of the block letters. For two distinct letters a, b, we have

$f(ab) = b + 1$ and $f(ba) = a + 1$. Reversing this pair changes f by the nonzero integer $b - a$. Removing a prefix gives

$$V(\sigma u) = \frac{f(\sigma) + V(u)}{\Pi(\sigma)}. \tag{8}$$

We apply this identity to $x_k = V(q_{k+1}q_{k+2}\cdots) = \sum_{j>k} Q_k/Q_j$. Every suffix contains every prime of P, so $0 < x_k < 1/(\min P - 1) \le 1$.

Remark 5.2 (integral tails). If $\mathcal{D}_P = N/K$ with integers N and $K \ge 1$, then Kx_k is an integer for every $k \ge 1$.

Proof. Each Q_i with $i \le k$ divides Q_k. Multiplication of the defining series by Q_k therefore gives

$$Kx_k = Q_k N - K\left(Q_k + \sum_{i \le k} \frac{Q_k}{Q_i}\right) \in \mathbb{Z}.$$

□

Two words with a long common prefix have close values. When the words differ only by rearrangements, equality of their values imposes a stronger condition on every rearranged block.

Remark 5.3 (rearranged blocks). Let $u = \sigma_1\sigma_2\cdots$ and $u' = \sigma_1'\sigma_2'\cdots$ be infinite words, where each σ_i is a nonempty finite word and σ_i' is a rearrangement of it, and suppose that every suffix of u or u' starting at a block boundary has value in $(0, 1)$.

(i) If $V(u) = V(u')$, then $f(\sigma_i) = f(\sigma_i')$ for every i.

(ii) If $\sigma_i = \sigma_i'$ for every $i < i_0$, then $|V(u) - V(u')| < 1/\Pi(\sigma_1\cdots\sigma_{i_0-1})$.

Proof. Let U_i and U_i' be the values starting at block i. Since the two blocks have the same product, (8) yields

$$\Pi(\sigma_i)(U_i - U_i') = f(\sigma_i) - f(\sigma_i') + U_{i+1} - U_{i+1}'.$$

If $U_i = U_i'$, the integer on the right equals a number in $(-1, 1)$, and hence both differences vanish. Induction proves (i). If the preceding blocks agree, repeated removal of their common prefix gives

$$V(u) - V(u') = \frac{U_{i_0} - U_{i_0}'}{\Pi(\sigma_1\cdots\sigma_{i_0-1})},$$

which proves (ii). □

Remark 5.4 (short rearrangements). The map f is injective on the orderings of any set of at most three distinct primes.

Proof. For two letters, $f(ab) = 1 + b$. For three, $f(abc) = 1 + c(1 + b)$. Suppose two orderings of $\{a, b, c\}$ have the same value. If their last letters agree, their middle letters agree too. Otherwise we may write $a(1 + s) = b(1 + t)$, with $s \in \{b, c\}$ and $t \in \{a, c\}$. The choices $(s, t) = (b, a)$ and (c, c) imply $a = b$; the other two imply $b \mid a$ or $a \mid b$. Each is impossible for distinct primes. □

There is a collision on four letters: $f(5,7,3,2) = f(7,2,3,5) = 51$. Consequently, a change in order alone will not suffice for larger P. We shall arrange an isolated interchange of two letters, for which $f(ab) - f(ba) = b - a \neq 0$.

The logarithmic flow. Consider

$$\mathbb{T} = \prod_{q \in P} \mathbb{R}/(\log q)\mathbb{Z}, \qquad \Theta(s) = (s \bmod \log q)_{q \in P}.$$

For $\theta \in \mathbb{T}$, a q-crossing is a time t at which $\theta_q + t \in (\log q)\mathbb{Z}$. At $\theta_k = \Theta(\log t_k)$ the positive crossings occur at $\log(q^j/t_k)$ with $q^j > t_k$, so their labels, read in time order, are $q_{k+1}q_{k+2}\cdots$. Crossings at this phase never coincide.

Let T_0 be the closure of $\Theta(\mathbb{R})$. Every forward orbit in T_0 is dense there. Indeed, the pigeonhole principle gives arbitrarily large $d > 0$ with $\Theta(d)$ arbitrarily close to 0, so negative times are limits of positive ones. Translation then proves the same assertion for every starting point and every later starting time.

In the normalised coordinates $\varphi_q = \theta_q/\log q$, Kronecker's theorem identifies T_0 with the image of the smallest rational subspace $L \subseteq \mathbb{R}^P$ containing $\omega = (1/\log q)_{q \in P}$. Any two coordinate forms φ_a, φ_b are linearly independent on L. To see this, their dependence would put a nonzero vector supported on $\{a, b\}$ in the rational annihilator of L. That intersection is rational, so it would contain a nonzero integer vector. Its pairing with ω would force $\log a/\log b \in \mathbb{Q}$, contrary to unique factorisation. In particular, $(\log b)\varphi_b - (\log a)\varphi_a$ is a nonzero form on L. We write it as $\theta_b - \theta_a$ when using real representatives.

To interchange just two jumps, we seek a phase at which exactly those two crossings coincide. The next argument finds such a phase inside T_0; it does not require this orbit closure to be the whole torus $\mathbb{T}$.

Remark 5.5 (two walls). If $|P| \geq 4$, there is $x \in T_0$ with $\theta_q(x) = 0$ for exactly two primes $q \in P$.

Proof. Choose a rational plane in L on which every pair of coordinate forms remains independent. Such a plane exists: for each pair, the determinant on $L \times L$ is a nonzero polynomial, and finitely many of these polynomials cannot vanish on all rational pairs. An integer basis of the plane identifies its image with a two-dimensional subtorus of T_0. On this subtorus each wall $\theta_q = 0$ has the equation $\langle \gamma_q, s \rangle \in \mathbb{Z}$, where $\gamma_q \in \mathbb{Z}^2 \setminus \{0\}$, and the vectors γ_q are pairwise nonparallel. A wall may have several components if γ_q is not primitive.

The walls form a finite arrangement of closed geodesics. Let k_v count the wall families through a vertex v. Each such family contributes two incident half-edges, so $E = \sum_v k_v$. Two families alone divide the torus into open parallelograms: their two integer linear forms give a finite covering of the standard torus. Adding the remaining straight lines subdivides these parallelograms into convex polygons, each with at least three sides. Thus $2E \geq 3F$, while Euler's formula gives $V - E + F = 0$. Together these imply

$$\sum_v (k_v - 3) = E - 3V \leq 0.$$

Every vertex belongs to at least two wall families, and the last inequality says that the average number of families through a vertex is at most three. The origin belongs to all $|P| \geq 4$ families. Another vertex must therefore belong to exactly two, as required. □

Comparison of two tails.

Proof of Remark 5.1. Suppose that $\mathcal{D}_P = N/K$, with $K \geq 1$. Fix $p \in P$ and a jump t_k which is a power of p. We shall compare the tails after t_k and $t_m = t_k p^n$. We first choose n so that the phase returns close enough to its initial value to preserve a long initial crossing word. After fixing this return, we shall find a later group of crossings whose order it changes. The common initial word will force equality of the two tails; the later rearrangement will contradict that equality.

For the common word to force equality, its reciprocal product must be at most $1/K$. Choose $T > 0$ so that the number ν of crossings in $(0, T]$ satisfies $2^\nu \geq K$, and let $g > 0$ be the minimum gap between distinct crossings in $[0, T+1]$. Put $\beta = \min_{q \neq p} \operatorname{dist}(\theta_{k,q}, 0) > 0$. The gap g will keep these early crossings in order, while β will prevent a crossing from moving past the starting time 0. When $|P| \geq 4$, also choose the point x and primes a, b from Remark 5.5, and set

$$\eta = \min\left(\frac{1}{10}, \min_{c \neq a,b} \operatorname{dist}(\theta_c(x), 0)\right) > 0.$$

In this case η leaves room to isolate the a- and b-crossings from all the others. Take $\varepsilon < \min(\beta, g/2, \log 2/(2|P|))$, and when $|P| \geq 4$ require also $\varepsilon < \eta/8$. Dirichlet's simultaneous approximation theorem gives arbitrarily large n and integers $m_q \geq 1$ for $q \neq p$ such that

$$\delta_p = 0, \qquad \delta_q = n \log p - m_q \log q, \qquad \max_q |\delta_q| < \varepsilon.$$

The numbers δ_q are pairwise distinct, since an equality would identify positive powers of distinct primes.

At $\theta_m = \theta_k + \Theta(n \log p)$ a q-crossing moves by $-\delta_q$. For $q \neq p$ every crossing of θ_k is at distance at least β from 0, so signs are preserved; the p-crossings do not move. We therefore obtain a bijection between the positive crossings of the two phases.

Cut the crossings of θ_k whenever successive times are more than 2ε apart, and call each resulting block a chain. Different chains keep their order after the shift. A chain contains at most $|P|$ crossings, all of different primes: otherwise its first $|P|+1$ crossings would span less than $2\varepsilon|P| < \log 2$, although some prime would have to occur twice. The same argument excludes a repeated prime in a shorter chain. Thus the two tail words are concatenations of the same chains, with the letters possibly rearranged inside each chain.

Every chain meeting $(0, T]$ is a singleton, by the choice of g and ε. The first ν letters consequently agree, and Remark 5.3(ii) gives $|x_k - x_m| < 2^{-\nu} \leq 1/K$. The integral-tail lemma forces $x_k = x_m$. By Remark 5.3(i), every chain C and its rearrangement C' now satisfy $f(C) = f(C')$.

The return, and hence every displacement δ_q, is now fixed. It remains to find a chain that violates this last equality. For at most three primes, any change of order will do; for larger sets we will isolate a chain of length two.

When $|P| \le 3$, choose $q \ne p$. If $\delta_q > 0$, density of $j \log p - i \log q + \log t_k$ for large positive i, j supplies crossings $s_p, s_q > 0$ with $s_q - \delta_q < s_p < s_q$. Their order reverses. The case $\delta_q < 0$ is the same with the inequalities reversed. The two crossings belong to one chain, which has at most three distinct primes and changes its order. This contradicts Remark 5.4.

For $|P| \ge 4$, put $D = \delta_a - \delta_b \ne 0$. Since the form $\theta_b - \theta_a$ is nonzero on L, choose a small $y \in L$ with $|\theta_q(y)| < \eta/8$ for all q and $\theta_b(y) - \theta_a(y)$ strictly between 0 and D. At the phase

$$z = x + y + \Theta(-\eta/2)$$

there is one a-crossing and one b-crossing in $[-\eta/4, 5\eta/4]$. Their times τ_a, τ_b lie in $(\eta/4, 3\eta/4)$, and $\tau_a - \tau_b$ is strictly between 0 and D. There are no other crossings in that interval: for $c \ne a, b$ the nearest one is at distance at least $7\eta/8$ from $\eta/2$.

The phase z thus has an isolated pair which the fixed displacement reverses. To obtain such a pair in the arithmetic crossing word, keep all the strict margins above and choose a neighbourhood U of z in T_0 where they still hold. Density gives $t^* > T + 2$ with $\theta_k + \Theta(t^*) \in U$. The two corresponding crossings of θ_k are less than $|D| < 2\varepsilon$ apart, whereas every other crossing is more than $\eta/2 > 4\varepsilon$ away. They form a chain of length two, beyond the initial word already used to equate the tails. After shifting, their time difference becomes $(\tau_a - \tau_b) - D$, which has the opposite sign. Thus $C = ab, C' = ba$ or conversely, contradicting $f(ab) - f(ba) = b - a \ne 0$. □

Reversals near the origin. A shorter route would take, at a return of the orbit near the origin of T_0, the chain of crossings there. That shortcut need not work when $\dim T_0 = 2$, which happens exactly when the numbers $1/\log q$ span a two-dimensional space over $\mathbb{Q}$; the proof makes no assumption excluding this case. Linear-form estimates for logarithms do not supply the required independence of their reciprocals. In that case every small vector of L, in the coordinates θ_q, has the form $\alpha(1, \dots, 1) + \beta w$ for one fixed w, so near the origin all the forms $\theta_b - \theta_a$ are proportional. The chain at such a return contains every prime of P, and the only rearrangement it can undergo is the full reversal. Full reversals can collide:

$$f(43, 13, 3, 53) = f(53, 3, 13, 43) = 2280,$$

and among the sets of four primes below 100 there are five such collisions. Remark 5.5 avoids the question by producing a point on exactly two walls for every P with $|P| \ge 4$; a point on three walls would also serve, by Remark 5.4. Swapping one adjacent pair always changes f, since $f(\sigma) - f(\sigma') = (\sigma_{i+1} - \sigma_i)\sigma_{i+2} \cdots \sigma_n$ when σ' swaps σ_i and σ_{i+1}; a rearranged chain of four or more primes can differ from its image by more than one swap. Nor can the argument stop at the first rearranged chain: for four primes that chain can collide, as $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$ shows, which is why Remark 5.3(i) is applied to every chain.

Exact illustrations. The program `two_tail.py` of the programs for sums over distinct values checks the finite facts above in exact arithmetic; the proof uses none of them. Among the sets of two or three primes below 200, f has no collision; among the 2380 sets of four primes below 60 it has 30. For $P = \{2, 3, 5, 7\}$, $t_k = 2^{60}$ and the return $n = 283526$,

with $\varepsilon \approx 0.0117$, the first 4000 crossings correspond exactly, every chain stays in place, and of the 39 rearranged chains 35 are transpositions; none has equal f-values. For the closer return $n = 2547450$, with $\varepsilon \approx 0.0034$, the difference $x_k - x_m$ computed exactly has $\log_{10}|x_k - x_m| = -501.1234$, the value predicted by its first rearranged chain, $(5,2)$ against $(2,5)$ after 1005 letters. On 400 random arrangements of four to six line families on $\mathbb{R}^2/\mathbb{Z}^2$, some with non-primitive normal vectors, $\sum_v (k_v - 3) \le -4$; the square with both diagonals attains 0 and still has a vertex on exactly two lines.

5.2 Blocks and affine maps

We now extend the calculation of Section 2 to an arbitrary finite prime set.

To obtain a second description of the sum over distinct running LCMs, group its jumps into $(p^a, p^{a+1}]$, where p is the least prime of P, and define

$$P_a = \mathrm{H}_P(p^a), \qquad Y_a = P_a \sum_{t_k > p^a} \frac{1}{\mathrm{H}_P(t_k)}.$$

Write $\tau_a = (r_1, \dots, r_v)$ for the labels of the jumps strictly inside block a, in increasing order of their powers. This is the block's *type*; write $\emptyset$ when there is no interior jump. The final jump at p^{a+1} always has label p.

Remark 5.6 (blocks, tails and integrality). Let P be a finite set of at least two primes with least element p.

(i) Block a ends with the jump at p^{a+1} and contains at most one power of each prime $q \in P \setminus \{p\}$, lying strictly inside it. The power q^f with $f \ge 1$ lies in block $\lfloor f \log_p q \rfloor$.

(ii) If block a has type $\tau = (r_1, \dots, r_v)$, then $Y_a = G_\tau(Y_{a+1})$, where

$$G_\tau(y) = \frac{\mu_\tau + y}{b_\tau}, \qquad b_\tau = p r_1 \cdots r_v, \qquad \mu_\tau = 1 + b_\tau \sum_{j=1}^{v} \frac{1}{r_1 \cdots r_j}.$$

Each G_τ is increasing with slope $1/b_\tau \le 1/p$ and maps $[0, 1/(p-1)]$ into itself, and $0 < Y_a \le 1/(p-1)$.

(iii) If $\mathcal{D}_P = N/K$ with integers N and $K \ge 1$, then KY_a is an integer for every $a \ge 0$.

Proof. (i) The endpoint p^{a+1} is a jump. Consecutive powers of q have ratio $q > p$, so the block contains at most one of them, and no power of q equals a power of p. The power q^f lies in (p^a, p^{a+1}) exactly when $a < f \log_p q < a + 1$.

(ii) The jumps of block a are the interior powers, with running LCMs $P_a r_1$, $P_a r_1 r_2$, ..., $P_a r_1 \cdots r_v$, followed by the jump at p^{a+1}, whose running LCM is $P_{a+1} = b_\tau P_a$. Splitting off the block,

$$Y_a = \sum_{j=1}^{v} \frac{1}{r_1 \cdots r_j} + \frac{1}{b_\tau} + \frac{P_a}{P_{a+1}} Y_{a+1} = \frac{\mu_\tau + Y_{a+1}}{b_\tau}.$$

Each jump multiplies the running LCM by at least p, so $0 < Y_a \le \sum_{k \ge 1} p^{-k} = 1/(p-1)$. Since every $r_j > p$, $G_\tau(1/(p-1)) \le \sum_{j=1}^{v} p^{-j} + p^{-v}/(p-1) = 1/(p-1)$, and $G_\tau(0) > 0$.

(iii) If $t \le p^a$, each exponent in $\mathrm{H}_P(t)$ is at most the corresponding exponent in P_a, so $\mathrm{H}_P(t) \mid P_a$. Hence $P_a \mathcal{D}_P - Y_a = P_a(1 + \sum_{t_k \le p^a} \mathrm{H}_P(t_k)^{-1})$ is an integer, and so is $KY_a = P_a N - K(P_a \mathcal{D}_P - Y_a)$. □

5.3 A decoding criterion for every finite set of primes

For other prime sets, a useful tail interval may depend on the recent occurrence of each prime. Put $Q = P \setminus \{p\}$ and choose the integer g_q by $p^{g_q} < q < p^{g_q+1}$. Successive blocks containing a q-power have indices $\lfloor f \log_p q \rfloor$ and $\lfloor (f+1) \log_p q \rfloor$, by Remark 5.6(i). Their difference is g_q or $g_q + 1$. The gaps give a finite memory of the last occurrences. We use it to enclose tails of every allowed path, a class containing all sufficiently late arithmetic paths but possibly others. Separation on this larger class suffices for recovery; overlap need not give equal arithmetic tails. The memory sharpens the enclosures, but the decoder must ultimately work without being told the state.

These constraints define a finite automaton. Its states are the tuples $s = (s_q)_{q \in Q}$ with $1 \le s_q \le g_q + 1$; the entry s_q counts the blocks since the last block containing a power of q. A *transition* from s is a pair (τ, s') in which τ is an ordering of a set $T \subseteq Q$ with

$$\{q : s_q = g_q + 1\} \subseteq T \subseteq \{q : s_q \ge g_q\},$$

and $s'_q = 1$ for $q \in T$, $s'_q = s_q + 1$ for $q \notin T$. A path from s is a sequence of transitions $(u_1, s_1), (u_2, s_2), \ldots$ with (u_1, s_1) a transition from s and each (u_{i+1}, s_{i+1}) a transition from s_i. The *value* of an infinite path is the limit of $G_{u_1} \circ \cdots \circ G_{u_n}(y)$ as $n \to \infty$; by Remark 5.6(ii) these numbers lie, for $y \in [0, 1/(p-1)]$, in nested intervals of length at most $p^{-n}/(p-1)$, so the limit exists and does not depend on y.

Every prime $q \in Q$ lies in block g_q. For $a > \max_q g_q$ let s_a record, for each q, the number $a - \ell$, where $\ell < a$ is the index of the last block before a containing a power of q. The next such block has index $\ell + g_q$ or $\ell + g_q + 1$, and it is at least a. Hence s_a is a state, block a contains a power of q when $(s_a)_q = g_q + 1$ and does not when $(s_a)_q < g_q$, and (τ_a, s_{a+1}) is a transition from s_a. Since $Y_a = G_{\tau_a} \circ \cdots \circ G_{\tau_{a+n-1}}(Y_{a+n})$ with $Y_{a+n} \in [0, 1/(p-1)]$, the tail Y_a is the value of the actual path $(\tau_a, s_{a+1}), (\tau_{a+1}, s_{a+2}), \ldots$ from s_a.

Remark 5.7 (a decoding criterion). Let P be a finite set of at least two primes, and for every state s let $J(s)$ be a closed interval containing the values of all infinite paths from s. For $k \ge 1$ and each type τ, let U_τ be the union of the intervals $G_{u_1} \circ \cdots \circ G_{u_k}(J(s_k))$ over all states s and all paths $(u_1, s_1), \ldots, (u_k, s_k)$ from s with $u_1 = \tau$. If the sets U_τ are pairwise disjoint, then $\mathcal{D}_P$ is irrational.

The union defining U_τ runs over *all* starting states. Thus membership can identify the first block from Y_a alone; separating types only within each known state would not give this conclusion.

Proof. Let $a > \max_q g_q$. The first k transitions of the actual path from s_a form a path of length k with first type τ_a, and $Y_a = G_{\tau_a} \circ \cdots \circ G_{\tau_{a+k-1}}(Y_{a+k})$, where $Y_{a+k} \in J(s_{a+k})$ because it is the value of an infinite path from s_{a+k}. Thus $Y_a \in U_{\tau_a}$, and by disjointness τ_a is the only type with this property. So Y_a determines τ_a, then $Y_{a+1} = b_{\tau_a} Y_a - \mu_{\tau_a}$, and by induction the word $\tau_a \tau_{a+1} \cdots$.

Suppose $\mathcal{D}_P = N/K$. By Remark 5.6(ii) and (iii), the numbers KY_a are integers in $[0, K/(p-1)]$, so $Y_a = Y_{a'}$ for some $\max_q g_q < a < a'$, and the word (τ_a) is eventually periodic. Fix $q \in Q$ and put $\beta = \log p / \log q$, which lies in $(0, 1)$ and is irrational because no power of p equals a power of q. By Remark 5.6(i), block a contains $\lfloor (a+1)\beta \rfloor -$

$\lfloor a\beta \rfloor$ powers of q, a number recorded by τ_a. As in the proof of Theorem 1.1, eventual periodicity with period ℓ forces $\ell\beta \in \mathbb{Z}$, a contradiction. □

Intervals $J(s)$ with the required property are easy to produce. The constant interval $[0, 1/(p-1)]$ qualifies by Remark 5.6(ii). If a family J qualifies, so does any family J' with

$$J'(s) \supseteq G_\tau(J(s')) \quad \text{for every transition } (\tau, s') \text{ from } s,$$

because an infinite path from s is a transition (τ, s') followed by an infinite path from s'. Starting from the constant family, take the interval hull of the images and round its endpoints outwards. Every round preserves containment, though a hull may fill genuine gaps. Keep the separate image intervals when forming the final unions U_τ. The smallest enclosing interval at state s is $J(s) = [L(s), U(s)]$, where $L(s)$ and $U(s)$ are the least and greatest infinite-path values. They satisfy $L(s) = \min G_\tau(L(s'))$ and $U(s) = \max G_\tau(U(s'))$ over the transitions from s, and these equations have unique solutions because each G_τ contracts by at least the factor $1/p$.

For $\{2, 3, 5\}$ the constant interval $I = [3/10, 13/15]$ has the required property: every path takes a transition with $5 \in T$ within any three steps, which is the only fact the bounds used. With $k = 1$ the criterion is then the proof given above.

5.4 Certificates for three and four primes

The saved computations apply the interval criterion in exact rational arithmetic to the following finite family. They accompany the ordinary proof of Remark 5.7 and have not been formalised. The exclusions in part (ii) concern these interval calculations; the ordinary torus argument also treats those sets.

Remark 5.8 (three and four primes up to 31). Let P be a set of primes, each at most 31.

(i) If $|P| = 3$, then $\mathcal{D}_P$ is irrational.

(ii) If $|P| = 4$ and P is not one of the 38 sets listed below, then $\mathcal{D}_P$ is irrational.

Proof. For each set, the program `certificate.py` of the programs for sums over distinct values builds the automaton, computes intervals $J(s)$ by 80 rounds of the operation above from $[0, 1/(p-1)]$, with endpoints rounded outwards to multiples of 2^{-96}, and tests the disjointness of the sets U_τ for $k = 1, 2, \ldots$. Every comparison is made in exact rational arithmetic, and each U_τ is computed as a union of intervals merged where they meet. The test passes for each of the 165 sets of three primes with some $k \le 4$, and for each set of four primes outside the list with some $k \le 5$. Remark 5.7 then gives irrationality. □

For three primes the least passing depth is $k = 1$ for 141 sets and $k = 2$ for the 17 sets $\{2,3,11\}$, $\{2,3,13\}$, $\{2,3,17\}$, $\{2,3,19\}$, $\{2,3,23\}$, $\{2,3,29\}$, $\{2,3,31\}$, $\{2,5,7\}$, $\{2,11,17\}$, $\{2,13,19\}$, $\{2,17,19\}$, $\{2,19,29\}$, $\{2,19,31\}$, $\{3,13,17\}$, $\{3,17,23\}$, $\{3,23,31\}$ and $\{3,29,31\}$. It is $k = 3$ for $\{2,3,7\}$, $\{2,11,13\}$, $\{2,17,29\}$ and $\{2,19,23\}$, and $k = 4$ for $\{2,17,31\}$, $\{2,23,29\}$ and $\{2,29,31\}$. The smallest gap between sets U_τ of different types, among all passing triples, is about $9.4 \cdot 10^{-6}$, for $\{2,17,31\}$; for $\{2,3,5\}$ it is $27983/963915 \approx 0.029$ at depth 1, much larger than the gap $1/900$ of the single interval used above. For

four primes the least passing depth is $1, 2, 3, 4, 5$ for 184, 34, 28, 42 and 4 sets respectively, and the smallest gap is about $1.8 \cdot 10^{-8}$, for $\{3, 7, 17, 19\}$. All these gaps are far above the rounding scale 2^{-96}. The 38 four-prime sets with no passing depth $k \leq 5$ are $\{2, 3, 5, 7\}$, $\{2, 3, 5, 11\}$, $\{2, 3, 5, 13\}$, $\{2, 3, 5, 17\}$, $\{2, 3, 5, 19\}$, $\{2, 3, 5, 23\}$, $\{2, 3, 7, 11\}$, $\{2, 3, 7, 13\}$, $\{2, 3, 7, 17\}$, $\{2, 3, 7, 23\}$, $\{2, 3, 11, 13\}$, $\{2, 3, 11, 17\}$, $\{2, 3, 11, 19\}$, $\{2, 3, 11, 29\}$, $\{2, 3, 11, 31\}$, $\{2, 3, 13, 19\}$, $\{2, 3, 13, 29\}$, $\{2, 3, 17, 19\}$, $\{2, 3, 17, 29\}$, $\{2, 3, 29, 31\}$, $\{2, 5, 7, 11\}$, $\{2, 5, 17, 29\}$, $\{2, 5, 23, 29\}$, $\{2, 5, 23, 31\}$, $\{2, 11, 13, 19\}$, $\{2, 11, 17, 23\}$, $\{2, 11, 17, 29\}$, $\{2, 11, 23, 31\}$, $\{2, 13, 19, 31\}$, $\{2, 19, 23, 31\}$, $\{3, 5, 13, 17\}$, $\{3, 5, 17, 19\}$, $\{3, 5, 23, 31\}$, $\{3, 7, 11, 13\}$, $\{3, 19, 23, 29\}$, $\{3, 19, 23, 31\}$, $\{3, 19, 29, 31\}$ and $\{7, 13, 29, 31\}$. Twenty of them contain both 2 and 3.

A second computation replaces the rounded intervals by the exact extremes $[L(s), U(s)]$, found by policy iteration and checked by substitution into their defining equations. It returns the same least depth for all 495 sets. A third check compares the automaton with the first 400 actual blocks of every set: every actual word is a path, and every exact enclosure of an actual tail lies in the interval of its state. This comparison is a consistency check only. The whole triple computation takes about three seconds, and the four-prime computation about seventy seconds, on one laptop; the saved logs record every depth and gap.

For $\{2, 3, 5, 7\}$ the automaton has 18 states and 16 types, and the test fails through depth 7. Its tail intervals range from about 0.188 to 0.785, so the largest exceeds three times the smallest, and the images of the empty block and of the block with one power of 3 overlap. The largest overlap between two first types is always of this pair. For each depth k the table gives the number of merged intervals in the sets U_τ and the length of that largest overlap.

k	intervals	largest overlap
1	16	$2.4 \cdot 10^{-2}$
2	58	$1.8 \cdot 10^{-3}$
3	386	$1.0 \cdot 10^{-4}$
4	1413	$4.6 \cdot 10^{-5}$
5	8395	$7.6 \cdot 10^{-7}$
6	38494	$2.4 \cdot 10^{-7}$
7	193905	$1.8 \cdot 10^{-8}$

At depth 7 two further pairs of types still overlap, $(5, 7, 3)$ with (7) by about $1.5 \cdot 10^{-9}$ and $(3, 5, 7)$ with $(3, 7)$ by about $5.8 \cdot 10^{-10}$.

The persistent overlap is not a rounding error or merely slow convergence. An exact collision explains why no depth succeeds. The two-block words $(5, 7, 3), \emptyset$ and $(7), (3, 5)$ have the same map,

$$G_{(5,7,3)} \circ G_\emptyset(y) = G_{(7)} \circ G_{(3,5)}(y) = \frac{103 + y}{420},$$

which is the collision $f(5, 7, 3, 2) = f(7, 2, 3, 5) = 51$ of Section 5.1 read in blocks: the letters are $5, 7, 3, 2, 2$ and $7, 2, 3, 5, 2$, and $f = 103$ for both. From the states $(1, 2, 2)$ and

$(1,2,3)$, written as (s_3, s_5, s_7), both words continue by the blocks (3), $(5,7)$ and end in the state $(2,1,1)$, so the four-block words have the same map and the same end state. Hence the sets $U_{(5,7,3)}$ and $U_{(7)}$ of Remark 5.7 meet at every depth, for any valid choice of intervals, and the criterion cannot decide $\{2,3,5,7\}$ with this automaton. Both four-block words occur in the actual block sequence, 12 and 40 times among the first 20000 blocks. Within this automaton, a tail value therefore need not determine its first block. The identity does not exhibit equal tails at two positions of the actual prime-power sequence: those tails have different later phases. It proves failure of this interval criterion. The proof of Remark 5.1 uses the isolated reversal of two actual jumps instead.

5.5 Repeated tails and prime-power subseries

We compare the two normalisations before considering the sums over pure prime powers. The subseries assertions in this subsection have ordinary proofs and remain unformalised. For $P = \{2,3,5\}$ the two tail recurrences have the same bases, $b_{\tau_a} = P_{a+1}/P_a = b_a$, but their coefficients behave differently:

$$Y_{a+1} = b_a Y_a - \mu_{\tau_a}, \qquad X_{a+1} = b_a X_a - m_a.$$

The first identity is Remark 5.6(ii); the second is Theorem 8.2, with the notation of Section 8. The five types give only the coefficients $1, 3, 9, 13$, whereas the weighted block counts satisfy $m_a = \Theta((a+1)^2)$ by Lemma 8.3. Consequently $X_a > m_a/30$ is unbounded. Under rationality the integral values BX_a need not repeat, and the maps $y \mapsto (m_a + y)/b_a$ form an infinite family depending on a. Even equality of two tails would not identify their later coefficients. The residue criterion in Section 10 compares an endpoint residue with a quadratic bound instead. The example after Proposition 13.2 shows the limit of an argument using just bases and growth: the same bases admit quadratic numerators $\widehat{m}_a$ with $\sum_a \widehat{m}_a/P_{a+1} = 9$.

The two-tail proof of Remark 5.1 stops at the same place. Over the prime powers the repeated sum is $\mathcal{R}_P = \sum_{k\geq 0} c_k/Q_k$, where c_k counts the smooth integers in $[t_k, t_{k+1})$, with $t_0 = 1$. The two tails after t_k and after $t_k p^n$ still have words that differ by rearranging chains, and a rational value still makes the scaled tails integers of one denominator. But the scaled tails are unbounded, as the X_a are, and Remark 5.3 needs every tail in $(0,1)$: the integer $f(\sigma_i) - f(\sigma_i')$ is shown to vanish because it equals a difference of two tails. With unbounded tails that step fails, the digits c_k move with the rearrangement as well, and equality of two tails no longer forces agreement on each rearranged run. The irrationality of $\mathcal{R}_P$ for $|P| \geq 3$ remains open.

Prime-power subseries. The comparison of two tails applies to each prime-power subseries. For $p \in P$ put

$$E_p = \sum_{\alpha\geq 0} \frac{1}{\mathrm{L}(p^\alpha)} = \sum_{\alpha\geq 0} \frac{1}{\mathrm{H}_P(p^\alpha)},$$

the sub-sum of $\mathcal{R}_P$ over the terms at which the smooth integer is a power of p. For $P = \{p,q\}$ with $p < q$ these are the numbers $A = E_p$ and $1 + B_* = E_q$ of the proof of Theorem 1.2.

Remark 5.9 (sub-sums and the catalogue sum). Let P be a finite set of primes with $|P| \geq 2$.

(i) $\mathcal{D}_P = \sum_{q\in P} E_q - (|P|-1)$.

(ii) If $P = \{p, q\}$, then $\mathrm{H}_P(p^i q^j) = \mathrm{H}_P(p^i)\,\mathrm{H}_P(q^j)$ for all $i, j \geq 0$, and $\mathcal{R}_P = E_p E_q$.

(iii) If $P = \{p, q, r\}$, then $\mathrm{H}_P(p^i q^j r^k) = \mathrm{H}_P(p^i)\,\mathrm{H}_P(q^j)\,\mathrm{H}_P(r^k)\,p^{\kappa_p(j,k)} q^{\kappa_q(i,k)} r^{\kappa_r(i,j)}$, where $\kappa_p(j,k) = \lfloor j\log_p q + k\log_p r\rfloor - \lfloor j\log_p q\rfloor - \lfloor k\log_p r\rfloor \in \{0,1\}$ and κ_q, κ_r are defined in the same way. For a general finite P the corresponding exponents lie in $\{0,\dots,|P|-2\}$.

Proof. (i) Each term $1/Q_k$ of $\mathcal{D}_P$ with $k \geq 1$ is a term of E_{q_k} and of no other E_q, and each E_q also contains the term 1. (ii) and (iii) The exponent of p in $\mathrm{H}_P(\prod_{q\in P} q^{e_q})$ is $e_p + \lfloor \sum_{q\neq p} e_q \log_p q\rfloor$, and its exponent in $\prod_{q\in P}\mathrm{H}_P(q^{e_q})$ is $e_p + \sum_{q\neq p}\lfloor e_q \log_p q\rfloor$. For m nonnegative reals the floor of the sum exceeds the sum of the floors by an integer in $\{0,\dots,m-1\}$. With two primes $m = 1$, so the running LCMs factor, and summing $\mathrm{H}_P(p^i)^{-1}\,\mathrm{H}_P(q^j)^{-1}$ over $i, j \geq 0$ gives $E_p E_q$. □

With three primes $\mathcal{R}_P$ is therefore the product of the three single-prime sums twisted by the carries κ, and the plain product $E_p E_q E_r$ differs from it.

Remark 5.10 (the single-prime sub-sums). Let P be a finite set of primes with $|P| \geq 2$ and let $p \in P$. Then E_p is irrational.

For $P = \{p, q\}$, E_p is A or $1 + B_*$ in the proof of Theorem 1.2, an affine function of the Hecke–Mahler value $F_\theta(1/p, 1/q)$ with nonzero slope; for $p < q$, $E_p = p/(p-1) - (q-1)F_\theta(1/p, 1/q)$ by (6). It is transcendental by the theorem of Bugeaud and Laurent [15, Theorem 1.1], in the case due to Loxton and van der Poorten [16, Theorem 8, p. 40]. We cite these sources for this case and have not re-verified its earlier attribution history.

Proof. We modify the word argument so that it counts only occurrences of p. For an infinite word u and a finite word $\sigma = \sigma_1 \cdots \sigma_n$, put

$$V_p(u) = \sum_{i:u_i=p} \frac{1}{u_1 \cdots u_i}, \qquad f_p(\sigma) = \sum_{j:\sigma_j=p} \sigma_{j+1} \cdots \sigma_n.$$

Then $V_p(\sigma u) = (f_p(\sigma) + V_p(u))/\Pi(\sigma)$. Writing $y_k = V_p(q_{k+1} q_{k+2} \cdots)$, we have

$$E_p = 1 + \sum_{k:q_k=p} \frac{1}{Q_k}, \qquad Q_k E_p = Q_k + \sum_{j\leq k,\ q_j=p} \frac{Q_k}{Q_j} + y_k.$$

In an actual suffix, the prefix product at the jth occurrence of p is at least p^j, with strict inequality after another prime has occurred. Hence $0 < V_p(u) < 1/(p-1) \leq 1$, and rationality $E_p = N/K$ would imply $K y_k \in \mathbb{Z}$. The proof of Remark 5.3 therefore applies to V_p, f_p as well. On a chain of distinct primes containing p, only one term remains in f_p: the product of the primes following p. Unique factorisation detects a change in that set, without recovering its order. This weaker requirement avoids the collisions of the full block numerator f.

For each $q \neq p$, the form $\theta_p - \theta_q$ is nonzero on the rational subspace L used in Section 5.1. We may therefore choose $v \in L$ with $|v_q| \leq 1$ for every q and $v_q \neq v_p$ for $q \neq p$,

where $v_q = \theta_q(v)$, and put $\mu = \min_{q\neq p} |v_p - v_q| > 0$. Suppose $E_p = N/K$. Choose a jump t_k labelled p, and choose T, ν, g, β as in the general irrationality proof, with $2^\nu \geq K$. Set $\eta = 1/10$ and first fix

$$0 < \varepsilon < \min\{\beta, g/2, \log 2/(2|P|), \eta/4, \eta\mu/4\}.$$

Dirichlet's theorem then supplies arbitrarily large n and integers $m_q \geq 1$ such that $\delta_p = 0$ and $0 < |\delta_q| = |n \log p - m_q \log q| < \varepsilon$ for $q \neq p$. We compare t_k with $t_m = t_k p^n$ and cut crossings at gaps greater than 2ε. As before, each chain consists of distinct primes and is merely rearranged by the phase shift. The common prefix of ν letters and the integral-tail argument give $y_k = y_m$, and hence $f_p(C) = f_p(C')$ for every chain and its rearrangement.

We shall contradict this equality near the origin of T_0. Put

$$b_0 = \min_{q\neq p} \frac{|\delta_q|}{|v_p - v_q|}, \qquad B = \{q \neq p : \delta_q < 0\},$$
$$A_+ = \{q \neq p : v_q < v_p\}, \quad A_- = \{q \neq p : v_q > v_p\}.$$

Since $0 < b_0 \leq \varepsilon/\mu < \eta/4$ and A_+, A_- partition the nonempty set $P \setminus \{p\}$, one of these two sets, say A_s with $s \in \{+1, -1\}$, differs from B. Take $b = sb_0/2$ and consider $z = bv + \Theta(-\eta/2)$. For each q, its crossing nearest $\eta/2$ occurs at

$$\tau_q = \eta/2 - bv_q \in (3\eta/8, 5\eta/8), \qquad \tau_q - \tau_p = b(v_p - v_q).$$

Every other q-crossing lies outside $(-\eta/4, 5\eta/4)$, and for $q \neq p$ the displayed difference is nonzero with absolute value at most $|\delta_q|/2$. We choose a neighbourhood U of z in T_0 on which there are no crossings in $[-\eta/4, \eta/4] \cup [3\eta/4, 5\eta/4]$, each prime has a crossing in $(\eta/4, 3\eta/4)$, and every difference $\tau_q - \tau_p$ retains its sign and has absolute value less than $|\delta_q|$. These conditions are open and hold strictly at z.

By density, $\theta_k + \Theta(t^*) \in U$ for some $t^* > T + 2$. The resulting crossings near t^* form one chain: each is within ε of the p-crossing and every other crossing is more than $\eta/2 > 2\varepsilon$ away. In this chain the followers of p form A_s. After the shift, q follows p precisely when $\tau_q - \tau_p > \delta_q$, which, by the bound $|\tau_q - \tau_p| < |\delta_q|$, is equivalent to $\delta_q < 0$. Its followers now form $B \neq A_s$, contradicting $f_p(C) = f_p(C')$. □

This argument uses a change of followers and so needs no two-wall lemma. Either of A_+, A_- may be empty, but they are distinct when $|P| \geq 2$. For $P = \{p\}$ the sum is $p/(p-1)$, and this is precisely where the choice of two distinct follower sets fails. The strict conditions defining U are needed to carry the construction to an actual orbit.

For $P = \{2, 3, 5\}$, $p = 2$, $t_k = 2^{20}$ and $n = 65$, the first rearranged chain is $(2, 3)$ against $(3, 2)$ after 131 letters, with f_2 equal to 3 and 1, and the difference of the two tails, computed exactly to 300 letters, has $\log_{10} |y_k - y_m| = -58.098$.

For the repeated sum $\mathcal{R}_P$, changing the scale from t_k to $t_k p^n$ brings in early terms from the rays $p^\alpha w$, with w supported on $P \setminus \{p\}$. The comparison above accounts only for the ray $w = 1$, so it gives no corresponding estimate for the difference of the full repeated tails. In the case $P = \{2, 3, 5\}$ those normalised tails are unbounded, as the lower bounds below show.

6 Words obtained from a linear flow

We apply the interchange argument to the order in which finitely many linear functions take integer values. Their slopes will be assumed linearly independent over $\mathbb{Q}$; this hypothesis is not assumed for reciprocal prime logarithms. Interchanging two late crossings then determines the rational relations among the corresponding subseries, without requiring recovery of every letter. The arguments in this section remain ordinary proofs awaiting formalisation.

Let A be a finite alphabet. Give $a \in A$ an integer radix $b_a \geq 2$ and a rational coefficient c_a. For $w = w_1 w_2 \cdots$, define

$$Q_0 = 1, \qquad Q_n = \prod_{j=1}^{n} b_{w_j}, \qquad V_c(w) = \sum_{n \geq 1} \frac{c_{w_n}}{Q_n}, \qquad T_a(x) = \frac{c_a + x}{b_a}.$$

Write $\sigma w = w_2 w_3 \cdots$ and $X_w = \overline{\{\sigma^n w : n \geq 0\}}$ in the one-sided product topology. A word belongs to X_w precisely when each of its finite prefixes begins some shifted word. It need not itself be a shift. The first letter acts by the outermost map: if $u = u_1 \cdots u_m$, then $V_c(uz) = T_{u_1} \circ \cdots \circ T_{u_m}(V_c(z))$. Put $B(u) = \prod_{j \leq m} b_{u_j}$, with $B(\emptyset) = 1$.

Remark 6.1 (interchanging two letters after a long common prefix). Suppose $V_c(w)$ is rational. If for distinct letters a, b and arbitrarily long words u there is a common infinite suffix z such that both $uabz$ and $ubaz$ belong to X_w, then

$$c_a(b_b - 1) = c_b(b_a - 1).$$

If such interchanges join every pair of letters along a connected graph, $V_c(w)$ is rational precisely when there is $\lambda \in \mathbb{Q}$ with $c_a = \lambda(b_a - 1)$ for all a; in that case $V_c(w) = \lambda$.

Proof. First clear the common denominator of the coefficients, so that all c_a are integers. The recurrence $V_c(\sigma w) = b_{w_1} V_c(w) - c_{w_1}$ shows that, if $V_c(w) = p/K$, then every shifted value belongs to $K^{-1}\mathbb{Z}$. For $\lambda_a = c_a/(b_a - 1)$, the positive weights

$$\frac{b_{w_n} - 1}{Q_n} = \frac{1}{Q_{n-1}} - \frac{1}{Q_n}$$

sum to one. Hence $V_c(v)$ lies between $\min_a \lambda_a$ and $\max_a \lambda_a$ for every word v. Sharing m initial letters changes a value by at most the width of this interval divided by $B(u)$. Thus V_c is continuous. Since $K^{-1}\mathbb{Z}$ is closed, the lattice condition passes from the actual shifts to all of X_w.

The order of the maps matters. A direct calculation gives, for every x,

$$T_a(T_b(x)) - T_b(T_a(x)) = \frac{c_a(b_b - 1) - c_b(b_a - 1)}{b_a b_b}.$$

The values of $uabz$ and $ubaz$ therefore differ by this number divided by $B(u)$. Their difference belongs to $K^{-1}\mathbb{Z}$. If its numerator were nonzero, its absolute value would be positive but smaller than $1/K$ for a sufficiently long u, a contradiction. On a connected interchange graph all λ_a agree. Conversely, for $c_a = \lambda(b_a - 1)$ the displayed positive weights telescope and give $V_c(v) = \lambda$ for every v. Dividing by the earlier common denominator restores the statement for rational coefficients. □

The common suffix is essential: distinct words can have the same affine map. For example, the blocks $(5,7,3,2)$ and $(7,2,3,5)$ have the same map $(51+x)/210$. Choosing the second block at block indices $1,2,4,8,\dots$ produces a non-eventually-periodic word with rational value $51/209$. Thus aperiodicity and bounded tails do not suffice, even with integer bases; the interchange hypothesis is essential to the argument. Nor does the lemma assert sufficiency when its graph is disconnected.

Now let $r_1,\dots,r_d > 0$, $d \geq 2$, be linearly independent over $\mathbb{Q}$. Record the letter i at each positive time t for which $r_i t + \theta_i \in \mathbb{Z}$, where $\theta_i \in [0,1)$. Assume that no two of these functions take integer values simultaneously at positive time. In chronological order, the crossings give an infinite word w on $\{1,\dots,d\}$.

Remark 6.2 (isolated interchanges in the crossing sequence). Every pair of distinct labels satisfies the late-interchange hypothesis of Remark 6.1 for X_w.

Proof. Fix a label a. Immediately after its sufficiently late crossings, the remaining coordinates of the flow are

$$\theta_i - \frac{r_i}{r_a}\theta_a + k\frac{r_i}{r_a} \pmod 1 \qquad (i \neq a),$$

where k runs through all sufficiently large integers. These points are dense in the $(d-1)$-torus. Indeed, 1 and the r_i/r_a for $i \neq a$ are rationally independent. For every nonzero integral character of the torus, its average along this orbit is a geometric sum divided by its length and tends to zero. Trigonometric approximation of continuous functions then shows that every nonempty open set is visited infinitely often. This also proves density after any finite initial segment is discarded.

Fix $b \neq a$ and an arbitrarily large integer L, and work on the section just after an a-crossing. Set $T = L/r_a$ and choose the phase of b so that the b-coordinate crosses at T. This phase is nonzero, since $Lr_b/r_a \notin \mathbb{Z}$. Choose the remaining phases outside the countable sets that cause a crossing at zero, a third crossing at T, or any other positive simultaneous crossing. The resulting singular configuration has exactly the one double crossing of a,b at T: a second meeting of these two coordinates would make r_b/r_a rational.

Perturb the b phase on opposite sides. On each bounded time interval, a sufficiently small perturbation changes only this pair's order. Taking one-sided limits gives $uabz$ and $ubaz$ with the *same* infinite suffix z. No fixed nonzero perturbation is required to preserve all later crossings: agreement on each finite interval suffices for these limiting words. Their common prefix has at least $L-1$ letters, from the earlier a-crossings. For each finite prefix of either limit, a small open set of nonsingular section phases realises that prefix. The dense actual a-section orbit meets that set, so the prefix occurs at the beginning of an actual shifted word. Consequently both limiting words belong to X_w. Letting L grow proves the assertion. □

Remark 6.3 (rational relations among the subseries). Under the preceding hypotheses on the linear flow, choose integers $b_i \geq 2$, put $Q_n = \prod_{k\leq n} b_{w_k}$ and

$$S_i = \sum_{n:\, w_n = i} \frac{1}{Q_n} \qquad (1 \leq i \leq d).$$

For every $c_i \in \mathbb{Q}$,

$$\sum_{i=1}^{d} c_i S_i \in \mathbb{Q} \quad \Longleftrightarrow \quad c_i = \lambda(b_i - 1) \quad \text{for all } i \text{ and some } \lambda \in \mathbb{Q}.$$

The rational value is λ. Equivalently, all rational relations $a_0 + \sum_i a_i S_i = 0$ are multiples of $(-1, b_1 - 1, \dots, b_d - 1)$. Thus the span of $1, S_1, \dots, S_d$ has dimension d; each S_i and each nonempty proper unweighted sub-sum is irrational. The full unweighted sum is irrational exactly when the radices are not all equal.

The following argument proves the assertion of the remark in ordinary mathematics; formalisation and external peer review remain outstanding.

Proof. Absolute convergence gives $\sum_i c_i S_i = V_c(w)$. The previous two remarks make the interchange graph complete, so they give both directions and the stated value. The relation space follows by moving the rational value to the constant coordinate. An indicator vector of a nonempty proper set has both zero and one entries and cannot be a multiple of the strictly positive vector $(b_i - 1)_i$. The all-ones vector is such a multiple precisely when all the b_i agree. □

For example, take

$$(r_1, r_2, r_3, r_4) = (1, \sqrt{2}, \sqrt{3}, \sqrt{6}), \qquad (b_1, b_2, b_3, b_4) = (2, 3, 5, 7), \qquad \theta_i = 0.$$

The four rates form the standard basis of the biquadratic field $\mathbb{Q}(\sqrt{2}, \sqrt{3})$ over $\mathbb{Q}$; the ratio of any two distinct rates is irrational, so positive crossings never coincide. Write $(d_1, d_2, d_3, d_4) = (1, 2, 3, 6)$. Counting crossings up to the nth crossing of the i-th coordinate gives the explicit series

$$S_i = \sum_{n \geq 1} \left(\prod_{j=1}^{4} b_j^{\lfloor n \sqrt{d_j / d_i} \rfloor} \right)^{-1}. \tag{9}$$

The only rational relation among $1, S_1, S_2, S_3, S_4$ is generated by $S_1 + 2S_2 + 4S_3 + 6S_4 = 1$. In particular their rational span has dimension four and $\sum_i S_i$ is irrational. This does not provide an irrationality measure or transcendence result. In the related two-letter Hecke–Mahler setting, stronger transcendence results exist [15]; the example's four independent rates do not satisfy that two-letter hypothesis. The ordinary two-wall argument for the distinct running-LCM sum avoids this assumption on reciprocal prime logarithms. The repeated running-LCM question remains open.

7 Separation of the three-prime kernel

We test an exact kernel identity: can finitely many products separate one exponent from the other two? A single binary carry rules this out for three primes. Diagonal rescaling leaves threshold columns; independent row and column choices produce staircase minors of every order. This says nothing yet about identities after summation or approximation, which we treat separately. The initial two-generator identity is valid for real generators; distinct primes enter the later density argument.

Proposition 7.1 (two generators separate)**.** *For real $p, q > 1$, with $L_{p,q}(t) = p^{\lfloor \log_p t \rfloor} q^{\lfloor \log_q t \rfloor}$ as above, and all integers $i, j \geq 0$, the two-prime kernel $\mathrm{K}_2(i,j) = 1/L_{p,q}(p^i q^j)$ is the outer product*

$$\mathrm{K}_2(i,j) = (p^i q^{\lfloor \log_q p^i \rfloor})^{-1} (p^{\lfloor \log_p q^j \rfloor} q^j)^{-1},$$

so every two-by-two minor of K_2 vanishes.

Proof. Since i and j are integers, $\lfloor \log_p(p^i q^j) \rfloor = i + \lfloor \log_p q^j \rfloor$ and $\lfloor \log_q(p^i q^j) \rfloor = j + \lfloor \log_q p^i \rfloor$. Hence $L_{p,q}(p^i q^j)$ is the product of $p^i q^{\lfloor \log_q p^i \rfloor}$, which depends on i alone, and $p^{\lfloor \log_p q^j \rfloor} q^j$, which depends on j alone. A matrix whose entries are a product of a row function and a column function has vanishing two-by-two minors. For distinct primes p, q the product $L_{p,q}$ is the running least common multiple by the argument of Theorem 3.1. □

At three primes the smallest rectangle already disproves a single product. A factorisation $f(i)g(j)h(k)$ would force $\mathrm{K}(0,0,0)\,\mathrm{K}(1,1,0) = \mathrm{K}(1,0,0)\,\mathrm{K}(0,1,0)$.

Proposition 7.2 (non-separability at $\{2,3,5\}$)**.** *With $(p,q,r) = (2,3,5)$,*

$$\det \begin{pmatrix} \mathrm{K}(0,0,0) & \mathrm{K}(0,1,0) \\ \mathrm{K}(1,0,0) & \mathrm{K}(1,1,0) \end{pmatrix} = \det \begin{pmatrix} 1 & 1/6 \\ 1/2 & 1/60 \end{pmatrix} = -\frac{1}{15} \neq 0.$$

Proof. The four values are computed from $\mathrm{H}(1) = 1$, $\mathrm{H}(2) = 2$, $\mathrm{H}(3) = 2 \cdot 3 = 6$ and $\mathrm{H}(6) = 4 \cdot 3 \cdot 5 = 60$, so the determinant is $1/60 - 1/12 = -1/15$. □

Theorem 7.3 (no finite separation of the kernel)**.** *Let p, q, r be primes with $p \neq q$, $p \neq r$ and $q \neq r$. For every $n \geq 1$ there are injective maps $I, J : \{0, \dots, n-1\} \to \mathbb{N}$ such that, for every $k \geq 0$,*

$$\det(\mathrm{K}(I(a), J(b), k))_{0 \leq a,b < n} \neq 0.$$

Consequently, for no finite d do there exist rational-valued functions $f_\ell : \mathbb{N} \to \mathbb{Q}$ and $G_\ell : \mathbb{N}^2 \to \mathbb{Q}$, $0 \leq \ell < d$, satisfying $\mathrm{K}(i,j,k) = \sum_{\ell<d} f_\ell(i) G_\ell(j,k)$ for all i, j, k.

Proof. For fixed k, the p- and q-exponents still split into row and column terms. The r-exponent also includes the carry from adding two fractional parts. To isolate it, put $\alpha = \log_r p$, $\beta = \log_r q$, $x_i = \{i\alpha\}$ and $y_j = \{j\beta\}$. Write $v_p(N)$ for the exponent of the prime p in the positive integer N. The three prime exponents are

$$\begin{aligned} v_p(\mathrm{H}(p^i q^j r^k)) &= i + \lfloor j \log_p q + k \log_p r \rfloor, \\ v_q(\mathrm{H}(p^i q^j r^k)) &= j + \lfloor i \log_q p + k \log_q r \rfloor, \\ v_r(\mathrm{H}(p^i q^j r^k)) &= k + \lfloor i\alpha \rfloor + \lfloor j\beta \rfloor + \lfloor x_i + y_j \rfloor. \end{aligned}$$

Every term except the last floor depends on i and k alone or on j and k alone, so with positive rational $R_i(k)$ and $C_j(k)$,

$$\mathrm{K}(i,j,k) = R_i(k)\, C_j(k)\, t^{\lfloor x_i + y_j \rfloor}, \qquad t = r^{-1}. \tag{10}$$

The remaining matrix is independent of k. Since $x_i + y_j \in [0,2)$, its entry is 1 when there is no carry and t when there is one.

Each of α and β is irrational, since a rational value would give an equality of positive powers of distinct primes, so each fractional-part orbit is dense in $[0,1]$ and each is injective. The row and column indices are chosen independently, so no density theorem for a single orbit of pairs is required. First choose indices with $0 < x_{I(0)} < \cdots < x_{I(n-1)} < 1$. We want the columns to cross their thresholds at successive selected rows, so that consecutive row differences will leave a triangular matrix. Write $s_b = 1 - y_{J(b)}$, and use density of the second orbit to choose

$$s_0 \in (0, x_{I(0)}), \qquad s_b \in (x_{I(b-1)}, x_{I(b)}) \quad (1 \le b < n).$$

The intervals are disjoint, so the $J(b)$ are distinct. Their strict endpoints ensure that no selected entry lies on a threshold, and $x_{I(a)} + y_{J(b)} \ge 1$ holds exactly when $b \le a$. Dividing row a by $R_{I(a)}(k)$ and column b by $C_{J(b)}(k)$ therefore leaves

$$C_n(t) = \begin{pmatrix} t & 1 & 1 & \cdots & 1 \\ t & t & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \ddots & \vdots \\ t & t & \cdots & t & 1 \\ t & t & \cdots & t & t \end{pmatrix}, \qquad \det C_n(t) = t(t-1)^{n-1} \neq 0,$$

the determinant following by subtracting from each row its predecessor, working upwards from the last. Before the division, the determinant equals

$$\det C_n(t) \prod_{a<n} R_{I(a)}(k) \prod_{b<n} C_{J(b)}(k),$$

which is nonzero for every k; this also explains why the same indices work in every layer.

For the last assertion, suppose that a representation with d summands exists and fix k. On the selected rows $I(0), \dots, I(d)$ and columns $J(0), \dots, J(d)$ the resulting matrix factors as

$$(f_\ell(I(a)))_{0 \le a \le d,\, \ell < d} (G_\ell(J(b), k))_{\ell < d,\, 0 \le b \le d}.$$

It has rank at most d and hence zero determinant, contradicting the nonzero minor of order $d+1$. □

Index selection is essential. In the $k = 0$ layer of the $\{2,3,5\}$ kernel, the leading 4×4 minor is singular because $\mathrm{K}(3,j,0) = \mathrm{K}(0,j,0)/120$ for $0 \le j \le 3$. This is not a proportionality of the full rows: at the next column,

$$\mathrm{K}(3,4,0) - \frac{\mathrm{K}(0,4,0)}{120} = -\frac{1}{19440000}.$$

A singular leading minor therefore does not settle the rank.

The selected minors put thresholds at distinct cuts among the row phases. A general sample need not do so: several columns can have the same cut, and two can be constant. Counting the distinct cuts gives its rank.

Proposition 7.4 (rank of threshold columns)**.** *Let $m \geq 1$ and let c lie in a field with $c \neq 0, 1$. For $0 \leq h \leq m$, let v_h be the length-m column whose first h entries are 1 and whose remaining entries are c. If the distinct columns of a matrix are the v_h with h in a nonempty set $E \subseteq \{0, \dots, m\}$, then its rank is*

$$|E| - \mathbf{1}_{\{0,m\} \subseteq E}.$$

Proof. Write $E = \{h_1 < \cdots < h_s\}$. The $s - 1$ consecutive differences are

$$v_{h_{a+1}} - v_{h_a} = (1 - c)\mathbf{1}_{\{h_a, \dots, h_{a+1}-1\}},$$

so they are linearly independent because their nonempty supports are disjoint. If $h_1 > 0$ or $h_s < m$, those supports miss a coordinate on which v_{h_1} is nonzero, and adjoining v_{h_1} gives rank s. If $h_1 = 0$ and $h_s = m$, the differences sum to $(1 - c)\mathbf{1}$ while $v_0 = c\mathbf{1}$, so v_0 is already in their span and the rank is $s - 1$. □

The restrictions $c \neq 0, 1$ exclude the zero column at $c = 0$ and the collapse of all columns at $c = 1$. They hold automatically at $c = 1/r$ over $\mathbb{Q}$. To apply the proposition, fix k and sort the chosen row phases x_i. Each normalised column is a threshold column v_h, where h is the number of sampled phases strictly below $1 - y_j$. Repeated threshold positions give proportional columns before column normalisation. The row and column factors in (10) are nonzero, so the formula gives the exact rank of every nonempty rectangular sample within this layer, independently of k. Repeated rows or columns do not change rank.

This also gives an exact algorithm. Order the sampled rows by the rational numbers $p^i / r^{\lfloor \log_r p^i \rfloor}$. For column j, count those strictly below $r^{\lfloor \log_r q^j \rfloor + 1} / q^j$, then apply the endpoint correction to the set of resulting counts. Indeed, raising $x_i < 1 - y_j$ to base r gives exactly this rational comparison, and $x_i + y_j = 1$ belongs to the c side of the threshold. The integer logarithms are found by comparing powers, so the whole calculation uses integer arithmetic rather than numerical logarithms.

The fixed-layer restriction is essential. With $(p, q, r) = (2, 3, 5)$, rows $i = 0, 1$ and columns indexed by $(j, k) = (0, 0), (0, 1)$ give

$$\begin{pmatrix} \mathrm{K}(0,0,0) & \mathrm{K}(0,0,1) \\ \mathrm{K}(1,0,0) & \mathrm{K}(1,0,1) \end{pmatrix} = \begin{pmatrix} 1 & 1/60 \\ 1/2 & 1/360 \end{pmatrix}, \qquad \det = -\frac{1}{180}.$$

Both columns have the same phase $y_0 = 0$, yet their rank is 2, not 1. When k varies with the column, the row factor also varies with that column and cannot be removed by one common diagonal rescaling. The threshold formula is not a rank formula for such mixed-layer samples.

Scope of the rank theorem. Fan's comment of 26 June 2026 [17] notes that the two-prime argument does not seem to generalise immediately to $|P| \geq 3$. The theorem above gives a precise obstruction: no finite exact separation of the stated form exists. The same finite-dimensional proof works for real or complex factors, without continuity or boundedness assumptions. This does not exclude approximation or imply irrationality or transcendence of the sum. The threshold construction needs $\log_r p$ and $\log_r q$ irrational, but not $p \neq q$. The latter condition is needed for the three-prime running-LCM interpretation, not for the two independent dense orbits.

7.1 Smith factors of the selected staircase

The preceding argument only uses that a determinant is nonzero. An irrationality argument would also need control of integer coefficients after denominators are cleared. We examine that separate question for the selected minors. For integers $N \geq 1, r \geq 2$, let $A_N(r)$ have entry 1 on and below the diagonal and r above it. This is the staircase used in Theorem 7.3. Its Smith factors determine the lattice index; they do not by themselves give a small integral linear form.

Remark 7.5 (Smith factors of the selected staircase). The Smith factors of $A_N(r)$ are $1, r-1, \dots, r-1$, and $\det A_N(r) = (1-r)^{N-1}$.

The Smith calculation and its consequence for integral coefficients are ordinary arguments here; the existing Lean result proves nonvanishing of the selected minors, not these arithmetic statements.

Proof. Working from bottom to top, subtract row $i-1$ from row i for every $i > 1$. Row i then has just $1-r$ in column i. The first row is $(1, r, \dots, r)$; subtract r times its first column from each later column. The resulting diagonal is $(1, 1-r, \dots, 1-r)$. These are unimodular integral operations, and 1 divides $r-1$, so this is its Smith form up to signs. Its determinant has the displayed sign. □

For the selected $\{2,3,5\}$ minor, the positive integers U_i, V_j give

$$K = \operatorname{diag}(U_i)^{-1} A_N(5) \operatorname{diag}(5V_j)^{-1}.$$

Consequently

$$\det K = \frac{(-4)^{N-1}}{W}, \qquad W = 5^N \prod_i U_i \prod_j V_j.$$

More generally, an entrywise-integral diagonal clearing $\operatorname{diag}(s) K \operatorname{diag}(t)$ has diagonal entries $z_i = s_i t_i / (5U_i V_i) \in \mathbb{Z} \setminus \{0\}$ and determinant $(-4)^{N-1} \prod_i z_i$. Its absolute determinant is therefore at least 4^{N-1}, attained by $s_i = 5U_i$, $t_j = V_j$. This optimality is restricted to diagonal entrywise clearings.

For the selected carry pattern, the injective nonnegative index families I, J satisfy $v_2(W) \geq \sum_i I(i) + \sum_j J(j) \geq N(N-1) \geq 2(N-1)$. Thus 4^{N-1} divides W, and the *reduced* determinant is $\pm 1/D$, where $D = W/4^{N-1} \in \mathbb{N}_{>0}$. Thus the factor 4^{N-1} in the integer determinant is absorbed entirely by its denominator. For a linear form in one target value, a small rational prefactor likewise disappears when the coefficients are reduced to coprime integers. Indeed, for $K \in \mathrm{GL}_N(\mathbb{Q})$ and rational u, v, the Schur complement gives

$$\det \begin{pmatrix} K & u \\ v^T & X \end{pmatrix} = \det K (X - v^T K^{-1} u).$$

Write $v^T K^{-1} u = s/t$ in lowest terms, with $t > 0$. If a nonzero rational scalar multiple of $\det K(X - s/t)$ has integer coefficients $aX + b$, then $b = -as/t$. Since $\gcd(s,t) = 1$, we have $t \mid a$. Dividing the two coefficients by their common factor therefore leaves $\pm(tX - s)$. This is the primitive integral coefficient polynomial, regardless of the smallness of

det K or a compatible rational change of basis. The conclusion concerns this construction with one target and the diagonal clearings above. It does not exclude new recurrences, multiple target values or stronger approximation data, and it does not settle the repeated running-LCM problem. Smith form is used here in its standard sense [14].

7.2 Uniform approximation of the rescaled kernel

Exact rank does not decide approximation. For the rescaled kernel, infinitely many columns remain a fixed distance apart, giving a uniform lower bound against every finite rank. For the original kernel, decay instead permits summable truncation. We prove these statements for the two different matrices. Write

$$C(i,j) = t^{\lfloor x_i + y_j \rfloor}, \qquad x_i = \{i \log_r p\}, \quad y_j = \{j \log_r q\}, \quad t = r^{-1}. \tag{11}$$

This is the remaining factor in (10) after removing the row and column factors. A matrix has *finite separated rank* if $A(i,j) = \sum_{\ell<d} f_\ell(i) g_\ell(j)$ for some finite d. Equivalently, its columns span a finite-dimensional space of real sequences. No continuity or boundedness of f_ℓ, g_ℓ is assumed.

Theorem 7.6 (distance from matrices of finite separated rank)**.** *Let p, q, r be pairwise distinct primes and let C be as in (11). Then*

$$\inf_A \sup_{i,j\geq 0} |C(i,j) - A(i,j)| = \frac{1-t}{2} = \frac{r-1}{2r},$$

the infimum being over all matrices A of finite separated rank, and it is attained by the constant matrix of value $(1+t)/2$.

Proof. We first show that any two distinct columns of C have supremum distance $1-t$. Every entry belongs to $\{1, t\}$, with $C(i,j) = t$ precisely when $x_i + y_j \geq 1$. Given $j \neq k$, irrationality of $\log_r q$ makes $y_j \neq y_k$, so suppose $y_j < y_k$. By density of (x_i) we can choose i with $1 - y_k < x_i < 1 - y_j$; the two entries in this row are t and 1.

Suppose a finite-rank matrix A approximated C with error $\varepsilon = \sup_{i,j} |C(i,j) - A(i,j)| < (1-t)/2$. Its columns A_j would be bounded sequences with $\|A_j\|_\infty \leq 1+\varepsilon$, and their span V would be finite-dimensional. Although no boundedness was assumed for the separated factors, the supremum norm is defined on this span because each column is bounded. For distinct columns the triangle inequality gives

$$\|A_j - A_k\|_\infty \geq \|C_j - C_k\|_\infty - 2\varepsilon = 1 - t - 2\varepsilon > 0.$$

Thus a bounded subset of V would contain infinitely many points at a fixed positive distance from one another, contrary to total boundedness in finite dimension. This proves the lower bound.

For equality, take the constant matrix $A(i,j) = (1+t)/2$. It has rank one, and its distance from every entry of C is $(1-t)/2$. □

The lower bound concerns the entire normalised matrix, not each finite restriction. For example, at $t = 1/5$ the matrices

$$T = \begin{pmatrix} 1/5 & 1 \\ 1/5 & 1/5 \end{pmatrix}, \qquad A = \begin{pmatrix} 3/10 & 9/10 \\ 1/10 & 3/10 \end{pmatrix}$$

satisfy $\det A = 0$ and $\|T - A\|_{\max} = 1/10 < 2/5$. Here $\|\cdot\|_{\max}$ is the largest absolute entry. This finite example has no bearing on the infinite family of separated columns used in the proof.

For the original kernel, the decaying row and column factors instead allow approximation in the summation norm. Let $K^{(N)}(i,j,k) = \mathrm{K}(i,j,k)$ for $i < N$ and $K^{(N)}(i,j,k) = 0$ otherwise. This is a sum of at most N terms separated between i and (j,k). Since $\mathrm{H}(x) > x^3/(pqr)$ for $x \geq 1$, geometric summation gives

$$\sum_{i,j,k\geq 0} |\mathrm{K}(i,j,k) - K^{(N)}(i,j,k)| \leq \frac{pqr\, p^{-3N}}{(1-p^{-3})(1-q^{-3})(1-r^{-3})}. \tag{12}$$

The same bound controls the supremum of the entrywise errors, so these finite separated-rank approximants converge both in ℓ^1 and uniformly to the original kernel. The positive uniform lower bound in Theorem 7.6 belongs to the rescaled matrix. Diagonal rescaling preserves exact rank, but the rescaling factors here are unbounded and do not preserve uniform error estimates. Neither the rank obstruction nor these approximation bounds decide whether the scalar sum is rational.

8 The recurrence for tails between powers of two

Return to $S = \mathscr{R}_{\{2,3,5\}}$ and $P = \{2,3,5\}$. In place of the three jumps in $(16,32]$, we now count all seven smooth integers in $[16,32)$: $16, 18, 20, 24, 25, 27, 30$. Their running LCMs are $720, 720, 720, 720, 3600, 10800, 10800$. Two changes need keeping separate: we retain multiplicities, and we include the left rather than the right endpoint. The first changes the coefficients; the second changes the normalisation.

We group the terms of S into $[2^a, 2^{a+1})$, including the left endpoint. The prefix to be cleared is therefore strictly before 2^a: it excludes the jump by 2 at that boundary. Put $P_a = \mathrm{H}(2^a)$ and $h_a = P_a/2$. Every running LCM in that prefix divides h_a. Here $h_0 = 1/2$, and h_a is an integer for $a \geq 1$. The powers of 3 and 5 selected at 2^a exceed $2^a/3$ and $2^a/5$, respectively, while the power of 2 is 2^a itself. Thus

$$\frac{8^a}{15} < P_a \leq 8^a \qquad (a \geq 0).$$

With this normalisation, define the block masses and tails by

$$s_a = \sum_{\substack{i,j,k\geq 0 \\ 2^a \leq 2^i 3^j 5^k < 2^{a+1}}} \frac{1}{\mathrm{H}(2^i 3^j 5^k)}, \qquad T_a = \sum_{j\geq a} s_j, \qquad X_a = h_a T_a. \tag{13}$$

For $a \geq 1$, the prefix has exponent of 2 at most $a-1$ and other exponents at most their boundary values. This proves the divisibility by $P_a/2$ needed in Lemma 9.2.

The jumps after 2^a and up to 2^{a+1} consist of any powers of 3 or 5 strictly inside that interval, followed by the factor 2 at its right endpoint. There is at most one power of each odd prime: successive powers have ratio greater than two. Let I_a list the pairs (p,e) with $p \in \{3,5\}$ and $2^a < p^e < 2^{a+1}$, ordered by the value p^e.

Proposition 8.1 (four possible bases)**.** *For every a,*

$$b_a = \frac{P_{a+1}}{P_a} = 2 \prod_{(p,e)\in I_a} p \in \{2,6,10,30\}, \qquad \text{so} \qquad 2 \le b_a \le 30. \tag{14}$$

Proof. A power of 3 may occur and a power of 5 may occur, each at most once. The resulting factor is 2, $2\cdot 3$, $2\cdot 5$ or $2\cdot 3\cdot 5$. □

The coefficient subtracted at step a is the block mass with its denominators cleared:

$$m_a = h_{a+1}s_a = \sum_{\substack{x \text{ smooth} \\ 2^a \le x < 2^{a+1}}} \frac{P_{a+1}}{2\,\mathrm{H}(x)}.$$

Each summand is an integer, by the same strict-cutoff argument. For example, $[2,4)$ contains just 2 and 3, so $m_1 = 6(1/2 + 1/6) = 4$ and $X_2 = 6X_1 - 4$. The first four pairs (b_a, m_a), starting at $a = 1$, are $(6,4)$, $(10,7)$, $(6,7)$ and $(30,65)$. In particular $m_3 > b_3$: these coefficients are not positional digits.

In the block $[16,32)$, the internal jumps are 25 and 27, in that order. Multiplication by $h_5 = 10800$ gives the weights for the seven points listed above. The four smooth numbers $16, 18, 20, 24$ precede both jumps and have weight 15; 25 precedes only the jump at 27 and has weight 3; and $27, 30$ have weight 1. Thus

$$m_4 = 4\cdot 15 + 1\cdot 3 + 2\cdot 1 = 7 + (5-1)\cdot 4\cdot 3 + (3-1)\cdot 5 = 65.$$

The second expression starts with the seven points and adds the corrections before 25 and 27. It is this form that extends to every block.

To compute m_a in general, count the smooth numbers before each prime-power jump. Let $\mathcal{N}(t)$ be the number of positive $\{2,3,5\}$-smooth integers strictly below t. First give every point in $[2^a, 2^{a+1})$ weight one. At an internal jump p^e, the points before that jump need an additional weight $p-1$, multiplied by the prime factors at all later internal jumps. Thus, for $a \ge 1$,

$$\begin{aligned} m_a &= \mathcal{N}(2^{a+1}) - \mathcal{N}(2^a) \\ &\quad + \sum_{(p,e)\in I_a} (p-1)\bigl(\mathcal{N}(p^e) - \mathcal{N}(2^a)\bigr) \prod_{\substack{(q,f)\in I_a \\ p^e < q^f}} q, \\ m_0 &= 1. \end{aligned} \tag{15}$$

The first difference counts the whole block; each later difference counts its points strictly before the indicated jump. The proof below justifies these weights, establishes convergence and derives the tail identities.

Theorem 8.2 (the tail recurrence)**.** *The block masses are summable and $S = \sum_{a\ge 0} s_a$. For every $a \ge 0$,*

$$m_a = h_{a+1}s_a \in \mathbb{N}_{>0}, \qquad X_{a+1} = b_a X_a - m_a, \qquad X_a = \sum_{j\ge a} \frac{m_j}{b_a b_{a+1} \cdots b_j}. \tag{16}$$

Proof. Every exponent of 3 or 5 in the ath block is at most a, and for each such pair Lemma 3.5 leaves at most one exponent of 2 placing the point in $[2^a, 2^{a+1})$. Each running LCM is at least 2^a and the block contains 2^a, so $0 < s_a \le (a+1)^2 2^{-a}$. The majorant is summable, which justifies every tail splitting below, and unique prime factorisation identifies $\sum_a s_a$ with the original repeated series.

Write the internal jumps as $t_\ell = p_\ell^{e_\ell}$, $1 \le \ell \le v$, and set $t_0 = 2^a$, $t_{v+1} = 2^{a+1}$ and $n_\ell = \mathcal{N}(t_\ell)$. On $[t_\ell, t_{\ell+1})$ the running LCM is $P_a \prod_{u \le \ell} p_u$, and the terminal jump has factor two, so

$$h_{a+1} s_a = \sum_{\ell=0}^{v} (n_{\ell+1} - n_\ell) \prod_{u>\ell} p_u = n_{v+1} - n_0 + \sum_{\ell=1}^{v} (p_\ell - 1)\Big(\prod_{u>\ell} p_u\Big)(n_\ell - n_0).$$

The second equality is finite summation by parts and is exactly (15). The count itself is finite: for $p = 2, 3, 5$, with q, r the other primes, counting by the exponent of p gives

$$\mathcal{N}(p^e) = \sum_{u=1}^{e} \#\{(i,j) \in \mathbb{N}^2 : q^i r^j < p^u\}.$$

At $a = 0$ the block is the single point 1, giving $m_0 = 1$. Positivity follows from $s_a > 0$. Splitting $T_a = s_a + T_{a+1}$ and multiplying by h_a gives the recurrence, and $b_a \cdots b_j = P_{j+1}/P_a$ with $m_j = h_{j+1} s_j$ gives $m_j/(b_a \cdots b_j) = h_a s_j$, whose summation is the last identity. □

For estimates, it is useful to count by the odd part $3^j 5^k$ instead. Each such part below 2^{a+1} has exactly one power-of-two multiple in $[2^a, 2^{a+1})$. This turns the same numerator into a two-dimensional count, with weights determined by the remaining odd-prime jumps.

Lemma 8.3 (the block numerator as a weighted lattice count)**.** *Put $\lambda_3 = \log_2 3$, $\lambda_5 = \log_2 5$ and $\theta_p = 1/\lambda_p$ for $p = 3, 5$. For $j, k \ge 0$ write $w_{j,k} = j\lambda_3 + k\lambda_5$ and $t_{j,k} = \{w_{j,k}\}$. The block numerator is*

$$m_a = \sum_{\substack{j,k \ge 0 \\ w_{j,k} < a+1}} 3^{\lfloor (a+1)\theta_3 \rfloor - \lfloor (a+t_{j,k})\theta_3 \rfloor} 5^{\lfloor (a+1)\theta_5 \rfloor - \lfloor (a+t_{j,k})\theta_5 \rfloor}.$$

Every summand is in $\{1, 3, 5, 15\}$. In particular, for $a \ge 0$,

$$(\lfloor a/(2\lambda_3) \rfloor + 1)(\lfloor a/(2\lambda_5) \rfloor + 1) \le m_a \le 15(a+1)^2.$$

Thus $m_a = \Theta((a+1)^2)$ and the numerator sequence is unbounded. These are integer numerators, not positional digits restricted to $\{0, \dots, b_a - 1\}$.

Proof. For each pair with $w_{j,k} < a + 1$, exactly one exponent $i = a - \lfloor w_{j,k} \rfloor \ge 0$ puts $2^i 3^j 5^k = 2^{a + t_{j,k}}$ in $[2^a, 2^{a+1})$. This pairs each point of the triangle with exactly one smooth integer in the block. Substituting in $P_{a+1}/(2\,\mathrm{H}(2^i 3^j 5^k))$ cancels the power of 2 and gives the displayed weight. Each remaining floor increment is zero or one since $0 < \theta_p < 1$. Finally $\lambda_3, \lambda_5 > 1$ implies $j, k \le a$ for every summation pair, proving the upper bound. For the lower bound, restrict to $0 \le j \le \lfloor a/(2\lambda_3) \rfloor$ and $0 \le k \le \lfloor a/(2\lambda_5) \rfloor$. Then $w_{j,k} \le a$ and each weight is at least one. □

The quadratic order comes from the size of the triangle, not from a polynomial formula for m_a. As a changes, new points enter and the weights of old points can change at floor crossings. A finite-difference argument must control both effects; polynomial growth alone cancels neither. Quadratic growth does not make this count a quadratic polynomial.

The recurrence also restricts how persistently a nonintegral tail can approach the integers. The following alternative uses only the integer coefficients and the bounds $2 \le b_a \le 30$.

Proposition 8.4 (integer tails or repeated separation from the integers)**.** *For every integer $B \ge 1$, either $BX_a \in \mathbb{Z}$ for some $a \ge 0$ and every later a, or for every a_0 there is $a \ge a_0$ with $|BX_a - z| \ge 1/31$ for every $z \in \mathbb{Z}$.*

Proof. If the second alternative fails, there are A and integers z_a such that $e_a = BX_a - z_a$ satisfies $|e_a| < 1/31$ for all $a \ge A$. The recurrence gives $z_{a+1} - b_a z_a + Bm_a = b_a e_a - e_{a+1}$. Since Bm_a is integral, the left side is an integer; the right side has absolute value less than $(30 + 1)/31 = 1$. Both sides therefore vanish, so $e_{a+1} = b_a e_a$. Hence $|e_{A+k}| \ge 2^k |e_A|$ for every k, whereas $|e_{A+k}| < 1/31$. Thus $e_A = 0$, and the integer recurrence propagates integrality from BX_A to every later BX_a. □

The proposition does not determine which alternative holds for the tails of S. For comparison, Erdős–Taylor [21, Theorem 1, p. 600] prove a countability statement for increasing integer sequences with bounded successive ratios; Fan [22, Lemma 3.1, arXiv v1] allows an unbounded, not necessarily increasing sequence, still with a uniform upper bound on successive ratios. For our divisibility chain the same error argument identifies the entire exceptional set, not just its cardinality. For each fixed integer $B \ge 1$,

$$\{\xi \in \mathbb{R} : \operatorname{dist}(Bh_a\xi, \mathbb{Z}) \longrightarrow 0\} = \frac{1}{B}\mathbb{Z}[1/30], \qquad \mathbb{Z}[1/30] = \{m/30^t : m \in \mathbb{Z},\ t \in \mathbb{N}\}.$$

Indeed, apply the proof above to $Bh_a\xi$, whose successive terms are related by multiplication by b_a. Convergence of the distances forces $Bh_A\xi \in \mathbb{Z}$ for some $A \ge 1$. Since h_A has no prime factors outside $2, 3, 5$, this gives membership in the right-hand side. Conversely, every fixed product of powers of $2, 3, 5$ eventually divides h_a, so every member of the right-hand side gives integral values for all large a.

Separation from the integers for one multiplier does not prove irrationality. For example, $\xi = 1/7$ satisfies $\operatorname{dist}(h_a\xi, \mathbb{Z}) \ge 1/7$ for every $a \ge 1$, because $7 \nmid h_a$; with $B = 7$ the values $Bh_a\xi$ are all integers. For S, the denominator clearing in Theorem 9.3 below shows which multipliers matter: the integral alternative for the actual tails must be excluded for every $B \ge 1$ coprime to 30, not merely for $B = 1$.

9 Bounding the tails and clearing a rational denominator

We sharpen the quadratic estimate by counting points of equal running LCM. As the cutoff increases, one prime exponent increases at each jump, so the sum of the exponents indexes the nonempty cells. At 2^a it is

$$n_a = a + \lfloor \log_3(2^a) \rfloor + \lfloor \log_5(2^a) \rfloor.$$

Counting the points in each cell and summing their geometric weights gives the polynomial

$$Q(n) = \frac{n^2 + 8n + 18}{9}.$$

Theorem 9.1 (a quadratic upper bound)**.** *For every $a \geq 0$, $0 < X_a \leq Q(n_a)$.*

Proof. Partition the smooth integers $x \geq 2^a$ by the exponent vector of their running LCM

$$(A, B, C) = (\lfloor \log_2 x \rfloor, \lfloor \log_3 x \rfloor, \lfloor \log_5 x \rfloor).$$

The cell of a vector is the interval $[\lambda, \eta)$, where

$$\lambda = \max(2^A, 3^B, 5^C), \qquad \eta = \min(2^{A+1}, 3^{B+1}, 5^{C+1}).$$

Thus $\eta \leq 2^{A+1} \leq 2\lambda$. By Lemma 3.5, fixing the exponents of 3 and 5 leaves at most one exponent of 2. Since $A \geq B \geq C$, the cell contains at most

$$(B + 1)(C + 1) \leq \frac{(A + B + C + 3)^2}{9}$$

smooth points: writing $u = B + 1 \geq v = C + 1$ and using $A + 1 \geq u$, the difference $(2u + v)^2 - 9uv = (u - v)(4u - v)$ is nonnegative. Unique factorisation identifies these exponent triples with distinct smooth integers.

As the cutoff increases all three prime exponents are nondecreasing, so two nonempty cells with the same exponent sum are the same cell, and there is at most one nonempty cell with each exponent sum. Every cell above 2^a has exponent sum at least n_a, and a cell with sum $n_a + k$ has running LCM at least $P_a 2^k$, since each of the k extra prime factors is at least 2. Nonnegative summation over exponent sums, allowing empty cells, gives

$$X_a \leq \frac{1}{18} \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{2^k} = \frac{n_a^2 + 8n_a + 18}{9},$$

the evaluation using the geometric moments $\sum 2^{-k} = 2$, $\sum k2^{-k} = 2$ and $\sum k^2 2^{-k} = 6$. Positivity follows from the block at 2^a. □

Each extra prime factor costs at least two in the denominator; each cell contains at most quadratically many points. This is a geometric sum of elementary projection counts, not an asymptotic smooth-number estimate or a Hecke–Mahler argument.

The quadratic order is also necessary. The positive next tail and the recurrence give

$$X_a = \frac{m_a + X_{a+1}}{b_a} > \frac{m_a}{b_a} \geq \frac{m_a}{30}.$$

The lower bound of Lemma 8.3, together with $n_a \leq 3a$, therefore gives $X_a = \Theta((a+1)^2)$. In particular, the normalised tails are not uniformly bounded. This order estimate does not assert an asymptotic constant or optimality of Q.

The bound is used at the endpoint of a segment, where the natural index is the positive prime-power jump count strictly below the cutoff. Put

$$j_a = \#\{p^e < 2^a : p \in \{2,3,5\},\ e \geq 1\} = n_a - 1 \qquad (a \geq 1), \tag{17}$$

the equality holding for $a \geq 1$ because the powers of 2 below 2^a number $a-1$ while the powers of 3 and of 5 below 2^a number $\lfloor \log_3 2^a \rfloor$ and $\lfloor \log_5 2^a \rfloor$; at $a = 0$ the count is $j_0 = 0$ and $n_0 - 1 = -1$. Substituting $n_a = j_a + 1$ into Q gives the integer bound used throughout the rest of the note:

$$K(B,a) = \lfloor B\,Q(n_a) \rfloor, \qquad K(B,a) = \left\lfloor \frac{B(j_a^2 + 10j_a + 27)}{9} \right\rfloor \quad (a \geq 1). \tag{18}$$

The strict cutoff in (17) is 2^a. There are two distinct uses of the bound: $BX_a \leq BQ(n_a)$ holds for the actual real tails, whereas $BX_a \leq K(B,a)$ follows only when BX_a is integral. For example, $Q(n_3) = 83/9$ gives the integer bound $K(1,3) = 9$, not an upper bound of 9 for an arbitrary real quantity below $83/9$. The short paper instead uses $K_{\text{short}}(B,a) = 90B(a+1)^2$, valid without integrality. Here K always means (18); either bound may be used in the residue contradiction.

Lemma 9.2 (finite denominator clearing)**.** *For all integers $0 \leq u \leq b$ the sum over the segment $h_b \sum_{a=u}^{b-1} s_a$ is a natural number. If $S = N/D$ with $N \in \mathbb{Z}$ and $D \in \mathbb{N}_{>0}$, then $DX_a \in \mathbb{Z}$ for every $a \geq 1$, and there are indices $1 \leq i < j \leq D+1$ for which $X_i - X_j \in \mathbb{Z}$.*

Proof. An empty segment has mass zero. Otherwise $b \geq 1$, and every integer $x < 2^b$ has exponent of 2 at most $b-1$ while its other prime exponents are at most those at 2^b, so $\mathrm{H}(x) \mid P_b/2 = h_b$ and every term of the finite segment clears at h_b. Since $h_a \sum_{u<a} s_u$ is an integer,

$$DX_a = h_a N - D h_a \sum_{u<a} s_u \in \mathbb{Z}.$$

Among $D+1$ of these integers two share a residue modulo D, and the corresponding tails differ by an integer. □

The strict upper endpoint is what permits division by two. A cutoff including 2^b would not clear its term at the normalising factor $h_b = P_b/2$. The pigeonhole argument identifies two tails modulo $\mathbb{Z}$, not two equal tails. Since X_a is unbounded, the lattice condition does not force the exact repetition used in the distinct-value proof.

Theorem 9.3 (rationality gives positive integer tails)**.** *Suppose $S = N/D$ with $N \in \mathbb{Z}$, $D \in \mathbb{N}_{>0}$, and write*

$$D = 2^u 3^v 5^w B, \qquad u,v,w \in \mathbb{N}, \quad B \in \mathbb{N}_{>0}, \quad \gcd(B,30) = 1, \qquad a_D = u + 1 + 2v + 3w.$$

Then for every $a \geq a_D$ the number $d_a = BX_a$ is a positive integer and

$$d_{a+1} = b_a d_a - B m_a, \qquad 1 \leq d_a \leq K(B,a) \leq 90B(a+1)^2.$$

Proof. Let $M = 2^u 3^v 5^w$. For $a \geq a_D$ we have $2^a \geq 2^{u+1}$, $2^a \geq 3^v$ and $2^a \geq 5^w$, using $3 < 2^2$ and $5 < 2^3$; hence $M \mid h_a$. By Lemma 9.2, X_a differs from $h_a N/D$ by an integer. Since $M \mid h_a$ and $D = MB$, multiplying by B shows that BX_a is an integer. Positivity and the recurrence come from (16), and the upper bound is Theorem 9.1 with the floor taken, since d_a is an integer at most $BQ(n_a)$. Finally $n_a \leq 3a$, so $Q(n_a) \leq a^2 + \frac{8}{3}a + 2 \leq 90(a+1)^2$. □

The factorisation restricts no possible denominator: remove its powers of 2, 3 and 5, leaving B coprime to 30. These parts have different roles. The removed part controls the onset of integrality; B scales the integer tails and their bound. The sufficient a_D need not be the first clearing index. For the exact answer, write $\mathrm{den}(x)$ for the positive reduced denominator of a rational x.

Proposition 9.4 (exact denominators and minimal clearing)**.** *Suppose $S = N/(MB)$ is in lowest terms, with $M = 2^u 3^v 5^w$ and $\gcd(B, 30) = 1$. For every $a \geq 1$,*

$$\mathrm{den}(X_a) = \frac{MB}{\gcd(M, h_a)}, \qquad \mathrm{den}(BX_a) = \frac{M}{\gcd(M, h_a)}.$$

Hence the first integral reduced tail occurs at

$$a_* = \min\{a \geq 1 : 2^a \geq \max(2^{u+1}, 3^v, 5^w)\} \leq a_D,$$

and BX_a is integral exactly for $a \geq a_*$. This onset is computable by integer powers; it is not an estimate obtained by rounding logarithms.

Proof. The finite prefix clears at h_a, so Lemma 9.2 gives $X_a - h_a N/(MB) \in \mathbb{Z}$. Since $\gcd(N, MB) = 1$ and every prime factor of h_a is in $\{2, 3, 5\}$, cancellation removes exactly $\gcd(M, h_a)$ from the denominator. Multiplication by B gives the second formula. Now $M \mid h_a$ means $a - 1 \geq u$, $\lfloor \log_3 2^a \rfloor \geq v$ and $\lfloor \log_5 2^a \rfloor \geq w$, precisely the three integer inequalities. All three persist when a increases, giving the first and every later integral reduced tail. The earlier bounds $3 < 4$ and $5 < 8$ give $a_* \leq a_D$. □

For example, a hypothetical reduced denominator $2^3 3^2 5 \cdot 7$ would require $2^a \geq \max(16, 9, 5)$, so $a_* = 4$ rather than the sufficient $a_D = 11$. Lowest terms matter for this exact answer: unreduced factors could cancel against the numerator sooner.

The recurrence preserves integrality forward. Its homogeneous equation also determines how fast two distinct solutions separate.

Proposition 9.5 (propagation of integrality and uniqueness of a small solution)**.** *For every a, $X_a = (m_a + X_{a+1})/b_a > 0$, and if $X_a \in \mathbb{Z}$ then $X_n \in \mathbb{Z}$ for every $n \geq a$. Moreover, fix A, a positive width function w with $w(A+k)/8^k \to 0$, and a real sequence $(y_n)_{n \geq A}$ satisfying $y_{n+1} = b_n y_n - m_n$. If y_n and X_n both lie in $(m_n/b_n,\ m_n/b_n + w(n)]$ for every $n \geq A$, then $y_A = X_A$.*

Proof. The identity is the recurrence solved for X_a, and positivity holds because every block contains its dyadic left endpoint. Integer coefficients preserve integrality at every later step. For the last assertion, $y_{A+k} - X_{A+k} = (P_{A+k}/P_A)(y_A - X_A)$. The dyadic running LCM bounds from Section 8 give $P_{A+k}/P_A > 8^k/15$, whereas the common interval bounds the absolute difference by $w(A+k)$. Thus $|y_A - X_A| < 15w(A+k)/8^k \to 0$. □

The width condition allows every positive polynomial width and widths ρ^n with $1 < \rho < 8$, but not 8^n itself. Both solutions must lie in the stated intervals; the recurrence alone does not supply that condition. More generally, fix an integer $B \geq 1$. Every real solution of $y_{a+1} = b_a y_a - B m_a$ starting at an index A satisfies

$$y_a = BX_a + \frac{P_a}{P_A}(y_A - BX_A) \qquad (a \geq A).$$

Subtracting the recurrence for BX_a proves the identity. Since $X_a = O((a+1)^2)$ and $P_a = \Theta(8^a)$, the unique solution with $y_a = o(8^a)$ is $y_a = BX_a$; every other solution has $|y_a| = \Theta(8^a)$. This classifies growth; integrality of BX_a remains a separate question. The same formula explains a numerical precaution: a starting error ε is multiplied by P_a/P_A under forward iteration. A decimal approximation propagated forwards is therefore not a certificate of the tail floors. The digit calculation in Section 12 uses rational intervals with an explicit infinite-tail bound instead.

9.1 A bound using the order of the jumps

The bound in Theorem 9.1 uses only the fact that every new prime factor is at least 2. Successive powers of 2 cannot continue indefinitely without a power of 3 between them. Incorporating that restriction improves the geometric majorant.

Proposition 9.6 (a smaller quadratic bound)**.** *For every $a \geq 0$,*

$$0 < X_a \leq \widetilde{Q}(n_a) < Q(n_a), \qquad \widetilde{Q}(n) = \frac{1210n^2 + 9130n + 18847}{11979}.$$

Proof. For a point $x \geq 2^a$, let d_p be the increase in the exponent of p in its running LCM from the boundary 2^a, and put $k = d_2 + d_3 + d_5$. The definition of d_3 gives $x < 3^{\lfloor \log_3 2^a \rfloor + d_3 + 1} \leq 2^a 3^{d_3+1} < 2^{a+2d_3+2}$, hence $d_2 \leq 2d_3 + 1$. It follows that $k \leq 3(d_3 + d_5) + 1$, so at least $j = \lfloor (k+1)/3 \rfloor$ of the k new prime factors are odd. Each of these contributes at least 3, and each remaining factor at least 2. Hence

$$\frac{\mathrm{H}(x)}{P_a} \geq 2^{k-j} 3^j =: d(k).$$

There is at most one interval of constancy with each exponent sum, with at most $(n_a + k + 3)^2/9$ supported points by the earlier projection bound. Consequently

$$X_a \leq \sum_{k \geq 0} \frac{(n_a + k + 3)^2}{18 d(k)}.$$

Now $d(3m) = 12^m$, $d(3m+1) = 2 \cdot 12^m$ and $d(3m+2) = 6 \cdot 12^m$. Grouping in threes and summing the quadratic geometric series gives $\widetilde{Q}(n_a)$. Finally $11979(Q(n) - \widetilde{Q}(n)) = 121n^2 + 1518n + 5111 > 0$ for $n \geq 0$. □

The estimate uses the original block multiplicities and the order of the prime-power jumps. It is not a bound for an arbitrary recurrence with the same bases. For an integral BX_a, the bound can be rounded down to $\lfloor B\widetilde{Q}(n_a) \rfloor$. This improves the bound available

in a rationality contradiction. Theorem 10.2 and the recorded finite tests use K, not this smaller bound. No optimality claim is made for either quadratic bound.

Section 10 explains how the smaller bound can also replace K in the residue argument.

Relation to Cantor-series criteria. Since $h_0 = 1/2$, the value $S/2 = X_0$ is the Cantor series $\sum_{a\ge 0} m_a/P_{a+1}$, with $P_0 = 1$ and $P_{a+1} = b_a P_a$. Its normalised tails X_a are the usual objects in the rationality criteria for Cantor series. Under a small-numerator hypothesis, Erdős and Straus characterise rationality by the existence of a positive integer B and integers c_a satisfying, eventually,

$$Bm_a = b_a c_a - c_{a+1}, \qquad |c_{a+1}| < b_a/2$$

[18, Theorem 2.1 and (2.4), pp. 85–86]. The second condition is part of the criterion, not a consequence of the recurrence alone. Their proof already chooses nearest integers. Hančl and Tijdeman make that choice part of the criterion: they prescribe c_a as a nearest integer to Bm_a/b_a and require the recurrence [19, §§2–3 and Theorem 3.1, pp. 372–375]. Koutsoukou-Argyraki and Li formalised the Erdős–Straus criteria in Isabelle/HOL [20].

Both cited criteria require $m_a/(b_{a-1}b_a) \to 0$ in our notation. Here the weighted-triangle count gives $m_a = \Theta((a+1)^2)$, and

$$\frac{m_a}{900} \le \frac{m_a}{b_{a-1}b_a} \le \frac{m_a}{4} \qquad (a \ge 1).$$

The ratio therefore has quadratic order and tends to infinity. There is also a direct lower count that does not use the real-logarithm triangle estimate. For $a \ge 1$, each pair $0 \le j,k \le \lfloor a/5 \rfloor$ has $3^j 5^k < 2^a$, since $15 < 2^5$, and hence a unique power-of-two multiple in $[2^a, 2^{a+1})$. Each such point contributes at least one to m_a. Thus $m_a \ge (\lfloor a/5 \rfloor + 1)^2$, an elementary lower bound sufficient to see directly that the small-numerator condition fails.

The actual scaled tails also satisfy

$$BX_a - \frac{Bm_a}{b_a} = \frac{BX_{a+1}}{b_a} = \Theta((a+1)^2)$$

for each fixed $B \ge 1$, by the recurrence, the quadratic tail bounds and $2 \le b_a \le 30$. Thus any eventual integer tails supplied by a hypothetical rational value could not be the prescribed nearest integers. In fact, those prescribed integers fail the recurrence at every sufficiently large index, regardless of how ties are resolved. If c_a is a nearest integer to Bm_a/b_a, then

$$|b_a c_a - Bm_a| \le 15, \qquad c_{a+1} \ge \frac{Bm_{a+1}}{30} - \frac{1}{2} \longrightarrow \infty.$$

Consequently $c_{a+1} = b_a c_a - Bm_a$ is impossible for all sufficiently large a. This is a failure of the proposed construction of integer coefficients, not an irrationality contradiction: the criterion's small-numerator hypothesis also fails.

The denominator clearing itself is standard; the calculations specific to this series retain its multiplicities, use the strict endpoint to divide the normalising factor by two, and bound the resulting positive tails.

For polynomial Cantor data, Hančl–Tijdeman [25, Theorems 2.2 and 3.1] give a polynomial cancellation criterion and a division mechanism. Their nonconstant polynomial radix is not our bounded (b_a). Their separate Theorem 4.2 uses a finite-product rearrangement. We do not use an unrestricted infinite reindexing from that argument: for the integer decompositions considered here, the terminal terms must be checked separately. In the first-order case $m_a = b_a c_a - c_{a+1}$, finite summation gives

$$\sum_{a=0}^{N-1} \frac{m_a}{P_{a+1}} = c_0 - \frac{c_N}{P_N}.$$

Every integer c_0 generates an integer solution; integrality alone places no restriction on the initial value. The terminal condition is the restriction: $c_N/P_N \to c_0 - S/2$, so decay to zero selects the actual initial tail $c_0 = S/2$. This is the same uniqueness mechanism as for the small solutions above, not a consequence of solving the recurrence.

For these actual numerators the terminal term cannot vanish for *any* integer c_0. Indeed, $P_3 = 120$, $n_3 = 5$ and $(m_0, m_1, m_2) = (1, 4, 7)$, so the tail bound gives

$$0 < \frac{S}{2} = \frac{1}{2} + \frac{4}{12} + \frac{7}{120} + \frac{X_3}{120} \le \frac{107}{120} + \frac{83}{1080} = \frac{523}{540} < 1.$$

Thus $c_0 - S/2 \neq 0$ for every integer c_0. Since $8^N/15 < P_N \le 8^N$, every such unscaled integer solution satisfies $|c_N| = \Theta(8^N)$. This is a concrete reason that quadratic growth of m_a does not imply polynomial growth of recursively defined carries. The starting index is essential: this does not exclude the actual tails becoming integral at a later index. After replacing m_a by Dm_a for an integer $D \ge 1$, however, a vanishing terminal term would require $DS/2 = c_0$. Excluding that possibility for every D would require irrationality itself; the unscaled calculation does not do so.

In the cited rearrangement the transformed integer numerator must be $o(b_a)$, which for bounded radices means eventual zero. Only after the terminal terms have been controlled does this cancellation determine the infinite sum. A polynomial bound on each individual decomposition coefficient would suffice for a fixed number of shifts: the corresponding series are absolutely convergent, since their denominators are products of at least 2 at each step. A polynomial bound on the combined numerators is not a substitute, as the exponentially growing integer solutions above demonstrate. Nor does quadratic growth force an integer solution to be a polynomial in the floor coordinates.

10 A residue criterion and the bounds it allows

Two iterations already show how the recurrence determines an endpoint residue:

$$X_2 = 6X_1 - 4, \qquad X_3 = 10X_2 - 7 = 60X_1 - 47.$$

Whenever BX_1 is integral, $BX_3 \equiv -47B \pmod{60}$. An integral starting tail therefore puts the terminal tail in an arithmetic progression. Positivity and the quadratic bound restrict its possible integer values. If the least positive representative exceeds the bound, the starting tail cannot have been integral. For the zero class, that representative is the

modulus, not zero. To exclude every rational denominator, the test must succeed for every eligible B beyond every prescribed clearing onset.

A segment starts at $\ell \geq 0$ and consists of $h \geq 0$ steps. Its product of bases and accumulated numerator are integers, even when the tails are not. Set

$$W_{\ell,0} = 1, \quad F_{\ell,0} = 0, \qquad W_{\ell,h+1} = b_{\ell+h} W_{\ell,h}, \quad F_{\ell,h+1} = b_{\ell+h} F_{\ell,h} + m_{\ell+h}.$$

Induction gives

$$X_{\ell+h} = W_{\ell,h} X_\ell - F_{\ell,h}, \qquad d_{\ell+h} = W_{\ell,h} d_\ell - B F_{\ell,h} \tag{19}$$

for any sequence satisfying $d_{n+1} = b_n d_n - B m_n$. Since $b_a = P_{a+1}/P_a$, the product telescopes to $W_{\ell,h} = P_{\ell+h}/P_\ell$. For the actual block masses, $m_a = h_{a+1} s_a$ and $h_a = P_a/2$ identify the two terms after division by $W_{\ell,h}$:

$$\frac{F_{\ell,h}}{W_{\ell,h}} = h_\ell \sum_{a=\ell}^{\ell+h-1} s_a, \qquad \frac{X_{\ell+h}}{W_{\ell,h}} = h_\ell T_{\ell+h}.$$

Thus the accumulated numerator records a finite part of the original reciprocal sum, in the same normalisation as the starting tail. No assumption of rationality or integrality is used in these identities. The previously proved bounds $8^a/15 < P_a \leq 8^a$ also give $W_{\ell,h} > 8^h/15$. For a positive integer C and an integer N, use $\operatorname{lpr}_C(N) = 1 + ((N - 1) \bmod C) \in \{1, \dots, C\}$. In particular $\operatorname{lpr}_C(0) = C$. Positivity of this representative is what allows comparison with a positive integer tail.

Proposition 10.1 (least positive residues)**.** *Let $C > 0$ and let c be an integer with $0 < c$ and $|c| \leq K$. If $c \equiv N \pmod{C}$ and $K < \operatorname{lpr}_C(N)$, then the hypotheses are contradictory.*

Proof. Every positive integer congruent to N is at least $\operatorname{lpr}_C(N)$. Thus $\operatorname{lpr}_C(N) \leq c = |c| \leq K$, contrary to the strict inequality. □

For a bound $G : \mathbb{N}_{>0} \times \mathbb{N} \to \mathbb{N}$ let $\mathsf{E}(G)$ be the statement

$$\begin{gathered} \text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a_0 \geq 1, \\ \text{there are } \ell \geq a_0 \text{ and } h \geq 1 \text{ with } \operatorname{lpr}_{W_{\ell,h}}(-B F_{\ell,h}) > G(B, \ell + h). \end{gathered} \tag{20}$$

The segment may depend on both B and a_0. All quantities in the inequality are finite integers. The quantifiers over denominators and prescribed onsets are nevertheless unbounded; a search over a finite rectangle of (B, ℓ) does not verify them. Every $b_a > 0$, so $W_{\ell,h} > 0$ is automatic.

Here K is the integer bound obtained above from the jump count. Under the hypothesis that BX_ℓ is integral, the recurrence makes $BX_{\ell+h}$ integral as well, so $0 < BX_{\ell+h} \leq K(B, \ell + h)$. Equivalently, the starting integer must lie in

$$\left(\frac{B F_{\ell,h}}{W_{\ell,h}}, \frac{B F_{\ell,h} + K(B, \ell + h)}{W_{\ell,h}} \right].$$

A residue exceeding K says exactly that this interval contains no integer. The floor in the definition of K is legitimate here because integrality is being assumed. The short paper

uses the coarser real-tail bound $90B(a+1)^2$ instead. The argument needs a valid bound on the possible integer tail and the stated growth condition, rather than a particular numerical choice.

The two hypotheses on G in the next theorem serve different purposes. Domination of K lets a residue above G exclude a positive integer tail. The limit $G(B,a)/8^a \to 0$ lets every sufficiently long segment from a fixed nonintegral start overtake G. It is a sufficient growth condition, not a necessary one: if only existence of a segment is required, a subsequence of small endpoint bounds can suffice, as we prove below. The stated theorem already covers every polynomial upper bound that dominates K, and also $\max\{K(B,a), \lceil B\rho^a \rceil\}$ for $1 < \rho < 8$. The zero bound fails to control the possible integer tail.

Theorem 10.2 (a residue criterion for every dominating bound of size $o(8^a)$)**.** *Let $G : \mathbb{N}_{>0} \times \mathbb{N} \to \mathbb{N}$ satisfy $K(B,a) \le G(B,a)$ for all B and a, and $G(B,a)/8^a \to 0$ as $a \to \infty$ for each fixed B. Then*

$$\mathsf{E}(G) \quad \Longleftrightarrow \quad S \notin \mathbb{Q}.$$

Both K of (18) and $K_0(B,a) = 90B(a+1)^2$ satisfy these hypotheses, so $\mathsf{E}(K)$, $\mathsf{E}(K_0)$ and irrationality of S are mutually equivalent. By contrast, $\mathsf{E}(0)$ holds automatically, since every least positive residue is at least 1; its truth alone therefore provides no contradiction to an integral tail.

Proof. Suppose $\mathsf{E}(G)$ and suppose $S = N/D$ were rational. Theorem 9.3 supplies $B \ge 1$ coprime to 30, an onset a_D, and positive integers $d_a = BX_a \le K(B,a) \le G(B,a)$ for $a \ge a_D$ satisfying the cleared recurrence. Apply (20) with $a_0 = a_D$ to obtain a segment (ℓ, h) with $\ell \ge a_D$. By (19), $d_{\ell+h} \equiv -BF_{\ell,h}$ modulo $W_{\ell,h}$, while $0 < d_{\ell+h} \le G(B, \ell+h) < \mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h})$. Proposition 10.1 is the contradiction, so S is irrational.

Conversely suppose $S \notin \mathbb{Q}$, and fix $B \ge 1$ coprime to 30 and $a_0 \ge 1$. Set $\ell = a_0$ and $\delta = \lceil BX_\ell \rceil - BX_\ell \in (0,1)$; the prefix identity makes BX_ℓ irrational. For all sufficiently large h, $0 < \delta + BX_{\ell+h}/W_{\ell,h} < 1$, because $X_{\ell+h} = O((\ell+h+1)^2)$ and $W_{\ell,h} > 8^h/15$ as shown above. The integer $\lceil BX_\ell \rceil W_{\ell,h} - BF_{\ell,h}$ is therefore exactly

$$\mathrm{lpr}_{W_{\ell,h}}(-BF_{\ell,h}) = \delta W_{\ell,h} + BX_{\ell+h}.$$

Its first term eventually exceeds $G(B, \ell+h)$, since $G(B,a) = o(8^a)$. Thus every sufficiently long segment from this fixed start satisfies the residue inequality, which is stronger than the required existence.

For the two stated bounds, $n_a \le 3a$ gives $K(B,a) \le BQ(3a) \le 90B(a+1)^2 = K_0(B,a)$. Both are $O((a+1)^2)$ for fixed B, so both satisfy the required limit. Finally, $\mathrm{lpr}_C(N) \ge 1 > 0$ always, independently of whether any tail is integral; a residue greater than zero cannot exclude a positive integer. □

The domination assumption is sufficient, not asserted to be necessary for equivalence. In particular, Proposition 9.6 already supplies the smaller valid bound $\lfloor B\widetilde{Q}(n_a) \rfloor$. Replacing K by that bound in the forward implication leaves the argument unchanged; the converse uses only $G(B,a) = o(8^a)$. The zero-bound example explains why a size

comparison with the possible integer tail is needed in the contradiction, not why this particular K is indispensable.

At a fixed start, the same identity distinguishes two behaviours. If BX_ℓ is nonintegral, the least positive residue eventually occupies a fixed positive fraction of the modulus. If BX_ℓ is integral, the residue eventually equals the normalised tail $BX_{\ell+h}$ itself. The next proposition gives both conclusions without assuming that S is irrational.

Proposition 10.3 (the fixed-start residue limit)**.** *For fixed integers $B, \ell \geq 1$, write*

$$R_h = \operatorname{lpr}_{W_{\ell,h}}(-BF_{\ell,h}).$$

For all sufficiently large h,

$$R_h = (\lceil BX_\ell \rceil - BX_\ell)W_{\ell,h} + BX_{\ell+h},$$

and consequently

$$\lim_{h\to\infty} \frac{R_h}{W_{\ell,h}} = \lceil BX_\ell \rceil - BX_\ell.$$

In particular, if BX_ℓ is integral, then $R_h = BX_{\ell+h}$ for all sufficiently large h.

Proof. The least positive residue is the modulus times the gap to the next strictly larger integer. Division with remainder gives

$$\frac{R_h}{W_{\ell,h}} = \left\lfloor \frac{BF_{\ell,h}}{W_{\ell,h}} \right\rfloor + 1 - \frac{BF_{\ell,h}}{W_{\ell,h}}.$$

By the iterated recurrence, $BF_{\ell,h}/W_{\ell,h} = BX_\ell - BX_{\ell+h}/W_{\ell,h}$. These finite sums approach BX_ℓ strictly from below, since $BX_{\ell+h} > 0$ and $BX_{\ell+h}/W_{\ell,h} \to 0$. Their floors therefore eventually equal $\lceil BX_\ell \rceil - 1$, including when BX_ℓ is an integer. Substitution proves both claims. The strict approach from below matters at an integer limit; the residue convention alone would assign a zero congruence class the value $W_{\ell,h}$, not zero. □

Thus a nonintegral BX_ℓ, even for a rational S, gives arbitrarily long successful segments against any bound $G(B,a) = o(8^a)$ for that fixed B. The rationality contradiction instead uses the denominator's specific multiplier and a start beyond its clearing onset. Success for one pair (B,ℓ) does not replace either quantifier.

The exact growth restriction on a dominating bound. A bound need not be small at every late endpoint. For example, set

$$G(B,a) = \begin{cases} K(B,a), & a \text{ even}, \\ K(B,a) + 8^a, & a \text{ odd}. \end{cases}$$

At an odd endpoint $a = \ell + h$, no segment with $\ell \geq 1$ can pass: its residue is at most $W_{\ell,h} = P_a/P_\ell \leq 8^a/2 < G(B,a)$. At even endpoints the test is unchanged. If S is irrational, the proof above therefore supplies successful segments ending at every sufficiently large even index, from each fixed start. Thus $G(B,a)/8^a \to 0$ is not necessary for the equivalence.

More precisely, for every integer-valued $G \geq K$,

$$\mathsf{E}(G) \quad \Longleftrightarrow \quad \begin{cases} S \notin \mathbb{Q}, \\ \liminf\limits_{a\to\infty} \dfrac{G(B,a)}{8^a} = 0 \quad \text{for every } B \geq 1 \text{ with } \gcd(B,30) = 1. \end{cases}$$

Suppose first that $\mathsf{E}(G)$ holds. The earlier rationality contradiction uses only $G \geq K$. For a successful segment starting at $\ell \geq a_0$, put $a = \ell + h$. Since a least positive residue is at most its modulus,

$$0 \leq \frac{G(B,a)}{8^a} < \frac{P_a}{P_\ell 8^a} \leq \frac{1}{P_\ell} \leq \frac{1}{P_{a_0}}.$$

Such endpoints satisfy $a > a_0$, and $P_{a_0} \to \infty$; hence the lower limit is zero. Conversely, suppose S is irrational and these lower limits vanish. Fix B and $\ell \geq 1$. By the fixed-start residue limit, with $\delta = \lceil BX_\ell \rceil - BX_\ell > 0$,

$$\frac{\operatorname{lpr}_{W_{\ell,h}}(-BF_{\ell,h})}{P_{\ell+h}} \longrightarrow \frac{\delta}{P_\ell} > 0.$$

The bounds $8^a/15 < P_a \leq 8^a$ imply $\liminf_a G(B,a)/P_a = 0$. Along a sufficiently late subsequence of endpoints the residue therefore exceeds G. This gives arbitrarily long successful segments from each fixed start, but need not give success at every sufficiently large length, as the parity example shows.

In particular, $G(B,a) = K(B,a) + 8^a$ allows no segment at all. More generally, an eventual lower bound $G(B,a) \geq c8^a$, for one eligible B and some $c > 0$, excludes every sufficiently late start: choose $P_\ell \geq 1/c$ and compare the bound with $W_{\ell,h} \leq 8^{\ell+h}/P_\ell$. These are restrictions on the test bound, not additional assumptions about the series. The original quadratic bounds satisfy the lower-limit condition, so the remaining problem is still the existence of the segments. Equivalence alone says nothing about the relative difficulty of the formulations; a useful arithmetic proof must exploit further information about the numerators m_a.

A fixed segment can have a lower bound greater than 1. Write W and F for its product and accumulated numerator. Every prime factor of W lies in $\{2,3,5\}$, so an eligible multiplier B is a unit modulo W. Congruence then gives

$$\gcd(\operatorname{lpr}_W(-BF), W) = \gcd(F, W).$$

For $(W,F) = (6,4)$, the possible residues are 2 and 4, attained at $B = 1$ and 11; thus 2 is a lower bound for this segment, and the residues are not units. For $(W,F) = (60,47)$, however, $\operatorname{lpr}_{60}(-37\cdot 47) = 1$, with 37 coprime to 30. Thus no lower bound greater than 1 works for all segments and eligible multipliers, although a particular segment may have one.

10.1 How fast the product of the bases grows

The growth scale in the preceding criterion comes from the running LCM formula. The elementary estimate $W_{\ell,h} \geq 2^h$ would suffice for bounds $G(B,a) = o(2^a)$ with B fixed; the exact formula gives $W_{\ell,h} \asymp 8^h$ uniformly in ℓ and allows $G(B,a) = o(8^a)$. For an irrational

S, the preceding lower-limit condition also covers dominating bounds outside these two classes. The same growth estimate will give a necessary lower bound on successful segment lengths.

Proposition 10.4 (growth of the product of consecutive bases)**.** *Put* $\theta_3 = \log_3 2$ *and* $\theta_5 = \log_5 2$*. For all* $\ell \geq 0$ *and* $h \geq 1$*,*

$$W_{\ell,h} = 2^h\, 3^{\lfloor(\ell+h)\theta_3\rfloor-\lfloor\ell\theta_3\rfloor}\, 5^{\lfloor(\ell+h)\theta_5\rfloor-\lfloor\ell\theta_5\rfloor}, \qquad \frac{8^h}{15} < W_{\ell,h} < 15\cdot 8^h.$$

Proof. Telescoping (14) gives $W_{\ell,h} = P_{\ell+h}/P_\ell$, and the displayed formula is that quotient written out. Each floor difference differs from $h\theta_p$ by less than one, and $3^{\theta_3} = 5^{\theta_5} = 2$, so the 3-factor lies strictly between $2^h/3$ and $3\cdot 2^h$ and the 5-factor strictly between $2^h/5$ and $5\cdot 2^h$. Multiplying the three ranges gives the bounds. □

Corollary 10.5 (a fixed maximum length cannot cover arbitrarily late starts)**.** *Fix* $B \geq 1$ *coprime to* 30 *and* $H \geq 1$*. Only finitely many starts* ℓ *admit a segment of length at most* H *satisfying the residue inequality against* K*.*

Proof. A least positive residue never exceeds its modulus, so the residue inequality at (ℓ, h) requires $K(B,\ell+h) < W_{\ell,h} < 15\cdot 8^h \leq 15\cdot 8^H$. On the other hand $j_a \geq a-1$, since the powers $2,\dots,2^{a-1}$ already lie below 2^a, so $K(B,\ell+h) \geq \lfloor B((\ell-1)^2 + 10(\ell-1) + 27)/9\rfloor$, which tends to infinity with ℓ. □

Corollary 10.5 is the exact reason a finite scan cannot approach the cofinal quantifier by widening its denominator range alone. For $\ell \geq 1$, the inequality $K(B,\ell+h) > B\ell^2/9$ shows that any segment satisfying the residue inequality must also satisfy

$$3h > \log_2 K(B,\ell+h) - \log_2 15 > \log_2 B + 2\log_2 \ell - \log_2 135.$$

This is only a necessary lower bound, not an asymptotic formula or an upper bound for the first successful length. An explicit threshold beyond which the residue inequality holds can be obtained from a positive lower bound for $\delta = \lceil BX_\ell\rceil - BX_\ell$ and for $1-\delta$, together with the displayed tail and product estimates. No such uniform information about the tails of S is proved here.

11 Finite computations and their limits

We first calculate individual residue certificates, then enclose the sum using twelve blocks to exclude small rational denominators. The larger scan and two larger exclusions are separate saved computations; their programs and outputs are in the large-computation directory. Each certificate excludes its stated finite family. None establishes the criterion for every denominator beyond every starting threshold.

11.1 Checking individual segments

We calculate each row using the integer-only integer-arithmetic checker. For block a, enumerate the pairs with $3^j5^k < 2^{a+1}$, choose the unique $i \geq 0$ for which $2^i3^j5^k \in$

$[2^a, 2^{a+1})$, and sum $P_{a+1}/(2\,\mathrm{H}(2^i3^j5^k))$. The recursions for W and F then give the residue in each row. Powers and comparisons are integral throughout. The columns give the denominator B, the start ℓ, the length h, the endpoint jump index $j_{\ell+h}$, the product W of the bases, the accumulated numerator F, the residue $R = \mathrm{lpr}_W(-BF)$ and the bound K of (18).

B	ℓ	h	$j_{\ell+h}$	W	F	R	K
1	1	2	4	60	47	13	9
7	1	3	6	360	289	137	95
16	1	4	9	10800	8735	640	352

The first row reads as follows. The segment starts at $\ell = 1$ and has length 2, so $W = b_1b_2 = 6 \cdot 10 = 60$; the accumulated numerator is $F = 47$; and $\mathrm{lpr}_{60}(-47) = 13$, since $-47 + 60 = 13$, which exceeds $K(1,3) = \lfloor (16 + 40 + 27)/9 \rfloor = 9$. The third row lies outside the domain of (20), since $\gcd(16, 30) = 2$, and is displayed to illustrate the finite calculation at greater depth.

The larger scan covers $B \le 5000$ coprime to 30 and $100 \le \ell \le 3000$: all 3,869,934 pairs satisfy the residue inequality, with first successful length at most 18 in a search to length 24 (saved run). Its histogram in Section 13.4 is a finite observation that the product-growth bound does not predict. The accompanying integer-only check reconstructs the blocks and tests all 2496 pairs with $1 \le B \le 97$, $\gcd(B, 30) = 1$ and $1 \le \ell \le 96$. Every pair satisfies the residue inequality; the first successful lengths range from 1 to 10, with a search limit of 24. This check uses the enumeration and finite recursions above, without floating-point logarithms. Neither computation proves the residue inequality for unbounded B or arbitrarily late starts; Corollary 10.5 rules out covering the latter quantifier with any fixed maximum length.

11.2 Finite denominator bounds

A denominator bound from twelve blocks. We obtain a denominator bound by enclosing $X_1 = S - 1$ between two rational numbers. Here $h_1 = 1$, and summation through block 11, followed by the bound $Q(n_{12}) = Q(24) = 262/3$ with $P_{12} = 27993600000$, gives

$$X_1 = 2\sum_{a=1}^{11} \frac{m_a}{P_{a+1}} + \frac{2X_{12}}{P_{12}},$$

$$\frac{11327881097}{13996800000} < X_1 \le \frac{33983643553}{41990400000}.$$

The enclosure is strictly inside the interval $(11498/14207,\ 6409/7919)$, as integer cross-multiplication verifies. The endpoint determinant is $14207 \cdot 6409 - 11498 \cdot 7919 = 1$. For any rational u/v strictly between them, with $v > 0$, both integers $14207u - 11498v$ and $6409v - 7919u$ are positive. Consequently

$$v = 7919(14207u - 11498v) + 14207(6409v - 7919u) \ge 22126.$$

Thus, if S is rational, its reduced denominator is at least 22126. This is the elementary denominator bound between two fractions with cross-determinant one, often expressed

using Farey neighbours. It is also the best lower bound on a possible denominator obtainable from this enclosure alone: the mediant

$$\frac{11498 + 6409}{14207 + 7919} = \frac{17907}{22126}$$

is in lowest terms and lies strictly inside the certified enclosure. Thus the interval alone remains compatible with denominator 22126; we have neither identified X_1 with this mediant nor determined a denominator for S.

The certificate uses the actual block coefficients and proved tail bound. A narrower certified enclosure excludes further denominators; the continued-fraction exclusion below uses one built from 15,500 blocks.

A denominator bound from 128 consecutive blocks. A finite lattice computation, outside the theorem inputs, excludes the following denominators. Using segments of length 128 and 64 starting indices, the first 10005, no rational value of the $\{2,3,5\}$ running-LCM series has reduced denominator MB with M a divisor supported on $\{2,3,5\}$ of $2^{10005}3^{6312}5^{4308}$, $\gcd(B,30) = 1$ and $1 < B \le B_{\max}$, where $B_{\max}$ is the 106-digit integer

$$B_{\max} = 113459967099968776734952084570709335925735302228655873936360023501610320756406337327030532417214528197172 9,$$

so that $\log_2 B_{\max} = 348.9846\ldots$.

The exponent triple $(10005, 6312, 4308)$ is that of $\mathrm{H}(2^{10005})$. The normalisation therefore uses the full running LCM at its first start, whereas Lemma 9.2 uses the half running LCM h_{10005}. This family of smooth denominators is the larger one.

The certificate and its exact checker are published. Its starts $a_1 < \cdots < a_{64}$ are the first $a \ge 10005$ with $W_{a,128} = W_{10005,128} = W = 2^{128}3^{81}5^{56}$, so $\log_2 W = 386.40993\ldots$; put $m_0 = W/2$, $F_j = F_{a_j,128}$ and $w_j = (a_j + 128)^2 + 6(a_j + 128) + 11$. The certificate lists an integer matrix $C = (c_{ij})$ with $\sum_j c_{ij}F_j \equiv 0 \pmod{m_0}$ and $\sum_j |c_{ij}|w_j \le 184{,}560{,}461{,}211$ for every row i, an integer matrix E with $CE = EC = m_0 I$, and $B_{\max} = \lfloor (W-1)/184{,}560{,}461{,}211 \rfloor$. If $B \le B_{\max}$ and integers $f_j = Ws_j - 2BF_j$ satisfy $0 < f_j \le Bw_j$, each $\sum_j c_{ij}f_j$ is a multiple of W of absolute value less than W, hence zero, and invertibility of C forces every f_j to vanish, a contradiction. The weights bound the tails: since $n_b \le 2.07b$, Theorem 9.1 gives $2X_b \le 2Q(n_b) \le b^2 + 6b + 11$ for every $b \ge 0$. The reduction from a rational value $N/(MB)$ to the 64 residue inequalities, with $s_j = 2BX_{a_j}$ and $f_j = 2BX_{a_j+128}$, uses the iterated recurrence at full running LCM and is not machine-checked.

Continued-fraction exclusion. A finite computation, outside the theorem inputs, shows that neither X_1 nor S is a fraction with reduced denominator at most 2^{22482}, about 10^{6768}. Let W and F be the product of the bases and accumulated numerator of blocks 1 to 15,500. The tail bound used for the twelve-block calculation, at $Q(n_{15501}) = Q(31956) = 340480534/3$, gives $F/W < X_1 \le (F+Q)/W$. Every block is built with integers, and blocks 1 to 30 are also checked against a direct enumeration of the smooth numbers. The two endpoints share their first 13,540 partial quotients, and

the numbers with a given prefix of partial quotients form an interval [24, Lemma 10.1.2, p. 106]. The first 13,109 digits, $a_0 = 0$ included, define a cylinder whose endpoints have cross-determinant 1 and strictly contain the enclosure, so every fraction in the enclosure has denominator at least the sum of the two endpoint denominators, a 22,483-bit integer. The enclosure, the common digits, the regeneration program and the exact checker are published. A separate continued-fraction probe (saved run) certifies 13,109 partial quotients after a_0 and records a largest denominator of 22,483 bits, a largest partial quotient of 129,114, a mean partial quotient of 23.4133, observed Gauss–Kuzmin frequencies 0.4208, 0.1665, 0.0917, 0.0575, 0.0391 against the predicted 0.4150, 0.1699, 0.0931, 0.0589, 0.0406, and a Lévy constant of 1.18869 against $\pi^2/(12\log 2) = 1.18657$. The predicted values are the almost-everywhere frequencies of Gauss's law and Lévy's almost-everywhere constant [23, pp. 288–289 and footnote 5]. These statistics describe a finite prefix and do not bear on whether X_1 is a Liouville number, algebraic, or rational with a larger denominator.

The two exclusions concern different finite families: one uses a lattice at a fixed starting index and fixed smooth part, the other a continued-fraction enclosure at $a = 1$. Both have published witnesses and exact checkers. Each excludes only its stated denominator range, and neither settles an instance of the problem.

12 The remaining arithmetic questions

The finite recursions are explicit; the missing step is to prove that successful segments exist with the required unbounded quantifiers. Weighted shifts preserve the value of the series, but preservation does not supply cancellation. We calculate the surviving boundary terms, then identify the extra hypotheses needed by integer-recurrence and function-theoretic approaches. The irrationality of S remains open here.

12.1 An exact weighted-shift identity for the repeated series

Set $\alpha = S/2$ and retain the boundary running LCMs P_a. To combine shifted tails, we first express them with the same denominators. After multiplication by P_t, the term with index $a + t$ becomes

$$\frac{P_t m_{a+t}}{P_{a+t+1}} = \frac{P_t P_{a+1}}{P_{a+t+1}} \frac{m_{a+t}}{P_{a+1}}.$$

The first factor on the right is the weight $\gamma_{a,t}$ below. It is forced by the change of denominator, rather than chosen to produce cancellation. The removed finite prefix supplies the rational correction. The identity concerns the repeated sum S only. In this subsection and the next, r is a positive integer shift; the prime-generator notation is no longer in use.

Lemma 12.1 (weighted shifts preserve the actual value)**.** *Fix integers $c_0, \dots, c_\sigma$, not all zero, independently of the positive integer shift r. Put*

$$\gamma_{a,t} = \frac{P_t P_{a+1}}{P_{a+t+1}}, \qquad D_{r,a} = 15 \sum_{j=0}^{\sigma} c_j \gamma_{a,jr} m_{a+jr}.$$

Then $\gamma_{a,t} \in \{1, 1/3, 1/5, 1/15\}$ and $D_{r,a} \in \mathbb{Z}$. Furthermore, with

$$A_r = 15 \sum_{j=0}^{\sigma} c_j P_{jr}, \qquad Z_r = 15 \sum_{j=0}^{\sigma} c_j P_{jr} \sum_{k<jr} \frac{m_k}{P_{k+1}} \in \mathbb{Z},$$

we have the absolutely convergent identity

$$A_r \alpha - Z_r = \sum_{a \geq 0} \frac{D_{r,a}}{P_{a+1}}. \tag{21}$$

One may take $C = 225 \sum_{j=0}^{\sigma} |c_j| \max(1, j)^2$ in $|D_{r,a}| \leq C(a + r + 1)^2$. If J is the largest index with $c_J \neq 0$, then $A_r \neq 0$ whenever $\sum_{j<J} |c_j| 2^{-(J-j)r} < |c_J|$; an empty sum is zero.

Proof. The exponent of 2 in $\gamma_{a,t}$ is $t + (a + 1) - (a + t + 1) = 0$. For $p = 3, 5$, the exponent is $\lfloor t\theta_p \rfloor + \lfloor (a + 1)\theta_p \rfloor - \lfloor (a + t + 1)\theta_p \rfloor$, which is 0 or -1 by floor addition. Thus the only possible denominator factors are one 3 and one 5; the prefactor 15 in $D_{r,a}$ clears both. For $k < jr$, P_{k+1} divides P_{jr}, so Z_r is an integer. For each t, absolute convergence gives

$$\sum_{a \geq 0} \frac{\gamma_{a,t} m_{a+t}}{P_{a+1}} = P_t \sum_{k \geq t} \frac{m_k}{P_{k+1}} = P_t \left(\alpha - \sum_{k<t} \frac{m_k}{P_{k+1}} \right).$$

The finite linear combination proves the identity. Lemma 8.3, $0 < \gamma \leq 1$ and $a + jr + 1 \leq \max(1, j)(a + r + 1)$ give the stated constant. For $j < J$, $P_{jr}/P_{Jr} \leq 2^{-(J-j)r}$. Dividing A_r by $15P_{Jr}$ and bounding the remaining terms proves the nonvanishing condition, which holds for all sufficiently large r. □

For any fixed nonzero coefficient vector, including $(1, -3, 3, -1)$, the nonvanishing condition is automatic for sufficiently large r: each term on its left tends to zero. No arithmetic assumption on S is needed. The identity still represents $S/2$, but gives no sparsity or cancellation of $D_{r,a}$.

12.2 Boundary terms in weighted finite differences

Follow one lattice point through four successive triangles. If it lies in all four and its weight is unchanged, the coefficients $(1, -3, 3, -1)$ sum to zero. If it first enters the second triangle, its coefficient is $-3+3-1 = -1$; entry in the third gives $3-1 = 2$, and entry in the fourth gives -1. Thus a finite difference depends both on which triangles contain the point and on changes of its weight. We separate these two effects below. Avoiding floor crossings controls the weights, but does not remove the newly added strips.

Recall that $w_{j,k} = j \log_2 3 + k \log_2 5$, $t_{j,k} = \{w_{j,k}\}$ and $\theta_p = 1/\log_2 p$ for $p = 3, 5$. To distinguish the shared points from the new ones, let $\mathcal{J}_a = \{(j, k) \in \mathbb{N}^2 : w_{j,k} < a + 1\}$ and, for $0 \leq t < 1$, set

$$\omega_a(t) = \prod_{p \in \{3,5\}} p^{\lfloor (a+1)\theta_p \rfloor - \lfloor (a+t)\theta_p \rfloor}.$$

This weight is defined even for pairs outside $\mathcal{J}_a$. For $\nu \geq 0$, put

$$\kappa_p(a, t, \nu r) = \lfloor (a + t + \nu r)\theta_p \rfloor - \lfloor (a + t)\theta_p \rfloor - \lfloor \nu r \theta_p \rfloor,$$
$$\chi_\nu(a, t) = 3^{-\kappa_3(a,t,\nu r)} 5^{-\kappa_5(a,t,\nu r)}.$$

Each κ_p is 0 or 1. A value $\kappa_p = 1$ records a floor-addition carry. This is different from a new lattice point entering $\mathcal{T}_a$ as a increases.

Proposition 12.2 (an exact interior-and-strip decomposition)**.** *For a fixed operator $c_0, \dots, c_\sigma$ and $r \geq 1$, let $E_0 = \mathcal{T}_a$ and $E_s = \mathcal{T}_{a+sr} \setminus \mathcal{T}_{a+(s-1)r}$ for $1 \leq s \leq \sigma$. Then*

$$\frac{D_{r,a}}{15} = \sum_{s=0}^{\sigma} \sum_{(j,k)\in E_s} \omega_a(t_{j,k}) \sum_{\nu=s}^{\sigma} c_\nu \chi_\nu(a, t_{j,k}).$$

For the cubic operator $(1, -3, 3, -1)$, if all these crossing bits vanish, then

$$\frac{D_{r,a}}{15} = -M_0 + 2M_1 - M_2, \qquad M_s = \sum_{(j,k)\in \mathcal{T}_{a+(s+1)r} \setminus \mathcal{T}_{a+sr}} \omega_a(t_{j,k}).$$

Thus absence of floor crossings cancels the common interior, but not necessarily the three boundary strips.

Proof. For each $p = 3, 5$, the exponent of p on either side of

$$\gamma_{a,\nu r}\omega_{a+\nu r}(t) = \omega_a(t)\chi_\nu(a, t)$$

is

$$\lfloor (a+1)\theta_p \rfloor + \lfloor \nu r \theta_p \rfloor - \lfloor (a+t+\nu r)\theta_p \rfloor.$$

On the left the upper-boundary exponents cancel; on the right the $\lfloor (a+t)\theta_p \rfloor$ terms cancel. This proves the pointwise identity. Insert the triangle formula for each $m_{a+\nu r}$ and group by the first triangle containing a pair. A pair in E_s occurs exactly in the terms $\nu \geq s$, proving the first formula. Under the no-crossing hypothesis, $\chi_\nu = 1$. The full cubic sum is zero, while its three successive suffix sums are $-1, 2, -1$, proving the second formula. □

Example 12.3 (cubic cancellation can fail with no floor crossings)**.** Take $a = 0$, $r = 35$ and $(c_0, c_1, c_2, c_3) = (1, -3, 3, -1)$. Exact integer comparisons give

u	0	35	70	105
$\lfloor \log_3 2^u \rfloor = \lfloor \log_3 2^{u+1} \rfloor$	0	22	44	66
$\lfloor \log_5 2^u \rfloor = \lfloor \log_5 2^{u+1} \rfloor$	0	15	30	45
m_u	1	195	723	1582

The first two rows imply $b_u = 2$ and $\lfloor (u+t)\theta_p \rfloor = \lfloor u\theta_p \rfloor$ for every $0 \leq t < 1$, $p = 3, 5$. Hence every relevant $\kappa_p(0, t, u)$ vanishes, $\gamma_{0,u} = 1$, and all triangle weights at these four indices are 1. The last row is therefore the count of pairs satisfying $3^j 5^k < 2^{u+1}$, not a floating-point estimate. The strips have sizes $194, 528, 859$, so

$$D_{35,0} = 15(-194 + 2 \cdot 528 - 859) = 45 \neq 0.$$

Exact enumeration reproduces all four counts and checks the endpoint power inequalities using integers. This example disproves the claim that avoiding floor crossings alone forces the third difference to vanish. It does not exclude a different fixed operator, different shifts, a boundary correction or a sparse-defect statement restricted to a suitable infinite subsequence.

What the Hecke–Mahler comparison does and does not supply. In the published Luca–Ouaknine–Worrell article [26, Definition 5 and Theorem 6], the same fixed integer coefficients are applied to shifts of one integer sequence. The resulting nonzero terms must have expanding gaps and uniform polynomial variation; the original sequence must also have polynomial growth. Theorem 6 then gives a fixed-base value criterion. Theorem 8 verifies that condition for $f(\lfloor m\vartheta + \rho \rfloor)$ when $f \in \mathbb{Z}[x]$ is nonconstant, $\vartheta, \rho \in (0,1)$ and ϑ is irrational. These interval restrictions belong to that normalised combinatorial statement. Their main value theorem, Theorem 1, allows every real ρ and every irrational real ϑ, with any algebraic base β satisfying $|\beta| > 1$. Claim 10 uses a finite difference of order $\deg f + 1$, which vanishes when no floor crossing occurs. This is a polynomial evaluated at a floor, not merely an integer sequence of polynomial growth. Our m_a is instead a weighted lattice count. Proposition 12.2 separates floor crossings from its moving boundary, and Example 12.3 shows that the latter can survive a cubic difference even with no crossings.

There is a second distinction. In our displayed shift identity the coefficient of m_{a+jr} is $15c_j\gamma_{a,jr}$, which can depend on a. For example, $P_1 = 2$, $P_2 = 12$ and $P_3 = 120$ give $\gamma_{0,1} = 1/3$ but $\gamma_{1,1} = 1/5$. These factors are required by the denominator chain. A support estimate for $D_{r,a}$ is therefore not by itself a verification of Definition 5 for the sequence (m_a). No fixed-coefficient representation to which that theorem applies has been established here.

For a variable-denominator analogue, a useful hypothesis would be a fixed choice of coefficients and shifts $r_n \to \infty$ for which $\Delta_n = \{a \geq 0 : D_{r_n,a} \neq 0\}$ is infinite and distinct members are at least ηr_n apart, for a fixed $\eta > 0$. Even this would leave the uniform polynomial-variation requirement, for example

$$|D_{r_n,a'}| \leq C\big((a'-a)^d + |D_{r_n,a}|\big) \quad (a < a',\ a, a' \in \Delta_n),$$

with C, d independent of n, a and a'. This is a condition on the gaps between nonzero terms, not merely on their density: even a set of density zero can contain adjacent pairs. It also requires infinitely many nonzero terms for each selected shift. A small but nonzero term at every index would fail it, and an identically zero difference would also fail the infinitude requirement. The cited paper constructs suitable shifts for the polynomial-floor sequences just described; no such result is proved for our weighted counts. The bound $O((a+r+1)^2)$ controls absolute size: it depends on the location a itself. The variation condition instead bounds a later nonzero term by the gap from an earlier one and the size of that earlier term, with constants uniform in the shift. The absolute-size bound alone gives no such comparison. The denominator chain itself is not a fixed-base power sequence. The next proposition shows what is lost in two direct fixed-base recodings. A value criterion for the original chain would need a separate proof, including non-cancellation and the Diophantine height estimates required by a Subspace-Theorem argument. These heights measure the numerators and denominators of the approximating algebraic quantities; they are unrelated to the running LCM H. The finite set of prime divisors alone does not provide those estimates.

Why a finite alphabet of bases is not enough. Kebis, Luca, Ouaknine, Scoones and Worrell [27, Definition 3, Theorem 6 and Claim 7] work with fixed-base series whose

coefficients lie in a finite algebraic alphabet. Their echoing condition requires long near-repetitions, separated intervals containing the mismatches, and nonzero weighted mismatch sums on at least two of those intervals. Claim 7 supplies the non-cancellation step in the proof of the value theorem. Merely having finitely many letters does not give these properties. The four-letter radix sequence is not the numerator sequence: the actual m_a is unbounded. A recoding would have to preserve the scalar value, identify its coefficient alphabet and verify the echoing conditions. No fixed-base recoding with verified echoing properties is proved here.

Two natural bases expose the tradeoff. The least integer base whose powers clear every P_n is 30, while base 8 matches the growth of $P_n = \mathrm{H}(2^n)$. The former keeps integrality; the latter keeps polynomial size.

Proposition 12.4 (what direct fixed-base recoding preserves)**.** *The following identities converge absolutely:*

$$\alpha = \sum_{a\ge 0} \frac{e_a}{30^{a+1}} = \sum_{a\ge 0} \frac{v_a}{8^{a+1}}, \qquad e_a = \frac{m_a 30^{a+1}}{P_{a+1}}, \quad v_a = \frac{m_a 8^{a+1}}{P_{a+1}}.$$

Here $e_a \in \mathbb{Z}_{>0}$ and $e_a \ge (15/4)^{a+1}$, whereas $v_a \in \mathbb{Z}[1/15]$ and $0 < v_a < 225(a+1)^2$. For an integer $q \ge 2$, the termwise divisibility $P_n \mid q^n$ for every $n \ge 1$ holds exactly when $30 \mid q$; that direct recoding then has coefficients at least $(q/8)^{a+1}$.

Proof. Each exponent in P_n is at most n, so $P_n \mid 30^n$. The estimate $8^n/15 < P_n \le 8^n$ was proved in Section 8. These facts, $m_a \ge 1$ and $m_a \le 15(a+1)^2$, give all the coefficient bounds and identities. Necessity of $30 \mid q$ follows because each of $2, 3, 5$ divides some P_n; sufficiency follows from $P_n \mid 30^n$. The general lower bound follows again from $P_n \le 8^n$. □

The base-30 coefficients are integral but grow exponentially, so the cited polynomial-growth criterion does not apply. The base-8 coefficients have polynomial size but are not all integral: already $v_1 = 4 \cdot 8^2/12 = 64/3$. In fact, their reduced denominators grow exponentially even after cancellation. Since $P_{a+1}/2^{a+1}$ is odd, the expression

$$v_a = \frac{m_a 4^{a+1}}{P_{a+1}/2^{a+1}}$$

can cancel an odd factor only through m_a. Consequently,

$$\operatorname{den}(v_a) \ge \frac{P_{a+1}}{2^{a+1} m_a} > \frac{4^{a+1}}{225(a+1)^2}.$$

The last inequality uses $P_{a+1} > 8^{a+1}/15$ and $m_a \le 15(a+1)^2$. Thus the bound on absolute value does not control Diophantine height even for these particular coefficients; fixed denominator-prime support is not enough.

More generally, no fixed integer base gives this termwise recoding both eventually integral coefficients and polynomial growth. If $30 \mid q$, the proposition gives exponential

growth. Otherwise choose $p \in \{2,3,5\}$ not dividing q. Cancellation by m_a removes at most a factor m_a, so

$$\operatorname{den}\left(\frac{m_a q^{a+1}}{P_{a+1}}\right) \geq \frac{p^{\lfloor (a+1)\log_p 2 \rfloor}}{m_a} > \frac{2^{a+1}}{15p(a+1)^2}.$$

This denominator tends to infinity. Thus matching the denominator growth by base 8 keeps the coefficients small but introduces large rational denominators; making the coefficients integral by base 30 makes them exponentially large. This obstruction concerns termwise rescaling only, not regrouping or carrying.

Carrying resolves a different problem: it bounds the digits but leaves the radices variable. It also requires the integer parts of the actual tails, not just the known block numerators. Set

$$\delta_a = m_a + \lfloor X_{a+1} \rfloor - b_a \lfloor X_a \rfloor = \lfloor b_a \{X_a\} \rfloor.$$

The equality follows from the tail recurrence and $m_a \in \mathbb{Z}$. Thus $0 \leq \delta_a < b_a$ and $\{X_a\} = (\delta_a + \{X_{a+1}\})/b_a$. Iterating gives the finite identity

$$\{X_0\} = \sum_{a=0}^{N-1} \frac{\delta_a}{P_{a+1}} + \frac{\{X_N\}}{P_N}.$$

The last term is nonnegative, is less than $1/P_N \leq 2^{-N}$, and tends to zero. This is the greedy mixed-radix expansion of the fractional part of $X_0 = S/2$, with digits in $\{0, \dots, 29\}$. For the actual series its first eight digits are

$$(\delta_0, \dots, \delta_7) = (1, 4, 8, 3, 10, 1, 8, 5).$$

To certify them, use the twelve-block enclosure from Section 11, either propagated by the recurrence or evaluated directly as

$$P_a \sum_{j=a}^{11} \frac{m_j}{P_{j+1}} < X_a \leq P_a \sum_{j=a}^{11} \frac{m_j}{P_{j+1}} + \frac{P_a}{P_{12}} Q(n_{12}) \qquad (0 \leq a < 12).$$

Both descriptions give the same rational endpoints. At indices 0 through 8, each interval lies strictly between consecutive integers, certifying all nine floors needed for the eight digits. Width alone would not suffice: an interval crossing an integer leaves its floor undetermined. For example, $\lfloor X_4 \rfloor = 2$ and $\lfloor X_5 \rfloor = 5$, giving $\delta_4 = 65 + 5 - 30 \cdot 2 = 10$, not the uncarried numerator $m_4 = 65$. This finite calculation makes no claim about the later digit complexity.

A rational number need not have eventually periodic digits when the radices vary. For example, take radix 5 at the indices $1, 2, 4, 8, \dots$ and radix 3 elsewhere. The constant tail fraction $1/2$ gives digit 2 at those indices and 1 elsewhere. These digits are not eventually periodic, but their mixed-radix expansion sums to $1/2$. This example uses different radices from our four possible bases; it shows why fixed-base eventual periodicity cannot be transferred to a variable-radix expansion. Our digits also depend on the actual tail fractions, not just on the four-letter radix sequence. The construction supplies

neither a fixed-base expansion with verified echoing properties nor a bound on its digit complexity.

Adamczewski–Bugeaud's complexity theorem [30, Theorem 1] concerns the actual digits of an integer-base expansion: for an algebraic irrational their length-n block complexity divided by n tends to infinity. Neither (b_a) nor the uncarried (e_a) is such a digit expansion of α.

Problem 12.5 (the repeated three-prime target)**.** Prove irrationality of $S = \mathcal{R}_{\{2,3,5\}}$. Equivalent forms are $\mathsf{E}(K)$ and

$$BX_a \notin \mathbb{Z} \quad \text{for every } B \geq 1 \text{ with } \gcd(B, 30) = 1 \text{ and every } a \geq 1. \tag{22}$$

The residue formulation uses finite integer computations, but asks for a success for every eligible denominator and beyond every prescribed start. The tail formulation requires that no positive integer multiplier coprime to 30 make any X_a, $a \geq 1$, integral. These are reformulations of the same irrationality assertion, not extra assumptions on the series, and neither follows from the rank obstruction. The next proposition proves the tail reformulation, complementing Theorem 10.2.

Proposition 12.6 (irrationality is equivalent to nonintegrality of every reduced tail)**.** *Statement* (22)*, quantified over every $B \geq 1$ coprime to 30 and every $a \geq 1$, is equivalent to irrationality of S.*

Proof. If $BX_a \in \mathbb{Z}$ for some such B and a, then $X_a \in \mathbb{Q}$, and since $S = \sum_{j<a} s_j + X_a/h_a$ with h_a a positive rational and the prefix a finite sum of rationals, $S \in \mathbb{Q}$. Conversely if $S \in \mathbb{Q}$, then Theorem 9.3 produces B coprime to 30 with $BX_a \in \mathbb{Z}$ for every $a \geq a_D$. □

Integrality at one index propagates forwards, but nonintegrality at one early index does not propagate forever: a rational denominator can clear later. Proposition 9.4 describes that onset exactly. The dichotomy in Proposition 8.4 also permits integral tails, so it does not settle this question.

12.3 Function-theoretic representations

The elementary proof of irrationality of $\mathcal{D}_{\{2,3,5\}}$ gives no transcendence conclusion. Recall that

$$\mathcal{D}_{\{2,3,5\}} = 1 + \sum_{t \in \{2^n, 3^n, 5^n : n \geq 1\}} \frac{1}{2^{\lfloor \log_2 t \rfloor} 3^{\lfloor \log_3 t \rfloor} 5^{\lfloor \log_5 t \rfloor}}.$$

Each running LCM is counted once at its first jump, unlike the repeated sum. The bases in (14) come from these same jumps. For two primes a representation by one Hecke–Mahler value proves transcendence (Section 4). A corresponding three-prime representation has not been constructed here.

Problem 12.7 (an exact function representing the distinct-height sum)**.** Transcendence of $\mathcal{D}_{\{2,3,5\}}$ is not proved here. For a functional approach to it, first specify the function, its coefficient field and convergence domain, then prove an exact identity for $\mathcal{D}_{\{2,3,5\}}$, and only then apply a stated value theorem. A conditional theorem must display the extra nondegeneracy assumption; a no-representation theorem must specify the class it excludes.

For $(p, q, r) = (2, 3, 5)$, the two slopes in the rank proof are $\log_5 2 = \theta_5$ and $\log_5 3 = \theta_5/\theta_3$, not θ_3 and θ_5. Each is irrational: a rational value would equate a positive power of 5 with a positive power of 2 or 3. The proof chooses the row and column indices independently. It therefore needs no rational independence of $1, \theta_3, \theta_5$ and no density assertion for the singly indexed orbit $(\{n\theta_3\}, \{n\theta_5\})$. Any functional approach that needs such a stronger hypothesis must establish it separately.

A finite-dimensional encoding is useful here only if its data determine the represented function: equal encodings must give equal functions. The Lean sources impose this through a factorisation of the linear map to functions through a finite-dimensional space. The resulting space of functions is then finite-dimensional (identity for the carry, map to functions, finite-dimensional conclusion). They also bound the carry residue and its digit in their stated intervals (residue bound, digit bound). These conditional linear-algebra statements neither construct a function representing $\mathcal{D}_{\{2,3,5\}}$ nor show that a corresponding function space must be infinite-dimensional.

The cited value theorems and their hypotheses. Pellarin's rank-one theorem concerns quadratic irrational slopes and algebraic evaluation points in the convergence domain [31, Theorem 1.1 and Remark 1.2, pp. 330–332]. The prime-logarithm slopes used here are not quadratic: each $\log_r p$ is irrational by unique factorisation, and the Gelfond–Schneider theorem then makes it transcendental, since $r^{\log_r p} = p$ is algebraic.[1] This rules out direct substitution of those slopes into Pellarin's quadratic-slope theorem. It does not rule out a different representation of the scalar sum, and it does not affect the two-prime argument above, which uses a theorem for general irrational slopes.

In Pellarin's reduced-slope normalisation, the rank-one condition says that the evaluation points arise from a common point by the quadratic order's monomial action, up to multiplication by torsion points. For a single evaluation point it is automatic: use that point itself and the identity action. For several points it is a condition to check, not a consequence of having two lattice coordinates. Under it, algebraic independence of the values is equivalent to $\mathbb{Q}$-linear independence of the specified auxiliary formal Laurent series; the rank-one condition alone does not imply independence. This module rank is not the rank of our reciprocal kernel. No exact representation suitable for this theorem has been constructed here, but infinite kernel rank alone does not preclude one.

For one variable, Adamczewski and Faverjon's proofs of Nishioka's theorem and the lifting theorem are regular-point results [32, Theorems 1 and 2]. Regularity means that the system matrix stays defined and invertible along the evaluation orbit. Their multivariate lifting theorem [33, Theorem 3.3] additionally assumes an admissible transformation–point pair. Its growth, decay and nonvanishing conditions are specified in Definition 5.1; none follows merely from writing a double sum. Lifting transfers a relation among values to one among functions; a transcendence application must exclude the latter. A Mahler system alone cannot do this, as the constant function 1 illustrates.

Regularity of a preselected system is not, however, a universal prerequisite for Mahler's method. The same paper's Theorem 1.1 treats univariate Mahler functions

[1] For this logarithmic form of the Gelfond–Schneider theorem, see Keith Conrad, *Transcendence of e*, p. 1, footnote 1, https://kconrad.math.uconn.edu/blurbs/analysis/transcendence-e.pdf (accessed 18 September 2026).

at algebraic points in the punctured unit disc where their values are defined, without that regularity hypothesis. The hypotheses must therefore be matched to the particular theorem being invoked. No exact Mahler representation or suitable evaluation data for this sum over distinct running LCMs are constructed here. The locators for this paper refer to its 68-page author manuscript, not the older 52-page arXiv version.

12.4 Hypotheses for auxiliary integer recurrences

For ordered primes $p_1 < \cdots < p_s$, the interval (p_1^a, p_1^{a+1}) contains at most one power of each $p_i > p_1$. The radix including the final p_1-jump consequently belongs to

$$\left\{ p_1 \prod_{i=2}^{s} p_i^{\varepsilon_i} : \varepsilon_i \in \{0,1\} \right\}.$$

This finite alphabet determines neither the frequency of the radices nor the error terms in the weighted block counts. Estimates for those quantities, or for specified shifted kernels, would be additional arithmetic inputs. The conditional statements about integer recurrences below keep such inputs explicit.

Some auxiliary results explain why bounded integer approximations alone are insufficient. Here are their hypotheses in explicit form.

Let $j(n) \in \{2,3,5\}$ label a sequence of jumps, and suppose each label occurs. A perturbation ε_n in an additive abelian group has sum zero on every interval $[a,b)$ whose endpoints have the same label exactly when

$$\varepsilon_n = u(j(n+1)) - u(j(n))$$

for three potential values $u(2), u(3), u(5)$. One direction follows by telescoping. Conversely, the zero-sum assumption makes $\sum_{k<n} \varepsilon_k$ the same at any two indices with the same label. Assign that common value to $u(j(n))$; every label occurs, so all three values are defined. Consecutive partial sums then give the displayed identity. In the usual terminology, the perturbation is a coboundary of the function u on the labels: it is the difference of successive potential values. This is a telescoping condition, not a bound on the size of the perturbation. It does not by itself force the perturbation to vanish: unequal potential values give nonzero differences when the label changes. If the perturbation also vanishes on a genuine $2 \to 3$ transition and on a genuine $2 \to 5$ transition, all three potential values agree, so every perturbation is zero.

For an integer sequence satisfying $c_{n+1} = b_n c_n - D m_n$, compare it with an integer sequence z_n. Define the error $e_n = D z_n - c_n$ and the perturbation $\varepsilon_n = b_n z_n - z_{n+1} - m_n$. Then

$$e_{n+1} = b_n e_n - D\varepsilon_n.$$

Under the preceding zero-sum and transition hypotheses, $\varepsilon_n = 0$ and hence $e_N = (\prod_{n<N} b_n) e_0$. If every $b_n \geq 2$ and the integer e_0 is nonzero, then $|e_N| \geq 2^N$. This contradicts even one bound $|e_N| < 2^N$, and therefore also excludes a uniform bound for all N. The nonzero initial error is essential: the identically zero error causes no contradiction.

A calculation at four consecutive indices gives a related obstruction. If $0 < T_i < 1$, $z_i \in \mathbb{Z}$ and $|z_i - T_i| < 1$, then z_i is 0 or 1. For bases $2, 3, 2, 5$, the equalities $2z_0 - z_1 - 1 = 0$

and $2z_2 - z_3 - 1 = 0$ force all four integers to be 1. The sum of the first two perturbations is then

$$(2z_0 - z_1 - 1) + (3z_1 - z_2 - 1) = 1,$$

not zero. Thus unit accuracy, the two exact equalities and a zero sum on that block cannot all hold. Unit accuracy alone does not give either exact equality.

No integer approximation to the tails of S is constructed here that satisfies all these conditions. Rationality supplies integer carries of at most quadratic growth, but not the additional zero-sum and transition identities. The carries obtained from S satisfy a weighted block identity instead. Applying the auxiliary results would require that identity to imply their hypotheses, or would require a different construction of the integer approximations.

There is also a conditional criterion for a sum supported on the powers of 2. Its elementary ingredients can be stated without introducing another formal vocabulary. From $y' = by - 1$ with $b > 0$ and $0 < y' < 1$ one gets $1/b < y < 2/b$. For integers $a > 0$ and $p \geq 3$, if $u > 1/a$ and $v < 2/(pa)$, then $u - v > 1/(3a)$. Finally, a real number x is irrational if, for every positive integer d, there are integers n, k with $0 < |nx - k| < 1/d$.

To use these observations for a series, the conditional result requires exact identities $y_M = H_M x - z_M$, with $H_M, z_M \in \mathbb{Z}$, and, for every positive d, two indices with different H_M, different y_M, and $|y_M - y_{M'}| < 1/d$. Their difference gives the required nonzero linear form. Equal tail values would give zero and would not suffice; a merely bounded gap would not suffice either. Indeed, for a rational x of denominator d, every nonzero difference of this form has size at least $1/d$. No concrete series in this record is shown to satisfy all these clearing and arbitrarily small nonzero-gap conditions.

12.5 The unresolved repeated-sum question

The distinct sum at $\{2, 3, 5\}$ is settled by Theorem 1.1. The general and subseries arguments remain ordinary proofs pending formalisation (Remarks 5.1 and 5.10). The finite-state calculations of Remark 5.8 have their own scope; the $\{2, 3, 5, 7\}$ automaton collision limits that interval criterion only.

For $\mathcal{R}_{\{2,3,5\}}$, the missing ingredient is not just another normalisation or one sharper finite enclosure. The coefficients, recurrence, quadratic bounds and denominator-clearing formula are explicit. We must exclude eventual integral reduced tails for every possible denominator, or equivalently prove the residue inequalities beyond every prescribed onset. The fixed-start formula explains why an early success does not suffice; the strip identity explains why avoiding floor crossings does not supply the proposed cancellation.

The rank and uniform-norm results concern the kernel, not the arithmetic of its sum. Finite tests, countability, radix recoding and denominator support do not close the remaining argument. The settled two-prime case uses Fan's earlier reduction to the cited Hecke–Mahler value theorem.

Verification and reproducibility

The verification concordance lists the formal proofs by statement. *Lean* links to the supporting declarations; a dagger identifies a proof that assumes a named input. *Comparator*

links to a recorded kernel check against a separately written statement; *pending* means that this comparison has not been recorded. The verification record gives the precise correspondence, dependencies and reproducible checks. A row with only a record link has no complete formal proof recorded.

Proof sources. The evidence record fixes the source versions and individual statement associations. These are earlier checks; this revision makes no fresh formal-verification claim.

The general distinct-value assertion and its four word-and-torus arguments are stated as remarks with complete ordinary proofs in Section 5.1; they have no Lean proofs. The source record reports that the general and prime-power subseries arguments were found on 26 September 2026 and checked by a second AI agent, whose corrections were incorporated. It records no human review; the present revision does not constitute a new independent review. No separate literature search for the prime-power subseries theorem at $|P| \geq 3$ is recorded, and no priority claim is made for it. The subseries identities also have ordinary proofs. The program `two_tail.py` checks their finite instances in exact arithmetic.

For $\{2, 3, 5\}$ the separate theorem for the sum over distinct values has a Lean proof; its source record says that Comparator had not been run on it. The general affine-block calculation, decoding criterion and finite-set application are also remarks with ordinary proofs; the last additionally uses exact computation. Their formal coverage is restricted to the $\{2, 3, 5\}$ case with one block and the interval of Section 2. The linear-flow and Smith-factor results likewise have the ordinary proofs printed here; no additional formal coverage is inferred for them.

The two-prime deduction is credited to Fan [17]. Its formal proof assumes the cited Hecke–Mahler value theorem, as specified in Section 14; its algebraic identities are unconditional. The weighted-triangle and shift identities, denominator formula, scaled dichotomy, uniqueness argument, residue limit, strip decomposition and recoding also have written proofs here. Their individual formal associations are retained, without a blanket assertion that every result in the record is formalised. Kernel acceptance uses the standard axioms allowed by the linked checks and is not an independent review of the paper-to-statement correspondence.

The twelve-block enclosure, denominator bound and eight mixed-radix digits have finite exact proofs. The two large exclusions have saved witnesses and exact checkers, and the larger scan has a saved run. None of these finite computations proves the residue inequalities beyond every starting index. No Comparator replay is claimed for the general or subseries arguments.

Artefact and data availability. The links in the source inventory and to the integer-arithmetic checker open a fixed revision of the repository. The witnesses and exact checkers for the two large numerical exclusions and the saved larger scan are in the large-computation directory. The programs for sums over distinct values recompute the constants of Theorem 1.1 and the certificates of Remark 5.8 in exact rational arithmetic, with two independent computations of the tail intervals, and store their outputs;

a fast test checks them against the stored outputs. The same directory checks the finite illustrations of Section 5.1 and the block identity for $\{2, 3, 5, 7\}$.

Funding and competing interests. This work received no external funding. The author declares no competing interests.

Authorship and review. Will Cook directed the work. AI agents did most of the research and drafting. Cook has not independently verified every mathematical claim, and these manuscripts have not had independent human mathematical review.

Acknowledgements. Steve Fan's forum post of 26 June 2026 supplied the two-prime factorisation and its Hecke–Mahler reduction before this manuscript; it also records the elementary running-LCM identity for finite prime sets [17]. The transcendence input is due to Yann Bugeaud and Michel Laurent and to the earlier work of Loxton and van der Poorten cited by them. The problem numbering and its status as of 28 July 2026 are taken from the Erdős Problems catalogue maintained by Thomas Bloom [4].

13 Further examples, conditional lemmas and source references

We collect the historical statements and then separate the lemmas on integer recurrences from their three-prime application. Their explicit divisibility and segment hypotheses specify what another recurrence must supply. Finite examples and the declaration inventory complete the record.

13.1 The historical and catalogue record

For a singleton $P = \{p\}$, we have $a_n = p^{n-1}$ and $[a_1, \dots, a_n] = p^{n-1}$, so both sums equal $p/(p-1)$. Erdős's 1988 account calls the infinite-prime case a simple exercise and presents irrationality for a finite number of primes greater than one as a probable extension [2, p. 106]. Bloom's catalogue records an open problem at the access date of 28 July 2026 [4]. These are dated source statements, not a fresh determination of the catalogue's present status.

In the letter written on 1 January 1973 and published in 1974, Erdős calls the irrationality of $\mathcal{R}_P$ a conjecture that he cannot prove, and says that he can show the sum over the distinct values of $[a_1, \dots, a_k]$ to be irrational [3, p. 335]. That sum is $\mathcal{D}_P$. The letter introduces its primes as $p_1, \dots, p_s$, later writing p_r, without restricting their number, so the singleton calculation forces the qualification $|P| \geq 2$. It prints no argument. The catalogue has cited the letter for Problem #269 since 28 December 2025, after the letter was quoted in the site's discussion of missing problems [5], and its page records Erdős's statement about the distinct values. The catalogue page says that Erdős proved the statement; the letter asserts it and prints no argument. Remark 5.1 proves it for every finite P with $|P| \geq 2$. For $|P| = 2$ it was already a consequence of the Hecke–Mahler transcendence theorem of Loxton and van der Poorten [16, Theorem 8, p. 40] and Bugeaud and Laurent [15, Theorem 1.1], through the affine identity of Section 4, by the route Fan used for $\mathcal{R}_{\{p,q\}}$. We do not know what argument Erdős had in mind.

The literature search recorded here did not locate a proof of the assertion in print. Erdős and Graham state only the question for $\mathcal{R}_P$: the sum is irrational for infinite P, and

they ask what happens for finite P with more than one element [1, p. 65]; the sum over distinct running LCMs does not appear there. Erdős's survey of 1975 on the irrationality of series [8] does not mention the running least common multiple of the P-smooth integers, his survey of 1980 on combinatorial number theory [6] does not discuss irrationality, and his survey of 1988 states only the question for $\mathcal{R}_P$ [2, p. 106]. The Hecke–Mahler transcendence theorems cited here, of Loxton and van der Poorten, Bugeaud and Laurent, and Luca–Ouaknine–Worrell [26], have a single slope; the identifications in Section 4 apply to $\mathcal{D}_P$ when $|P| = 2$. The rationality criterion of Diananda and Oppenheim [7], which condenses a Cantor series into blocks of consecutive terms, has the form of the dyadic argument of Section 2; it restates rationality and does not decide $\mathcal{D}_P$. The discussion thread of Problem #269, as inspected for that search, supplied no argument for $\mathcal{D}_P$ with $|P| \geq 3$, and the *Formal Conjectures* statement [9] concerns $\mathcal{R}_P$ only. The Cantor-series criteria of Erdős–Straus [18, Theorem 2.1] and Hančl–Tijdeman [19, Theorem 3.1] do not apply to $\mathcal{D}_P$, as explained in Section 2. The search did not reach the sections on irrational series in Guy's *Unsolved Problems in Number Theory*, or MathSciNet reviews. The unresolved target of this record is the repeated series $\mathcal{R}_P$; a transcendence statement for $\mathcal{D}_P$ with $|P| \geq 3$ is a separate question (Problem 12.7).

On 26 June 2026 Steve Fan posted the two-prime factorisation, the Hecke–Mahler reduction and the transcendence conclusion in the discussion thread of the problem's page [17]; the comment itself notes that the argument does not seem to generalise immediately to $|P| \geq 3$, and a reply there observes that it applies to arbitrary coprime pairs. The first public version of this note is dated 22 July 2026, after Fan's post. We retain the calculation as exposition and make no priority claim for it.

The statement of the problem has been formalised as a conjecture with an unfilled proof in the *Formal Conjectures* collection [9]. In the cited source, the rational, irrational and infinite-prime assertions end in `sorry`. Its Nat-indexed series includes the empty-prefix least-common-multiple term, so its value differs from the conventional one by a rational constant; transporting a theorem across that boundary needs an explicit series-identification lemma, which is not supplied here.

13.2 Integer recurrences in their general form

Consider an arbitrary integer recurrence with positive integer radices. Divisibility and the existence of segments satisfying the residue inequality must now be supplied as hypotheses. For the actual three-prime radices, the criterion below shows that each integer solution is eventually divisible by the smooth part of a denominator. It still leaves the residue-inequality hypothesis to be proved. The comparison with Cantor-series criteria is in Section 9.

Let D and B be positive integers with $D = D_{\mathrm{sm}}B$, where $D_{\mathrm{sm}} = 2^u 3^v 5^w$, $u, v, w \in \mathbb{N}$ and $\gcd(B, 30) = 1$. Let (c_n) be an integer sequence satisfying $c_{n+1} = b_n c_n - Dm_n$ for integer sequences (b_n) and (m_n). For these general sequences, form W and F by the recursions in Section 10; a segment satisfying the residue inequality must have $W \neq 0$, and its modulus is $|W|$. In the factorisation below the quotients d_n are required to be integers; writing $c_n = D_{\mathrm{sm}} d_n$ asserts divisibility, not just an identity in $\mathbb{Q}$.

Proposition 13.1 (conditional denominator reduction)**.** *If $c_n = D_{\mathrm{sm}} d_n$ for every n, with $D_{\mathrm{sm}} > 0$, then the recurrence and iterated recurrence for (d_n) have multiplier B in place of D. Moreover, for every n and every real t,*

$$0 < c_n \le Dt \quad \Longleftrightarrow \quad 0 < d_n \le Bt.$$

Proof. Substitute $c_n = D_{\mathrm{sm}} d_n$ and $D = D_{\mathrm{sm}} B$ in the recurrence and divide by D_{sm}. Dividing $0 < c_n \le D_{\mathrm{sm}} Bt$ by this positive factor gives the stated bound equivalence. Dividing the iterated recurrence $c_{\ell+h} = W_{\ell,h} c_\ell - D F_{\ell,h}$ likewise gives $d_{\ell+h} = W_{\ell,h} d_\ell - B F_{\ell,h}$. □

For a general bound $c_n \le G(D, n)$, division and integrality give only

$$d_n \le \left\lfloor \frac{G(D,n)}{D_{\mathrm{sm}}} \right\rfloor,$$

not automatically $d_n \le G(B, n)$. The bounds used for the actual tails are linear in the multiplier before rounding. For any real t,

$$\left\lfloor \frac{\lfloor Dt \rfloor}{D_{\mathrm{sm}}} \right\rfloor = \lfloor Bt \rfloor.$$

Indeed, an integer z satisfies $D_{\mathrm{sm}} z \le \lfloor Dt \rfloor$ exactly when $D_{\mathrm{sm}} z \le Dt$, or $z \le Bt$. Thus rounding does not change the reduced bound in this case.

Growth of the radix product is not enough: $b_n = 3$, $m_n = 1$, $D = 2$ and $c_n = 1$ satisfy the recurrence, but 2 never divides c_n. The precise condition follows by reducing the recurrence modulo D_{sm}. For any starting index A and every $n \ge A$,

$$c_n \equiv \left(\prod_{j=A}^{n-1} b_j \right) c_A \pmod{D_{\mathrm{sm}}},$$

$$D_{\mathrm{sm}} \mid c_n \quad \Longleftrightarrow \quad \frac{D_{\mathrm{sm}}}{\gcd(D_{\mathrm{sm}}, c_A)} \mid \prod_{j=A}^{n-1} b_j.$$

The congruence is an induction, since $D_{\mathrm{sm}} \mid D$. For the equivalence, cancel the positive greatest common divisor; the remaining factor of c_A is coprime to the remaining modulus. Thus it is growth of the required prime valuations, not growth of the product as a real number, that ensures divisibility.

For the actual radices the product is P_n / P_A. Its valuations at $2, 3, 5$ all tend to infinity, so every fixed D_{sm} eventually divides c_n, for any integral initial value c_A. Under rationality $S = N/D$ in lowest terms, one may start at $A = 1$: $c_1 = D X_1 = N - D$ is integral and coprime to D_{sm}. Since $P_1 = 2$, the criterion becomes $D_{\mathrm{sm}} \mid P_n / 2 = h_n$, exactly the onset already obtained from the finite-prefix identity in Proposition 9.4. This is an alternative proof of the divisibility step, not an additional arithmetic hypothesis left to verify for S. The finite-prefix proof in Theorem 9.3 remains valid.

Proposition 13.2 (segments satisfying the residue inequality exclude a positive bounded integer solution)**.** *Let (b_n) and (m_n) be sequences of nonnegative integers, let $G : \mathbb{N}_{>0} \times \mathbb{N} \to \mathbb{N}$, and assume the residue condition (20) for these sequences and G, using $|W_{\ell,h}| > 0$ as the modulus. Fix $B > 0$ coprime to 30. There is no integral sequence (d_n) satisfying simultaneously $d_{n+1} = b_n d_n - B m_n$, $d_n > 0$ and $|d_n| \le G(B, n)$ for every $n \ge 0$.*

Proof. Choose one segment satisfying the residue inequality (ℓ, h). The iterated recurrence gives $d_{\ell+h} \equiv -BF_{\ell,h}$ modulo $|W_{\ell,h}|$. The endpoint value is positive and at most $G(B, \ell+h)$, whereas the least positive residue of the right-hand side exceeds that bound, so Proposition 10.1 applies. □

Product growth gives room for a large residue; it does not put the actual residue there. For the constant data $b_n = 2$, $m_n = 1$, the positive solution is $d_n = B$. Every segment has $W = 2^h$, $F = 2^h - 1$ and $-BF \equiv B \pmod{W}$. Its least positive residue is therefore at most B, however large W becomes, so no bound $G(B, n) \geq B$ can pass the test. For the actual three-prime coefficients and $G = K$, Theorem 10.2 makes the residue-inequality hypothesis equivalent to irrationality of S. That equivalence does not hold for an arbitrary choice of G.

Coprimality with 30 selects the denominators covered by the hypothesis; once a segment is fixed, the finite contradiction does not use it. The conventions cover the edge cases: a zero product of the bases is excluded, a zero residue is represented by the full modulus, and positivity prevents the endpoint value from vanishing. The version stated with a nonzero smooth factor assumes the exact factorisation $c_n = D_{\mathrm{sm}} d_n$, the corresponding recurrence for (c_n) and the positive upper bound for (d_n), and gives the same contradiction.

Even retaining the actual radix sequence and quadratic growth does not force irrationality if the numerators are changed. For the same b_a, prescribe the positive integer carries $(a+3)^2$ and set

$$\widehat{m}_a = b_a(a+3)^2 - (a+4)^2.$$

The shift by three ensures positivity even at $a = 0$: since $b_a \geq 2$,

$$\widehat{m}_a \geq 2(a+3)^2 - (a+4)^2 = a^2 + 4a + 2 > 0.$$

Both these integer coefficients and the prescribed carries have quadratic order, because $2 \leq b_a \leq 30$. Nevertheless, finite telescoping gives

$$\sum_{a=0}^{N-1} \frac{\widehat{m}_a}{P_{a+1}} = 9 - \frac{(N+3)^2}{P_N} \longrightarrow 9.$$

This example keeps the exact radices, not the actual block multiplicities or their numerical upper bound Q. It shows why the arithmetic of the specific m_a, rather than the radix alphabet and growth orders alone, must enter a proof for S.

13.3 Worked finite examples

At the first ten integer cutoffs, the running LCM changes by a factor 2 at $2, 4, 8$, by 3 at $3, 9$, and by 5 at 5. The table in Section 3 also shows constant values on $\{5, 6, 7\}$ and $\{9, 10\}$, as in Proposition 3.2. In particular, $\mathrm{L}(10) = 8 \cdot 9 \cdot 5 = 360$ is the LCM of $1, 2, 3, 4, 5, 6, 8, 9, 10$, the supported integers at most 10.

For Proposition 3.4 take $(p, q, r) = (2, 3, 5)$ and the box $\mathcal{B}(1, 1, 1)$, whose eight points carry the smooth values $1, 2, 3, 5, 6, 10, 15, 30$ and the running LCMs

$p^i q^j r^k$	1	2	3	5	6	10	15	30
H	1	2	6	60	60	360	360	10800

Six running LCMs occur, two of them twice: the points 5 and 6 share the running LCM 60, and 10 and 15 share the running LCM 360. The identity reads

$$1+\tfrac{1}{2}+\tfrac{1}{6}+\tfrac{1}{60}+\tfrac{1}{60}+\tfrac{1}{360}+\tfrac{1}{360}+\tfrac{1}{10800}=1+\tfrac{1}{2}+\tfrac{1}{6}+\tfrac{2}{60}+\tfrac{2}{360}+\tfrac{1}{10800}=\tfrac{18421}{10800},$$

and the two coefficients 2 carry the whole content of the regrouping on this box. Where the running LCMs are pairwise distinct the identity is a relabelling.

Multiplying block radices along a run of blocks gives the product of the prime multipliers at the jumps in that run: for instance $b_1 b_2 = 60$ is the product of the four multipliers at $3, 4, 5, 8$. Block 4 is the only one among the first six containing an internal power of each odd prime, whereas block 5 contains neither, so its radix falls back to the terminal factor alone.

13.4 The finite-scan histogram

For completeness we reproduce the first-success counts from the larger scan in Section 11. It tested $B \le 5000$ coprime to 30 and $100 \le \ell \le 3000$, stopping each search at depth 24. The saved output (scan-5000-100-3000.json) gives 3,869,934 pairs, a segment satisfying the inequality in every case, and the following first-success histogram:

h	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18
#	1	104	812	5437	51409	237423	735450	1431226	1132756	236752	34910	3076	521	49	8

The first case attaining the maximal observed length 18 has $B = 917$ and $\ell = 2980$. The smaller run (scan-1000-100-500.json) over $B \le 1000$ coprime to 30 and $100 \le \ell \le 500$ gives 106,666 pairs and maximal first successful length 14, first attained at $B = 359$ and $\ell = 291$. The two cases have the following data.

	Larger scan	Smaller scan
Denominator B	917	359
Start ℓ	2980	291
Length h	18	14
Endpoint jump index	6179	627
Product of bases	18139852800000000	5038848000000
Accumulated numerator	13196471407660025821045	25864575212865807
Least positive residue	76322101735	213175287
Upper bound	3896420420	15932659

These histograms concern a bounded region. The product-growth law gives a necessary lower bound on a possible length of such a segment, not a distribution law for residues or first successful lengths. No generic-residue model is used as evidence for the residue inequality at arbitrarily late starts.

13.5 Source inventory

The table gives declaration locations at their recorded revisions. Section 12.5 explains the assumptions attached to each kind of association. Before using the table, two conventions require attention: integer division at the initial index and removal of the first

block. The Lean version of the normalising factor h_a is a natural number computed by integer division: its value at 0 is 0, whereas $h_0 = 1/2$. It agrees with h_a for $a \geq 1$, as the identity for half the running LCM assumes. The Lean version of the tail X_a divides after casting to $\mathbb{R}$, so agrees with $X_a = h_a T_a$ at every index.

The denominator-clearing lemmas and the lemmas on division of an integer recurrence assume T_1 rational. The first block consists of 1 alone, so $T_1 = S - 1$; thus $S = N/D$ supplies

$$T_1 = \frac{N - D}{D}, \qquad \gcd(N - D, D) = \gcd(N, D).$$

Lean proves this translation, which preserves the denominator, its smooth factor and the clearing onset.

informal statement	linked Lean source
integer represented by an exponent triple	smooth3 val
pure-power running LCM	three prime height
lattice kernel	three prime kernel q
exponent triples below the cutoff	smooth prefix exponents
running least common multiple	smooth prefix lcm
prefix value divides the running LCM	smooth3 val dvd three prime height of mem
divisibility into the running LCM	smooth prefix lcm dvd three prime height
first pure-power membership	pure first mem smooth prefix exponents
second pure-power membership	pure second mem smooth prefix exponents
third pure-power membership	pure third mem smooth prefix exponents
running-lcm identity	smooth prefix lcm eq three prime height
cell relation	same three prime log cell
cell constancy of the running LCM	three prime height eq of same log cell
cell constancy of the running value	smooth prefix lcm eq of same log cell
cell constancy of the kernel	three prime kernel q eq of same log cell
positive power sets	positive prime powers
counting positive prime powers	positive prime powers card

informal statement	linked Lean source
exclusion of the origin	one not mem positive prime powers
disjoint positive prime powers	positive prime powers disjoint
union of positive prime powers	three prime positive jump set
positive jump count	three prime positive jump set card
jump set with the origin	three prime jump set with origin
jump count with the origin	three prime jump set with origin card
first running LCM step	three prime height first log step
second running LCM step	three prime height second log step
third running LCM step	three prime height third log step
first coordinate step	smooth prefix lcm first log step
second coordinate step	smooth prefix lcm second log step
third coordinate step	smooth prefix lcm third log step
exponent box	smooth exponent box
point running LCM	smooth point height
running LCM fibre	smooth height fiber
fibre sum	smooth height fiber kernel sum
grouping equal denominators in a finite sum	finite smooth kernel sum grouped by height
upper bound by the cube of the cutoff	three prime height le cube
origin value	kernel 235 origin
value at two	kernel 235 two
value at three	kernel 235 three
value at six	kernel 235 six
example excluding a product of two factors	kernel 235 not rank one
variable-base tail step	tail state step
expanded tail step	tail state step eq
smooth exponent block	smooth exponent shell
short-interval uniqueness	exponent unique in short interval
first-coordinate projection	smooth exponent shell card le drop first
third-coordinate projection	smooth exponent shell card le drop third
sorted quadratic estimate	sorted pair quadratic

informal statement	linked Lean source
quadratic block bound	smooth exponent shell card quadratic
dyadic internal power	dyadic internal power
internal-power uniqueness	dyadic internal power exponent unique
dyadic block base	dyadic block base235
exact dyadic radix alphabet	dyadic block base235 cases
bounded-radix consequence	dyadic block base235 mem interval
a nonzero 2×2 determinant	kernel 235 minor eq neg one fifteen
no integer rotation orbit	no integer orbit logb of prime
running LCM factorisation	three prime height factorisation
kernel factorisation	three prime kernel q factorisation
two-prime outer product	two prime kernel q eq outer product
two-prime vanishing minors	two prime kernel q minor two eq zero
uniform minors, general form	exists uniform nonsingular three prime kernel minor
uniform minors for primes	exists uniform nonsingular three prime kernel minor of prime
no finite separation	not finite separable three prime kernel
second-order minor	kernel 235 minor2 eq neg one fifteen
third-order minor	kernel 235 minor3 eq one over 81000
misleading proportional row	kernel 235 row three eq smul row zero
failure of that proportionality	kernel 235 row three ne smul row zero at four
integer numerator for one dyadic block	dyadic ordered block digit235
recurrence for normalised tails	dyadic normalized tail state r235 succ
block summability	summable dyadic shell mass r235
infinite tail recurrence	dyadic normalized shell tsum tail r235 succ
integer tails or separation from the integers	dyadic shell tsum tail integer or cofinal far
boundary divisibility	smooth height mul prime dvd boundary height
half the running LCM for $a \geq 1$	two mul height normalizer235
segment clearing identity	height normalizer235 mul window mass eq int
clearing a rational T_1 at $a \geq 1$	qsmul normalized tail state eq int of value eq rat
congruence between normalised tails	exists normalized tail state collision of value eq rat

informal statement	linked Lean source
least positive residue	least positive residue
positive representative range	least positive residue pos le
representative congruence	least positive residue mod eq
residue inequality	residue escapes window
excluding a bounded positive natural number	no bounded positive state of residue escape
contrapositive form	residue le bound of bounded positive state
excluding a bounded positive integer	no bounded positive int state of least positive residue
identifying the least positive representative	least positive residue eq nat abs of pos le mod eq
product of the bases	window base
accumulated numerator in a segment	window forcing
affine iterated recurrence	affine recurrence window
scaling the accumulated numerator	window forcing const mul
iterating an integer recurrence	integral carry window
endpoint residue identification	least positive residue window forcing eq carry
divisibility by the smooth part of the denominator	smooth3 val dvd three prime height of le
common-factor cancellation	integral carry cancel common factor
bound after dividing out the common factor	reduced carry pos le of common factor
iterated recurrence after division	reduced integral carry window
cofinal segment hypothesis	cofinal local window escape
the residue inequality excludes a positive integer solution of the reduced recurrence	no positive reduced carry of cofinal local window escape
the same contradiction with a common factor	no positive absorbed carry of cofinal local window escape
real iterated recurrence	true normalized state window

informal statement	linked Lean source
irrationality implies the residue inequality, general bound	cofinal local window escape of irrational of beaten
irrationality implies the residue inequality, quadratic family	cofinal local window escape of irrational of quadratic
irrationality implies the residue inequality, actual bound	cofinal local window escape of irrational
residue criterion equivalent to an irrational tail	actual cofinal local window escape iff
residue criterion equivalent to an irrational sum	actual cofinal local window escape iff irrational value
rational T_1 gives eventual integer carries	exists reduced carry of value eq rat
residue condition for the actual series	actual cofinal local window escape
irrationality from the residue condition	irrational value of cofinal local window escape
bounded-radix alternative	bounded radix zero or cofinal far
rational value from an integral scaled tail	rational of scaled tail integer
zero block sums as differences of potentials	channel block null iff channel potential
a nonzero error exceeds the bound at one index	no carry lift of error bound below two pow
a nonzero error cannot stay uniformly bounded	no bounded carry lift of block null two anchors
first-block sum	binary lift two anchors force first block sum one
four consecutive values cannot meet all three conditions	no unit accurate lift with two anchors and first two block null
carry as a residue plus a potential difference	carry eq residue digit add coboundary
carry interval	carry residue mem interval
digit interval	residue digit mem interval
map from formal expressions to functions	cartier realisation

informal statement	linked Lean source
finite-dimensional span of the resulting functions	finite realised span of factorisation
interval containing a Cantor-series tail	state mem ioo
lower bound for a difference of tails	state gap lower bound
small-form criterion	irrational of small nonzero forms
irrationality from clearing and small nonzero gaps	irrational of clearing and small gaps
indices giving the threshold staircase	staircase indices
finite threshold-column rank, including the extreme-column defect	finite cut rank
uniform admissible modular minors, with indices chosen before modulus and layer	uniform modular minors

14 Dependencies and remaining questions

The table separates the established special cases from the ordinary generalisations awaiting formalisation and the unresolved repeated-sum condition. Exact kernel rank concerns separation of a function; the residue criterion concerns a scalar sum and still needs arithmetic existence for its segments.

Result	Dependency and limit
Running LCMs and intervals of constancy	Prime-power divisibility and unique factorisation. Their geometry alone gives no irrationality statement.
Two-prime values	Earlier factorisation plus the cited external Hecke–Mahler value theorem. The reduction to that theorem is checked in Lean; the theorem itself is a cited input (Section 14).
Distinct values, every P	Integrality of the scaled tails, rearranged chains at a return of the logarithmic flow, Kronecker's theorem and an Euler characteristic count on a two-dimensional subtorus. Ordinary proof checked by a second agent; no Lean proof. It stops for the repeated sums because their scaled tails are unbounded.
Prime-power subseries	The same two-tail comparison with the integer f_p and a chain chosen near the origin of the torus; no two-wall lemma. Ordinary proof checked by a second agent; no Lean proof. Says nothing about the whole repeated sum.
Distinct values, blocks	Dyadic blocks, finitely many affine maps with disjoint images, integrality of the scaled tails and the irrationality of $\log p/\log q$. The theorem for $\{2,3,5\}$ is checked in Lean; the general criterion has an ordinary proof, and Remark 5.8 adds exact finite computations. No statement about the repeated sums.
Arbitrary-order rank	Diagonal rescaling, separate one-dimensional densities and a staircase determinant; checked in Lean.
Tail recurrence	Counting the original multiplicities, normalising by half the boundary running LCM, and convergence; an abstract recurrence would not identify the sum.
Denominator reduction	A finite prefix with cleared denominators and an upper bound for positive integer tails. The first valid index is determined exactly (Proposition 9.4); checked in Lean.
Residue criterion	The actual recurrence, a dominating bound and segment products comparable to 8^h; the inequality at arbitrarily late starts remains unproved.
Weighted differences	Exact reindexing and the contributions of the new boundary strips; integer enumeration checks the displayed counterexample to third-difference cancellation.
Finite computations	Direct checks settle only their individual segments. The two large exclusions have published witnesses and exact checkers, and the larger scan a published saved run; none is a theorem input.

None of these results proves irrationality of the repeated sum $\mathcal{R}_P$ for three or more primes.

The formal two-prime reduction

The source note in Section 12.5 distinguishes ordinary proofs from individual formal associations. For Theorem 1.2, the Lean development verifies the reduction of that theorem to the transcendence of the Hecke–Mahler boundary sum at an irrational $\theta \in (0,1)$ and nonzero algebraic β, α with $|\beta| < 1$ and $|\beta||\alpha|^{\theta} < 1$, which is Theorem 1.1 of Bugeaud and Laurent [15] in the case $\rho = 0$, due to Loxton and van der Poorten [16, Theorem 8, p. 40]. Lean does not prove that input; the Lean theorem assumes it. The formal development checks the following parts of the reduction: the running least-common-multiple identity, the selection of the single power of q between consecutive powers of p, the factorisation of the repeated sum, the identification of the distinct running values, the telescoping identity, the boundary identity (6) expressing A through the Hecke–Mahler sum, the irrationality of $\log p / \log q$, and each side condition verified before the value theorem is applied. The two identities expressing $\mathcal{D}_{\{p,q\}}$ and $\mathcal{R}_{\{p,q\}}$ as polynomials in A are unconditional and carry no additional input.

A theorem whose own statement is conditional is formalised exactly as stated. Theorem 1.2 is asserted unconditionally in this record, and its Lean counterpart carries the additional named input described here.

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Verification concordance

Statement	Formal proof	Statement	Formal proof
Theorem 1.1	Lean · Comparator pending	Theorem 1.2	Lean†
Theorem 3.1	Lean · Comparator	Proposition 3.2	Lean · Comparator
Proposition 3.3	Lean · Comparator	Proposition 3.4	Lean · Comparator
Lemma 3.5	Lean · Comparator	Proposition 3.6	Lean · Comparator
Remark 9, p. 10	Verification record	Remark 10, p. 11	Verification record
Remark 11, p. 11	Verification record	Remark 12, p. 12	Verification record
Remark 13, p. 12	Verification record	Remark 14, p. 15	Verification record
Remark 15, p. 17	Verification record	Remark 16, p. 17	Verification record
Remark 17, p. 20	Verification record	Remark 18, p. 20	Verification record
Remark 19, p. 22	Verification record	Remark 20, p. 23	Verification record
Remark 21, p. 24	Verification record	Proposition 7.1	Lean · Comparator
Proposition 7.2	Lean · Comparator	Theorem 7.3	Lean · Comparator
Proposition 7.4	Lean · Comparator	Remark 26, p. 28	Verification record
Theorem 7.6	Lean · Comparator	Proposition 8.1	Lean · Comparator
Theorem 8.2	Lean · Comparator	Lemma 8.3	Lean · Comparator
Proposition 8.4	Lean · Comparator	Theorem 9.1	Lean · Comparator
Lemma 9.2	Lean · Comparator	Theorem 9.3	Lean · Comparator
Proposition 9.4	Lean · Comparator	Proposition 9.5	Lean · Comparator
Proposition 9.6	Lean · Comparator	Proposition 10.1	Lean · Comparator
Theorem 10.2	Lean · Comparator	Proposition 10.3	Lean · Comparator
Proposition 10.4	Lean · Comparator	Corollary 10.5	Lean · Comparator
Lemma 12.1	Lean · Comparator	Proposition 12.2	Lean · Comparator
Proposition 12.4	Lean · Comparator	Proposition 12.6	Lean · Comparator
Proposition 13.1	Lean · Comparator	Proposition 13.2	Lean · Comparator